



Hardware User Manual



JAKA Mini Series

Original Instructions (en)

File Version: V03

Robot: MiniCobo, Mini 2

Control Cabinet: MiniCab

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The pictures in this manual are for reference only.

If the robot body is transformed or disassembled, JAKA will not be responsible for after-sales services.

JAKA reminds users that they must use safety equipment when using and maintaining JAKA robots and must comply with the safety terms.

Programmers of JAKA robots, designers and debugging personnel of robot systems must be familiar with the way to program JAKA robots and install system applications.

More Information

For more product information, scan the QR code on the right to visit our official website: www.jaka.com.



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Foreword

The robot pioneers in the control mode with a mobile smart terminal plus an App connected to the robot, so that one mobile terminal can match multiple robots. Operators do not need to master professional programming technology, but only need to manually guide the robot to complete the programming. In this way, human-machine collaboration is easier, which greatly improves the productivity.

JAKA Mini are intelligent, lightweight, 6-DOF modular collaborative robots with MiniCobo and Mini 2 available for light industry, education, new retail and other scenarios.

Product List

When purchasing a complete set of the JAKA Mini robot, the delivery will include the following items as shown in [Table 0-1](#).

Table 0-1: JAKA Mini series item list

No.	Name	Quantity
1	Robot	1
2	Control cabinet	1
3	Control stick	1
4	24 V DC power adapter / 48 V DC power adapter	1
5	Control cabinet power cord	1
6	Robot connection cable	1
7	Product qualified certificate	1
8	Quick start guide	1
9	After-sales service warranty card	1

Important Safety Notice

According to 2006/42/EC Machinery Directive, the JAKA robot is a **partly completed machine**, and a risk assessment must be performed for each robot installation. All safety instructions in chapter 2 must be followed.

1 Manual Instruction

1.1 About the Manual

The manual contains:

- Precautions for the use of JAKA Mini series robots.
- Installation and commissioning of JAKA Mini series robots.
- Cleaning and maintenance of JAKA Mini series robots.

1.2 Manual Reader

The manual is intended for:

- Operators.
- Commissioning staff.
- Maintenance staff.
- Integrators.

1.3 References

Documentation referred to in the manual:

- JAKA App Software User Manual.
- JAKA Mini Series Service Manual.



NOTE

All documents can be found via JAKA official website www.jaka.com.

1.4 Prerequisites

The reader should:

- Be trained and certified by JAKA and have the required knowledge of mechanical and electrical installation/repair/maintenance work.
- Be trained to respond to emergencies or abnormal situations.

2 Safety Standard

2.1 Introduction

This chapter mainly introduces the safety principles and standards that should be followed while using the JAKA robot. Users should carefully read and strictly abide by the content related to safety in this manual. Operators should fully recognize the complexity and danger of the robot system, and pay special attention to the content related to warning signs.

2.2 Safety Signals and Symbols

The danger level in this manual is described with the following safety symbols. Contents related to safety should be strictly observed.

Table 2-1: Symbols

Symbol	Description
	WARNING: ELECTRICITY This symbol indicates a potentially dangerous power consumption situation which, if not avoided, may result in injuries to personnel or serious damage to equipment.
	WARNING This symbol indicates a potentially dangerous situation which, if not avoided, may result in injuries to personnel or serious damage to equipment.
	WARNING: HOT SURFACE This symbol indicates a potentially dangerous hot surface that may result in injuries to personnel if touched.
	NOTICE This symbol is used to indicate important facts and conditions.
	NOTE This symbol is used to indicate additional information.

2.3 Warnings and Cautions

This section focuses on the protection of operators and the relevant precautions of the first installation. Users need to read the safety warnings in this manual carefully.

Table 2-2: General warnings

Symbol	Description
	WARNING: ELECTRICAL SHOCK - All JAKA hardware and software must be installed/configured in strict accordance with the instructions and cautions provided in this manual. - The installation of the power cut-off switch should be positioned within the height range of

Symbol	Description
	0.6~1.9 m (23.622~74.803 in) to facilitate prompt and convenient power disconnection in the event of an emergency. - Prior to the initial use of any JAKA products, a comprehensive inspection of all electrical components and safety protection systems must be conducted to verify their integrity and absence of prior damage. - Operators possessing the necessary qualifications for robot operation are instructed to perform a thorough assessment of all safety functions and ensure the accuracy of parameters and programs before initiating the power supply to the robot.
	WARNING: HOT SURFACE Both the robot and the control cabinet generate heat during operation. Avoid contact with the robot and the control cabinet during operation and after shutdown. In both instances allow approximately 1 hour for the equipment to cool down.
	WARNING - Technical personnel are instructed to carry out the installation and commissioning procedures for any JAKA products in strict accordance with the provided specifications. - Adjustment and alteration of any JAKA product parameters must be executed exclusively by authorized personnel to safeguard against unauthorized modifications by individuals lacking appropriate operating expertise. - Frequently toggling the power on/off is not recommended. Each joint of the JAKA robot is equipped with a brake mechanism to maintain its pose for safety reasons in the event of a power failure. Brake mechanisms can be damaged during unexpected power outages. - In the event that the applied force on the robot surpasses a predefined threshold, JAKA's collision detection feature will be triggered and robot motion will be ceased to prevent potential harm to the robot itself or injury to operators. Associated risks of the use of control cabinets not supplied by JAKA is solely the responsibility of the operator.
	WARNING - Ensure the proper installation of both the robot and its associated tools to the manufacturer's specifications. - Verify that there is adequate space available for unobstructed movement of the robot. - To prevent damage, refrain from connecting safety equipment to the standard I/O interface (use safety I/O interface). - Confirm the accuracy of the mounting settings, including mounting angles, Tool Center Point (TCP) position, tool mass, TCP offset and the robot's safety configuration. - Ensure that tools and obstacles do not possess sharp corners and maintain a safe distance between all personnel and the robot. - Connecting the robot to different machinery may increase existing hazards or introduce new ones. Conduct a comprehensive risk assessment for the entire system installation. - Do not make modifications to the robot, as such alterations may create unforeseen hazards for which JAKA cannot be held liable. If the robot is changed or modified in any form, JAKA assumes no liability.

Symbol	Description
	WARNING - When connecting external equipment that may pose a threat to the robot, it is advisable to independently check all robot functions and programs. Utilize temporary waypoints located outside the mechanical workspace to verify the robot's program. - Exposure to strong magnetic fields can damage the robot; hence, avoid exposing it to permanent magnetic fields. - Operators who use the robot system are strictly prohibited to wear loose clothes and jewelry. Long hair should be tied up. - During operation, even if the robot appears to have stopped, it may be in a state of imminent action because it is waiting for a start signal. In such state, the robot should be regarded as running. - During the operation, ensure proper connection of robot power cable and control cabinet. It is strictly forbidden to plug or unplug the power cables and terminals while the robot is running. - Warning lines should be drawn on the floor to clearly mark the workspace of the robot. - Ensure that safety measures (e.g., guardrails, ropes, or protective screens) are in place near the operation area of the robot to protect operators and surrounding persons. Locks should be provided if necessary to make the robot power inaccessible to anyone other than the operator in charge. - In emergency situations, the robot arm may be manually repositioned by gently pushing or pulling the joints. - Avoid excessive force to prevent potential damage to the joints.

2.4 Intended Use

The JAKA robot is intended to be integrated into a manufacturing line and/or assembled with other components to build up a machine or process in an industrial environment. The JAKA robot is either an open type robot that is intended to be installed into an enclosure, or a collaborative operation product where it works with an operator within a collaborative workspace.

JAKA robot intended use cases:

- New retail
- Education
- 3C manufacturing
- Loading and unloading.
- Machine tending.

The JAKA robot is equipped with built in safety-related functions when the robot operates without fences and/or together with an operator.

For JAKA robots, the following terms apply for different types of interactions:

1. Coexistence

As shown in figure 1, operators and robots work simultaneously in close proximity but have separate workspaces and do not work towards a common goal. There is no direct contact between operators and robots.

2. Cooperation

As shown in figure 2, operators and robots work at different times in the same workspace to achieve a

common goal. There is no direct interaction between operators and robots.

3. Collaboration

As shown in figure 3, operators and robots work simultaneously in the same workspace to achieve a common goal, such as the assembly of a product. There is direct interaction between the operator and the robot.

Coexistent, cooperative, and collaborative operations are only intended for non-hazardous applications, where the complete application, including tool, workpiece, obstacles and other machines is without any significant hazards according to the risk assessment of the specific application.

Any other use is not an intended use.

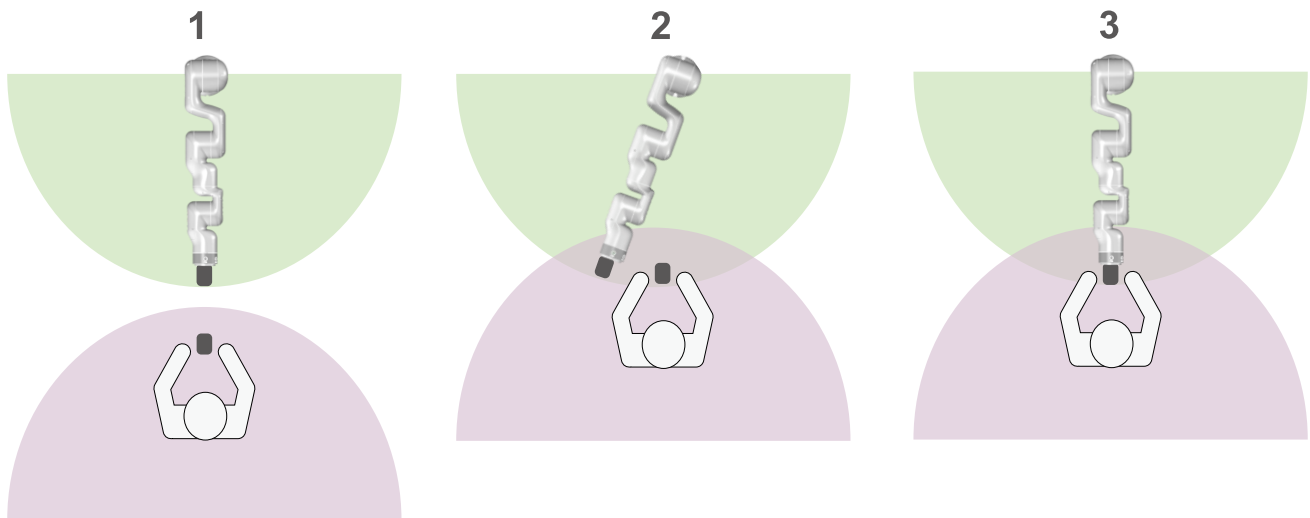


Figure 2-1: Operator interactions with robot



WARNING:

Personal injury:

- Do not use the JAKA robot in a coexistent, cooperative, or collaborative operation if the application could cause potential injuries.
- Please operate the robot in accordance with the requirements of the manual. If the robot is operated in the collaborative mode, additional safety input detection units must be used (such as scanners, light curtains, cameras, etc. with PLr=d or PLr=e), and a risk assessment must be conducted.

Failure to follow these instructions can result in equipment damage, serious or fatal injury.

2.5 Liability and Risk

Liability

This manual does not contain full application examples, complete solution designs, or information on peripheral equipment that could affect the robot system's security.

It is the responsibility of the user to ensure that relevant practical national laws and regulations are followed, and that there are no significant hazards in the workspace.

The safety information in this manual does not constitute a guarantee from JAKA. Even when all safety instructions are followed, operator actions may still result in injury or damage.

JAKA is committed to continuously improving the performance and reliability of our robots. However, we are not responsible for any errors or omissions in this manual and reserve the right to the final interpretation of its contents.

Risk

Operators engaging with the robot will inevitably experience direct or indirect physical contact. It is essential to

maintain awareness of self-protection and adhere to safe operating procedures. The following hazards should be considered:

- Robot brake failure causing it to drop.
- Loose bolts and screws causing vibrations.
- Robot collision with operator.
- Lack of prompt repair.
- Danger when sharp-end effectors are used.
- Toxic or corrosive environments.
- Strong magnetic environment.

2.6 Provide Protective Measures

Before installing the JAKA robot, provide appropriate protective devices in compliance with local and national standards. Do not commission components without appropriate protective devices. After installation, commissioning, or repair, test the protective devices used.

Other standards are applicable as guidelines for a JAKA robot integration into machinery such as (non-exhaustive list):

- Directive 2006/42/EC on machinery.
- Directive EMC 2014/30/EU.
- Standard ISO 10218-1 Robots and Robotic devices - Safety requirements for industrial Robots - Part 1: Robots.
- Standard ISO 10218-2 Robots and Robotic devices - Safety requirements for industrial Robots - Part 2: Robot systems and integration.
- Standard ISO 13849-1 Safety of machinery - Safety related parts of control systems - Part 1: General Principles for Design.
- Standard ISO/TS 15066 Robots and Robotic devices - Collaborative Robots.
- Standard ISO 13857 Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs.
- Standard ISO 14120 Safety of machinery - Guards - General requirements for the design and construction of fixed and movable guards.
- Standard EN ISO 13854 Safety of machinery - Minimum gaps to avoid crushing of parts of the human body.
- Standard ISO 13855 Safety of machinery - Positioning of safeguards with respect to the approach speeds of parts of the human body.
- Standard NFPA 79 Electrical Standard for Industrial Machinery.
- Standard NFPA 70 National Electric Code.
- Standard UL 1740 Standard for Robots and Robotic Equipment.
- Standard UL 2011 Standard for Factory Automation Equipment.

Perform a risk evaluation concerning the specific use before operating the JAKA robot and take appropriate security measures.

**WARNING:****Unintended:**

- Use appropriate protective devices (functional safety devices) in compliance with local and national standards.
- Ensure that a risk assessment is conducted and respected according to EN/ISO 12100 during

the design process.

- Apply all measures from the hazard and risk analysis before deployment.

Failure to follow these instructions can result in equipment damage, serious or fatal injury.

2.7 Risk Assessment

Robots are partially completed machines, and their safe installation depends significantly on the integration process, including components such as end effectors and communication equipment. Conducting a risk assessment is essential and a critical responsibility of the integrator. It is recommended that the integrator performs the risk assessment in accordance with ISO 12100 and ISO 10218-2, while also considering ISO/TS 15066 as additional guidance. The risk assessment should cover all aspects of the robot's lifecycle, including but not limited to:

- Teaching of the robot during installation, set-up and development.
- Operations of the robot installation.
- Troubleshooting and maintenance.

The risk assessment must be conducted before powering on the robot for the first time. It is the integrator's responsibility to determine the appropriate safety configuration settings and evaluate the need for additional emergency stop buttons or other protective measures based on the specific robotic application.

Cobots have specific safety functions, which can be configured through settings. These functions are especially important when integrators conduct risk assessments:

1. **Force limit:** the force set by the controller to limit the servo from exceeding the threshold. When the actual force exceeds the force limit, the robot will enter collision protection mode.
2. **Momentum limit:** the momentum set by the controller to restrict the momentum of the robot during motion. The limit will directly affect the speed of the robot. When the momentum of the robot exceeds the limit, the robot speed will be reduced.
3. **TCP speed limit:** refers to limiting the absolute speed of the TCP during the robot motion. The path speed will be kept within the TCP speed limit.
4. **Power limit:** refers to limiting the mechanical power during the robot movement. This limit will directly affect the speed of the robot. When the mechanical power of the robot exceeds the limit, the robot speed will be reduced.

Integrators must prevent unauthorized personnel from modifying safety configurations.

When performing a risk assessment, the integrator should consider the following:

- Potential collision scenarios.
- How to avoid potential collisions
- The severity of the potential collision.

If the robot is installed in a non-collaborative robot application and the risk cannot be eliminated by configuring the robot's safety functions, integrators should consider adding additional protective measures when conducting assessments.

JAKA identifies the following significant dangers that integrators must consider.

- Sharp edges and points on end effectors.
- Sharp edges and points on obstacles in and near the robot workspace.
- Bruises due to contact with the robot.
- Sprain or fracture caused by the impact between the heavier load on the end of the robot and the hard surface.
- Consequences caused by loose bolts or screws used to fasten robots or end effectors.
- Consequences of items falling from end effectors.

- Mis-operation due to different emergency stop buttons on different machines.
- Error due to unauthorized changes to safety configuration parameters.

**NOTICE**

Specific robotic applications may present other significant hazards.

2.8 Pre-use Assessment

After first use or any modifications, perform the following tests:

- Verify that all safety inputs and outputs are correctly connected.
- Test the functionality of all safety inputs and outputs.
- Ensure the payload is configured correctly.

The following tests are required:

- Test if the emergency stop button and emergency stop input (P8) can stop the robot and engage brakes.
- Test whether the safeguard input can stop the robot motion. If safeguard reset is configured, check if activation is required before resuming motion.
- Check whether the reduced mode input can switch the motion to the reduced mode.
- Test whether the 3-position enabling device must be pressed to enable motion in manual mode and if the robot is under deceleration control.
- Test whether the emergency stop output of the system can bring the entire system into a safety state.
- Test whether the system connected to the robot moving output, robot non-stop output, reduced mode output, or non-reduced mode output can detect output changes.
- Test whether the payload configuration matches the current actual payload of the robot.

2.9 Emergency Stop

In the event of an emergency, press the emergency stop button on the control stick to stop all the movements of the robot immediately. Emergency stop cannot be used as a risk reduction measure but it can be regarded as secondary protection equipment. If the robot needs to be stopped under normal circumstances, please adopt other measures. After risk assessment, if an emergency stop button needs to be installed, the button must meet the requirements of IEC-60947-5. The emergency stop of JAKA's robot has been tested. See [Appendix 1: Stopping Time and Distance](#) for the test data.

**WARNING**

When the emergency stop button is pressed, the cabinet power will cut off. In this case, even though the joints lock automatically, there will still be a slight downward movement of the robot under gravity, and therefore a risk of pinching or collision.

2.10 Emergency Release of the Brake

**WARNING**

- If the brake is released manually, the joint of the robot may move under gravity, so it is necessary to effectively support the robot, tools and workpieces installed on the robot before manually releasing the brake.

- At least two persons should be present when releasing the brakes.

When the robot is in emergency stop state, the power supply will cut off and the robot joints can be forced to move by unscrewing the joint lid and pressing the brake pin as shown in [Figure 2-2](#).

1. Remove the screws on the joint lid.
2. Remove the joint lid.
3. Press the pin in the small electromagnet to manually release the brake.

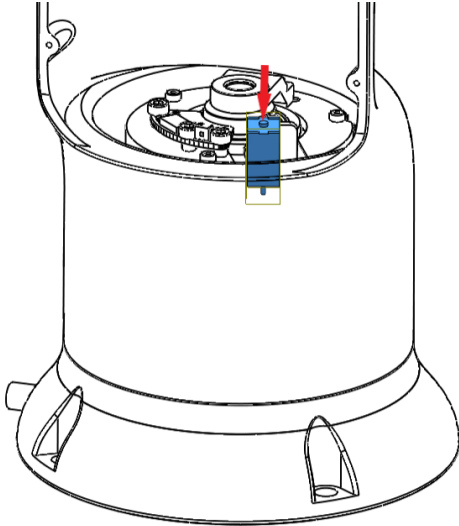


Figure 2-2: Brake pin

When the robot is powered on, the steps to release the brake by the JAKA App are as follows:

1. Ensure that the robot is powered on and disabled.
2. Open the JAKA App and go to the “Manual Operation” interface.
3. Press and hold the “Backdrive” button to release the joint brake (shown in [Figure 2-3](#) below).

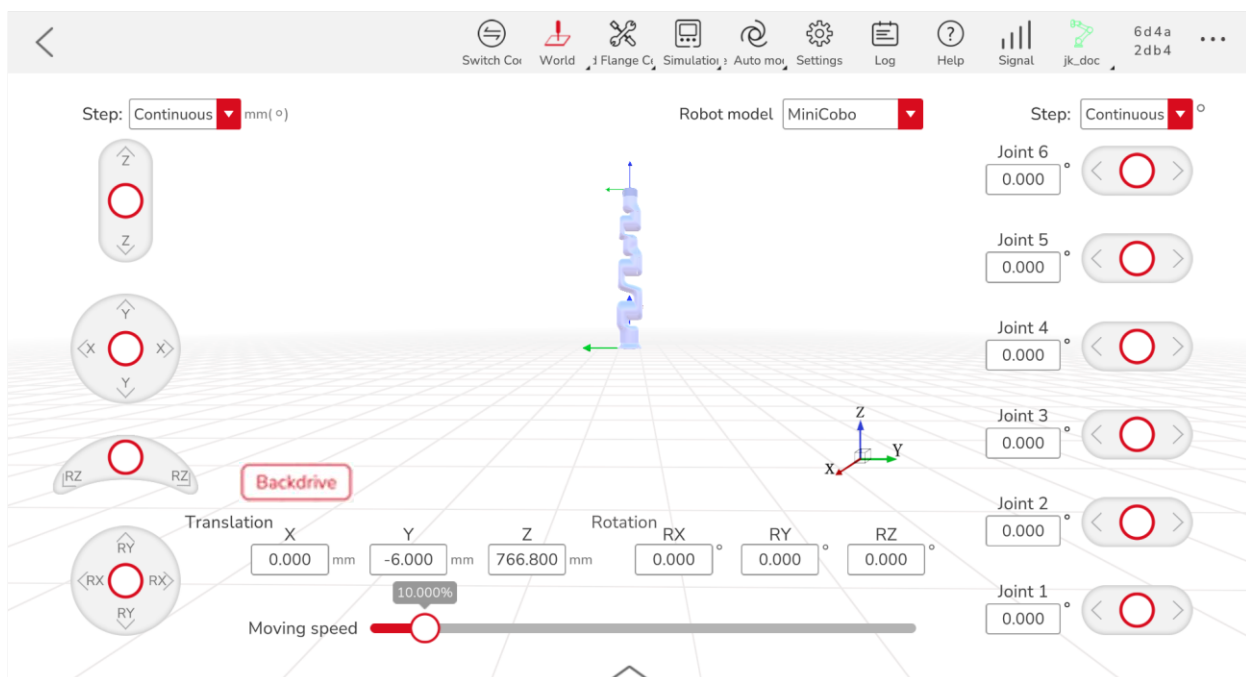


Figure 2-3: Backdrive button on JAKA App

4. Press and hold the FREE button on the flange (refer to [5.3.2 Buttons](#) for its position) to release the joint brake, and then this joint can be dragged.



NOTE

This function only supports JAKA App versions 1.7.2 and above and servo versions R3214 and above.

2.11 Labels

The following labels are safety warnings and product labels on the robot and the control cabinet. During operation, be sure to follow the instructions and warnings on the labels to ensure safety. Do not remove the labels casually. Handle labeled parts or units and their surrounding area with caution to avoid damage to the labels. Product labels and robot model figures are only for reference. Please figures and tables below for detailed description on product labels.

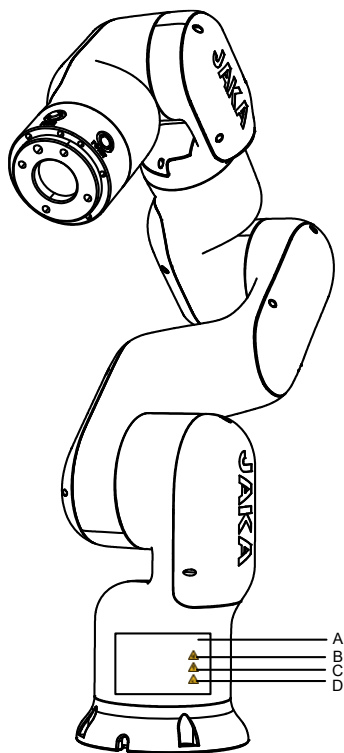


Figure 2-4: Robot label and symbols

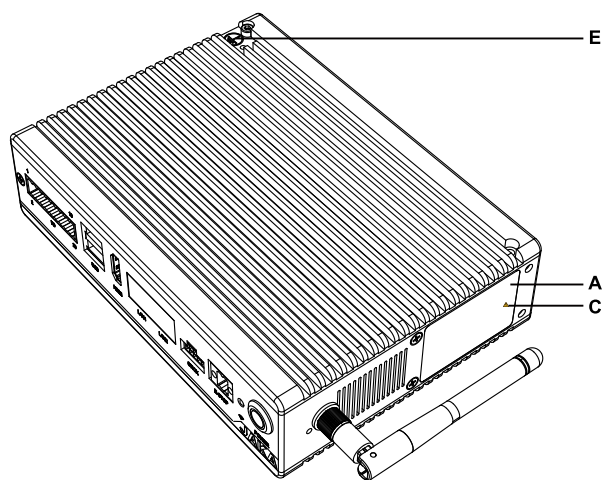
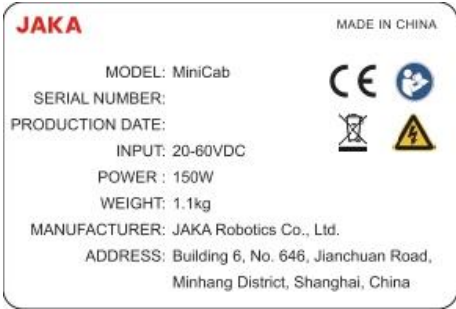


Figure 2-5: Control cabinet label and symbols

	Labels	Description
A	 JAKA MADE IN CHINA MODEL: JAKA MiniCobo SERIAL NUMBER: PRODUCTION DATE: INPUT: 24VDC POWER: 150W WEIGHT: 9.4kg MAX. PAYLOAD: 1.0kg MAX. REACH: 580mm MANUFACTURER: JAKA Robotics Co., Ltd. ADDRESS: Building 6, No. 646 Jianchuan Road, Minhang District, Shanghai	Robot label
	 JAKA MADE IN CHINA MODEL: MiniCab SERIAL NUMBER: PRODUCTION DATE: INPUT: 20-60VDC POWER : 150W WEIGHT: 1.1kg MANUFACTURER: JAKA Robotics Co., Ltd. ADDRESS: Building 6, No. 646, Jianchuan Road, Minhang District, Shanghai, China	Control cabinet label
B		Beware of collision
C		Beware of electrical shock
D		Beware of pinching
E		Grounding

NOTE

- The serial numbers of both the robot and control cabinet are on the respective labels. The labels of robots are on the upper part of the base, and the label of control cabinet is located beside the antenna.
- Product labels and robot figures are only for reference, labels vary by robot and control cabinet model.

3 System Overview

3.1 System Architecture

Before reading this chapter, ensure that you have read and understand chapter 2 [Safety Standard](#).

This chapter will introduce the basic components and use methods of JAKA robots.. For detailed mechanical and electrical specifications, refer to other chapters.

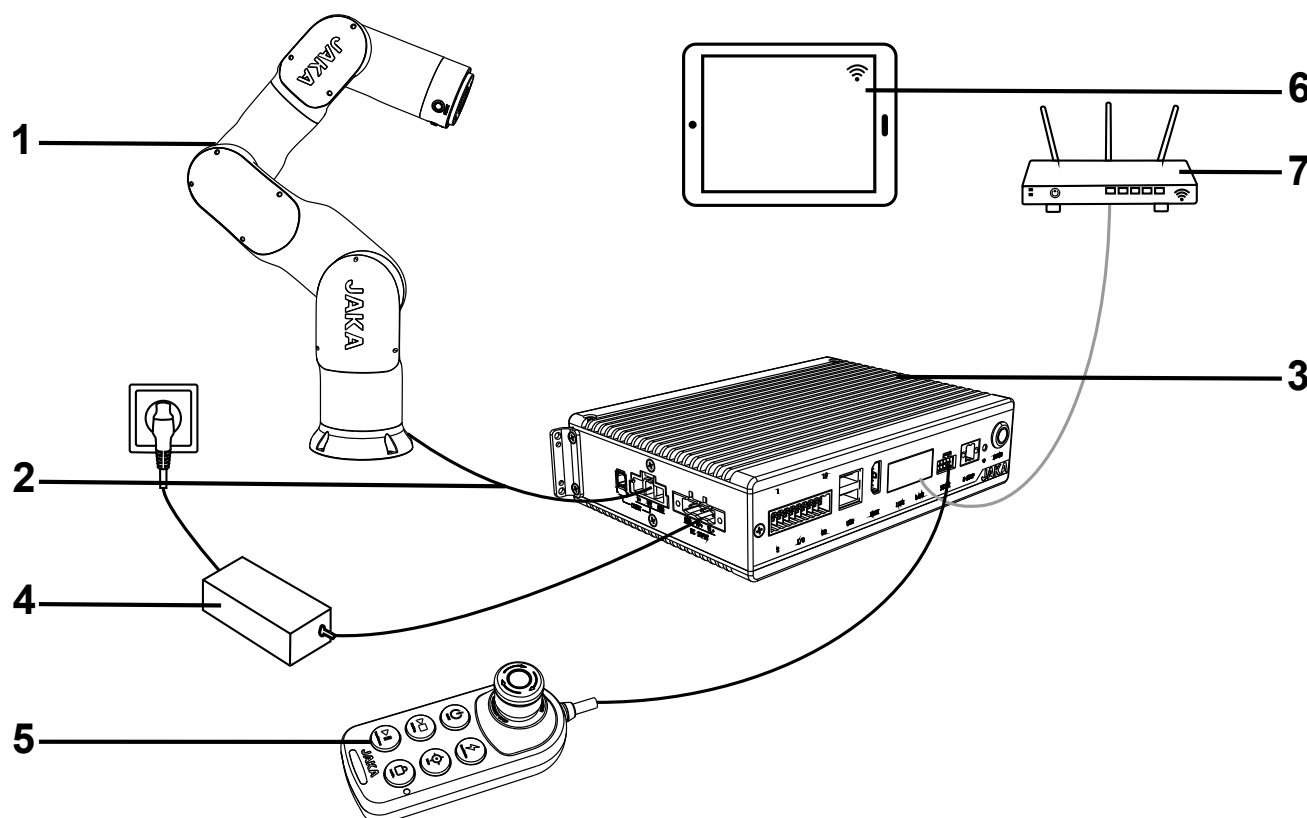


Figure 3-1: Robot system components

As is shown in the figure above, the JAKA robot system contains following components:

1 Robot: A six axes robotic arm which moves as programmed by the user. It consists of buttons for dragging and point teaching and a I/O interface for connecting tools (the TIO interface) on joint 6.

2 Robot connection cable: Cable connection robot to control cabinet.

3 Control cabinet: The control cabinet includes the core computing components and various electrical interfaces.

4 Control cabinet power cord and power adapter: Provide power supply to the control cabinet. The 24 V DC power adapter is for MiniCobo, and the 48 V DC is for Mini 2.

5 Control stick: Power on/off and enable/disable the robot, etc.

6 Operation terminal: The users desired teaching device.

7 Router and network cable: Each control cabinet has a built in Wi-Fi module which is accessible by any android, IOS or windows device. This allows the user to connect the JAKA Mini App to the controller wirelessly and continue programming. It is also possible to connect the cabinet network port of the control cabinet to the router and the operation terminal to the wireless network of this router at the same time. It is recommended to configure a specialized router for the robot to prevent conflicts with other devices.

3.2 Components Overview

The robot contains six joints and two connection arms. The base is used to connect the robot to the foundation, and the tool end flange is used to connect the robot to the end effector. The end effector can move and rotate in the workspace of the robot. This chapter will introduce the basic precautions during the installation of each component of the robot system.

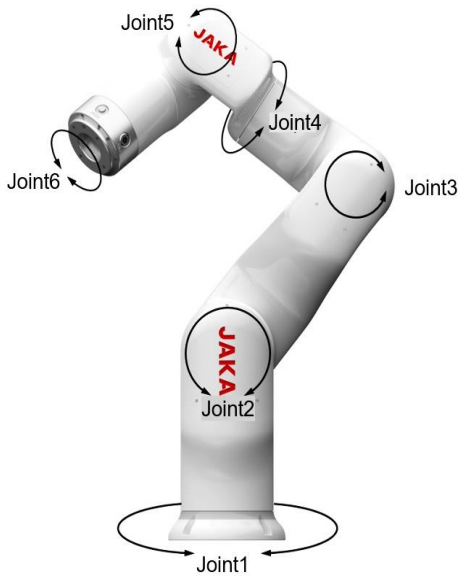


Figure 3-2: Robot

The JAKA control cabinet provides:

- Built in Power supply for the robot
- Interface to connect the control stick
- Interface for the operator terminal either via cable or Wi-Fi connection
- Interfaces for several inputs and outputs of different kinds

A control stick is included in the delivery of the control cabinet, and it can be used for various operations.



Figure 3-3: Control cabinet

4 Safety Functions

The JAKA robot has a series of safety functions to guarantee the safety of human-machine collaboration. This chapter introduces these safety functions, and users should strictly abide by the requirements and precautions.

4.1 Overview of Safety Functions

4.1.1 Specific Safety Requirements

Safety functions comply with EN ISO 10218-1 in general and specifically comply with the following requirements:

When safety related control systems are required, the safety related parts shall be designed so that:

- A single fault in any of these parts shall not lead to the loss of the safety function.
- Whenever reasonably practicable, the single fault shall be detected at or before the next demand upon the safety function.
- When the single fault occurs, the safety function is always performed and a safe state shall be maintained until the detected fault is corrected.
- All reasonably foreseeable faults shall be detected.

This requirement is considered to be equivalent to structure category 3 as described in ISO 13849-1. Category 3 is normally fulfilled by redundant circuits, such as dual channels. Safety functions together with the robot controller also comply with performance level (PL) d according to ISO 13849-1.



NOTICE

It is the responsibility of the integrator to do a risk assessment for the final application and also for a collaborative application. The safety needs to be configured depending on the application, to fulfill the requirements in ISO/TS 15066.

4.1.2 Limitations

Version 1.5 of the JAKA App only supports certain safety functions and some safety function specifications may differ slightly from those in version 1.7. Version 1.7 and above support all safety functions. The safety functions described in this manual are those supported by software version 1.7. For descriptions of safety functions in other versions, please refer to the corresponding version's software user manual.

4.1.3 Stop Category

According to the IEC 60201 standard, three stop categories have been set for JAKA's robot: stop category 0 (Cat. 0), stop category 1 (Cat.1), and stop category 2 (Cat.2). Among them, Cat.0 is uncontrollable stop, while Cat.1 and Cat.2 are controllable stop.

If Cat. 0 is triggered, the power supply of the robot is immediately cut off and the robot stops running immediately.

If Cat.1 is triggered, the robot immediately decelerates to stop, and the power supply of robot will be cut off when all joints enter stand-still condition.

If Cat.2 is triggered, the robot decelerates following the programmed trajectory, eventually all joints enter standstill condition (while the robot remains enabled).

4.1.4 Overview of Safety Function

JAKA robotic system features 27 safety functions, designed and tested with reference to the following standards:

- IEC 60204-1: 2016
- ISO 13849-1: 2023

- ISO 10281-1: 2020

These safety functions can be categorized into the following 8 classes:

- Emergency stop functions
- Safeguard stop functions
- Motion status functions
- Hand-guided mode safety functions
- Speed monitoring functions
- Torque and power limitations functions
- Collision protection safety functions
- Position monitoring functions

The following table provides an overview of the safety functions per group. For detailed information on the specific function descriptions, specific reaction sequence, and the fallback function, refer to the description of the respective group of safety functions.

Table 4-1: Safety function list

	SF	Name	PLr d	Structure Category	Response Time	Safe State
Emergency Stop Functions	SF1	Emergency Stop	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF2	External Emergency Stop	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF15	Additional Emergency Stop Input	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF19	Emergency Stop Button State Output	PLr d	PL d/Cat. 3	250 ms	The safe state is output as "low" or "high impedance" in dual channels.
	SF20	System Emergency Stop State Output	PLr d	PL d/Cat. 3	250 ms	The safe state is output as "low" or "high impedance" in dual channels.
Safeguard Stop Functions	SF3	Safeguard (Protective) Stop	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop
	SF16	Additional Safeguard Stop Input	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop
	SF17	Safeguard Reset Input	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop
	SF21	System Safeguard State Output	PLr d	PL d/Cat. 3	350 ms	The safe state is output as "low" or "high"

	SF	Name	PLr	Structure Category	Response Time	Safe State
						impedance” in dual channels.
Motion Status Functions	SF22	Robot Moving Output	PLr d	PL d/Cat. 3	100 ms	The safe state is output as “low” or “high impedance” in dual channels.
	SF23	Robot Not Stopping Output	PLr d	PL d/Cat. 3	100 ms	The safe state is output as “low” or “high impedance” in dual channels.
Hand-Guided Mode Safety Functions	SF13	Hand-Guided Mode TCP Speed Limit	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop
	SF27	3-Position Enable Input	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop
Speed Monitoring Functions	SF5	Joint Speed Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF9	TCP Speed Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF18	Reduced Mode Input	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF24	Robot Reduced Mode Output	PLr d	PL d/Cat. 3	250 ms	The safe state is output as “low” or “high impedance” in dual channels.
	SF25	Robot Not in Reduced Mode Output	PLr d	PL d/Cat. 3	250 ms	The safe state is output as “low” or “high impedance” in dual channels.
Torque and Power Limitations Functions	SF6	Joint Torque Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF7	Joint Power Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF8	Power Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
Collision Protection Safety Functions	SF12	TCP Position Mismatch Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF14	Collision Protection	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop
	SF26	TCP Force Limit	PLr d	PL d/Cat. 3	350 ms	Cat. 2 Stop

	SF	Name	PLr	Structure Category	Response Time	Safe State
Position Monitoring Functions	SF4	Joint Position Limits (Soft Axis Limits)	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF10	Tool Orientation Limit	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop
	SF11	TCP Position Limit (Safety Planes)	PLr d	PL d/Cat. 3	250 ms	Cat. 1 Stop



NOTICE

The safety function is only a part of the robot system, it is the responsibility of the user to do a risk assessment of the robot system. It is also the responsibility of the user of safety function to ensure that the robot system is designed and installed in accordance with the safety requirements set forth in the standards and regulations of the country where the robot system is installed.

4.2 Descriptions of Safety Functions

4.2.1 Emergency Stop Functions

The JAKA robot system provides the following options to implement an emergency stop:

- Press the emergency stop button on the control stick (SF1)
- Connect the external emergency stop device (for example, light curtains, safety doors, emergency stop buttons or other such device) to the E-STOP interface on the front panel of the control cabinet to perform emergency stop (SF2)
- Configure the “Additional emergency stop input” of the dedicated safety I/O in the JAKA App, and connect external devices to the Pin7&9 or Pin11&13 of the I/O interfaces on the MiniCab to provide emergency stop input, in order to perform the emergency stop (SF15)

When the safety digital input is HIGH, the emergency stop is triggered, the robot stops, disables and powers off, terminating the running program. When the robot is in an emergency stop state, powering on the robot is prohibited. When the emergency stop button is reset, and the safe digital input is LOW, the robot automatically exits the emergency stop state.



NOTE

In the “Add-on” interface of the JAKA App, you can enable the “Emergency stop program paused”, if it is enabled, when the emergency stop is triggered, the running program will be paused. After the robot exits the emergency stop status, re-power on and enable the robot, and the robot continues to run the program and return to the position where the emergency stop occurred to move along the programmed trajectory.

The Cat.1 Stop triggered by the emergency stop conforms to the IEC 60204-1:2009.

The maximum stopping time is 250 ms. If the robot does not decelerate to stop within this period, the deceleration is considered to have failed, and a Cat. 0 Stop will be triggered.

When the inconsistency of the safe digital input signals persists for more than 1 second, the App indicates a fault with the safety DI, the control stick indicator turns red, and the safety DI fault fallback function is activated. After the fault is repaired, the robot must be re-powered on to exit the fault state.



NOTE

The safety signals are designed for dual redundancy, so the input signals of the two input ports in pairs should be the same. Refer to the hardware user manual for the pairing relationship.

The following emergency stop-related output functions can also be used to control other systems:

- Configure the “Emergency stop button state output” of the dedicated safety I/O in the JAKA App. When the emergency stop button on the control stick is pressed, a dual digital output signal will inform other devices that the robot has entered an emergency stop state (SF19).



NOTE

The emergency stop function of the external emergency stop button and the additional emergency stop input do not affect this output.

- Configure the “System emergency stop state output” of the dedicated safety I/O in the JAKA App. When the emergency stop button on the control stick is pressed, an external emergency stop device is triggered, or an external emergency stop signal is input, a dual digital output signal will inform other devices that the robot has entered an emergency stop state (SF20)

When the robot enters an emergency stop state, the dual digital output signals will be HIGH. All safety digital output signals are designed with dual redundancy.

The following graphic presents the emergency stop sequence.

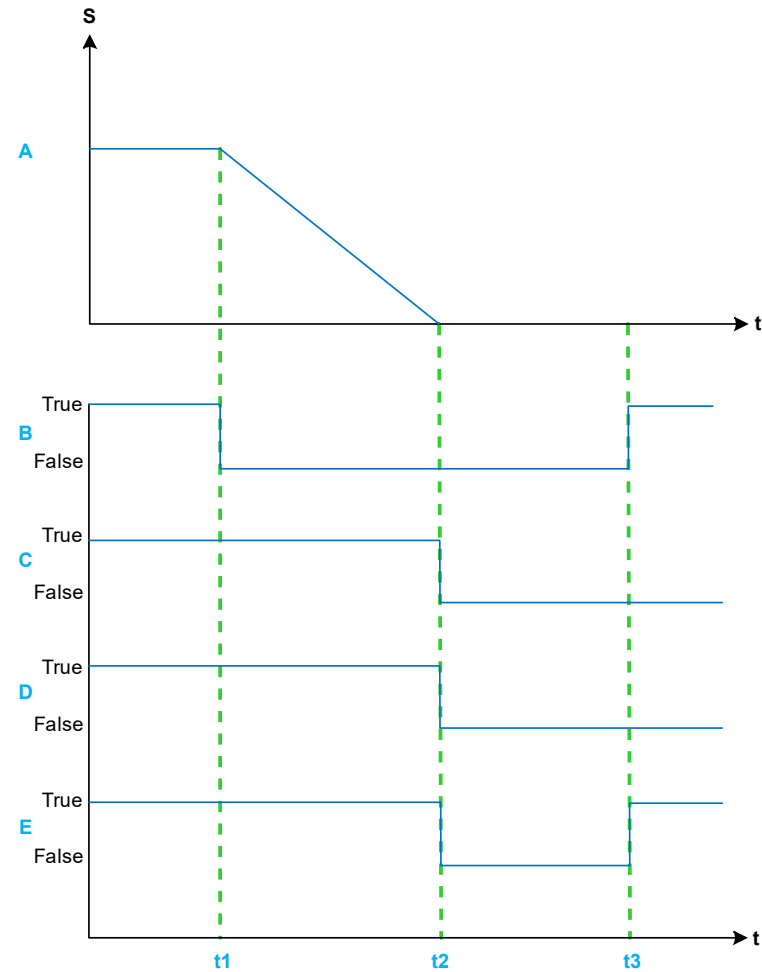


Figure 4-1: Emergency stop sequence

S: Motion speed

t: Time

A: Robot motion speed

B: Emergency stop input signal

C: Robot is disabled (True) or enabled (False)

D: Robot is powered on (True) or power off (False)

E: Emergency stop output signal

t1: Emergency stop button is pressed

t2: Maximum stopping time (250 ms)

t3: Emergency stop button is released

4.2.2 Safeguard Stop Functions

The JAKA robot system provides the following options to implement a safeguard stop:

- Connect the external safeguard stop device to the Pin10&12 of the I/O interfaces on the control cabinet to perform safeguard stop (SF3)
- Configure the “Additional safeguard stop input” of the dedicated safety I/O in the JAKA App, and connect external devices to the Pin7&9 or Pin11&13 of the I/O interfaces on the MiniCab to provide safeguard stop input, in order to perform the safeguard stop (SF16)

When the safety digital input is HIGH, the safeguard stop is triggered, the robot decelerates following the programmed trajectory, eventually all joints enter stand-still condition (while the robot remains enabled), pausing the running program. When the “Safeguard reset input” is not configured in the “Dedicated Safety I/O” interface (in automatic reset mode), and the safe digital input is LOW, the robot automatically exits the safeguard stop state, and the robot resumes the original speed to run the program.

The Cat. 2 Stop triggered by the safeguard stop conforms to the IEC 60204-1:2009.

The maximum stopping time is 350 ms. If the robot does not decelerate to stop within this period, the deceleration is considered to have failed, and a Cat. 1 Stop will be triggered.

If the two safety digital input signals differ, the Cat. 1 Stop will be triggered.

The following safeguard stop-related output can also be used to control other systems:

- Configure the “System safeguard stop state output” of the dedicated safety I/O in the JAKA App. When the safeguard stop function is triggered, a dual digital output signal will inform other devices that the robot has entered a safeguard stop state (SF21)

When the robot enters a safeguard stop state, the dual digital output signals will be HIGH. All safety digital output signals are designed with dual redundancy.

The following graphic presents the safeguard stop sequence in automatic reset mode.

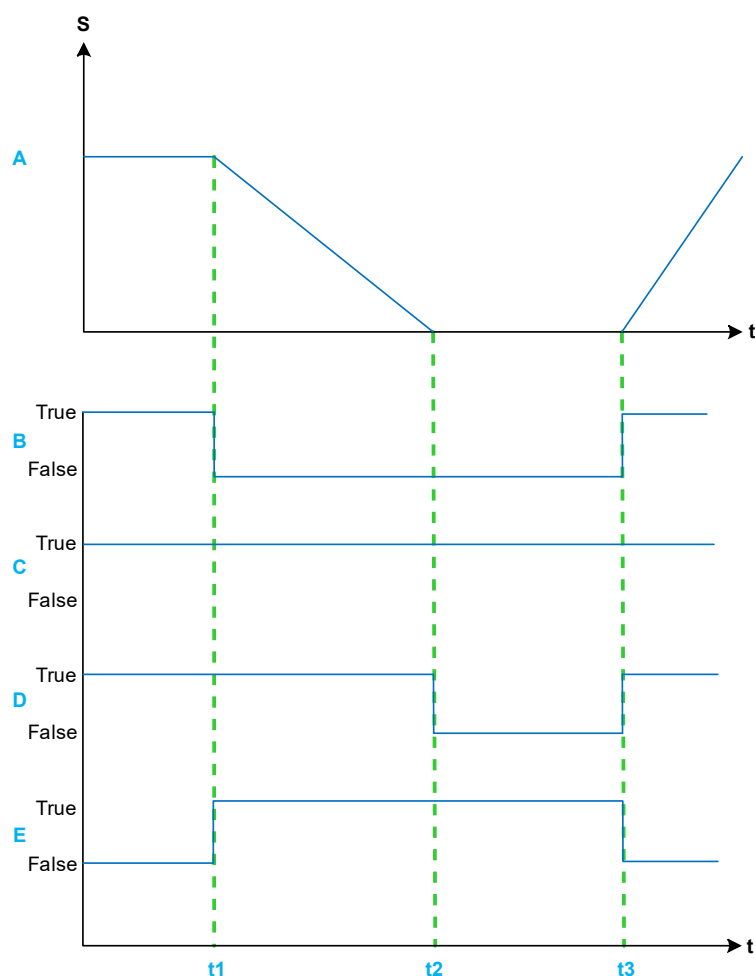


Figure 4-2: Safeguard stop sequence in automatic reset mode

S: Motion speed

t: Time

A: Robot motion speed

B: Safeguard stop input signal

C: Robot is powered on and enabled

D: Safeguard stop output signal

E: Program pause (True) and resume (False)

t1: Safeguard stop is triggered

t2: Maximum stopping time (350 ms)

t3: Safeguard stop reset

The default state of the safeguard stop function is automatic reset, but when configuring the “Safeguard reset input” (SF17) on the “Dedicated Safety I/O” interface of the JAKA App, the safeguard stop reset switches from automatic reset mode to manual reset mode. In manual reset mode, exiting the safeguard stop requires the safeguard stop safety digital input signal is LOW, the safeguard stop reset safety digital input changes from HIGH to LOW. After exiting the safeguard stop state, the robot resumes its original speed and continues running the program.

If the two safety digital input signals differ, it will fail to reset the safeguard stop and the Cat. 1 Stop will be triggered.

The following graphic presents the safeguard stop sequence in manual reset mode.

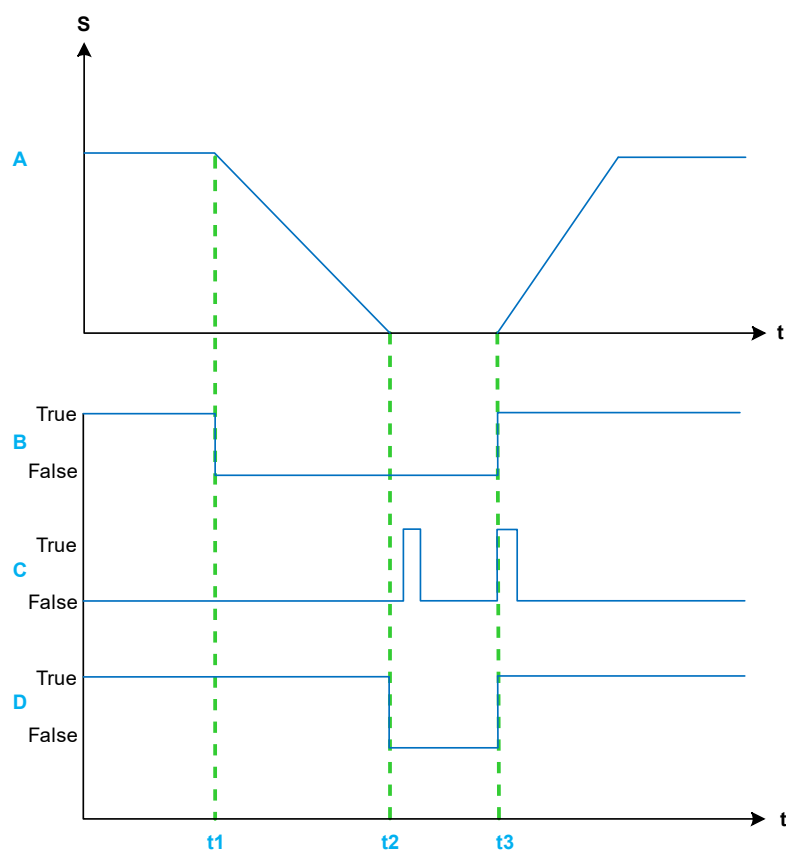


Figure 4-3: Safeguard stop sequence in manual reset mode

S: Motion speed

t: Time

A: Robot motion speed

B: Safeguard stop input signal

C: Safeguard stop manual reset signal input

D: Safeguard stop output signal

t1: Safeguard stop is triggered

t2: Maximum stopping time (350 ms)

t3: Safeguard stop reset



NOTE

Safeguard stop alternatively referred to as protective stop in this manual.

4.2.3 Motion Status Functions

The robot system has several ways to monitor the robot's motion status.

- Configure the “Robot moving output” of the dedicated safety I/O in the JAKA App, When the robot is in motion, the dual digital output signals are HIGH to inform other devices that the robot is in motion (SF22).



NOTE

As long as the robot TCP speed is greater than 10 mm/s, the control system considers the robot to

be in motion.

- Configure the “Robot not stopping output” of the dedicated safety I/O in the JAKA App. When the robot is powered on and enabled, and the robot has stopped or is stopping, the dual digital output signals are LOW to inform other devices that the robot is at standstill (SF23).

The following graphic presents the motion status sequence.

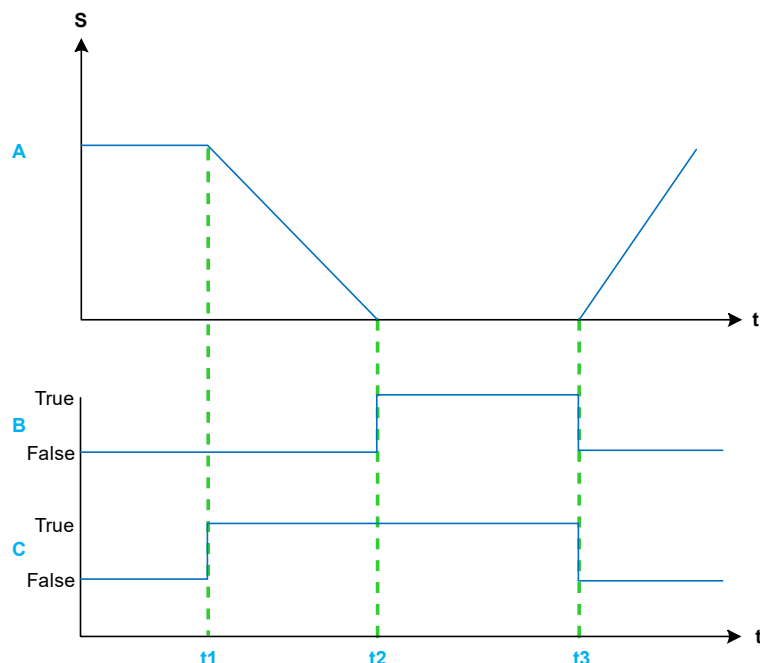


Figure 4-4: Motion status sequence

S: Motion speed

t: Time

A: Robot motion speed

B: Robot motion signal

C: Robot not stopping signal

t1: Emergency stop or safeguard stop causes the robot decelerates

t2: Maximum stopping time (100 ms)

t3: Robot is moving again

4.2.4 Hand-Guided Mode Safety Functions

4.2.4.1 Freedrive Mode TCP Speed Limit

Freedrive mode includes:

- Press and hold the FREE button on the robot end flange.
- Configure and activate function DI of “Enter freedrive mode”

To activate the TCP speed limit in freedrive mode, the TCP freedrive speed limit should be set in “Freedrive Limit” interface in the JAKA App. Refer to the software user manual's “Safety Settings” chapter for operation steps. The maximum TCP freedrive speed limit is 1500 mm/s, and the default value is 1000 mm/s. If the TCP speed exceeds this limit, the Cat. 2 Stop will be triggered. The robot exits freedrive mode, and remains powered on and enabled.



NOTE

Generally, TCP speed is the speed of the origin of tool coordinate system.

The following graphic presents the freedrive mode TCP speed limit sequence.

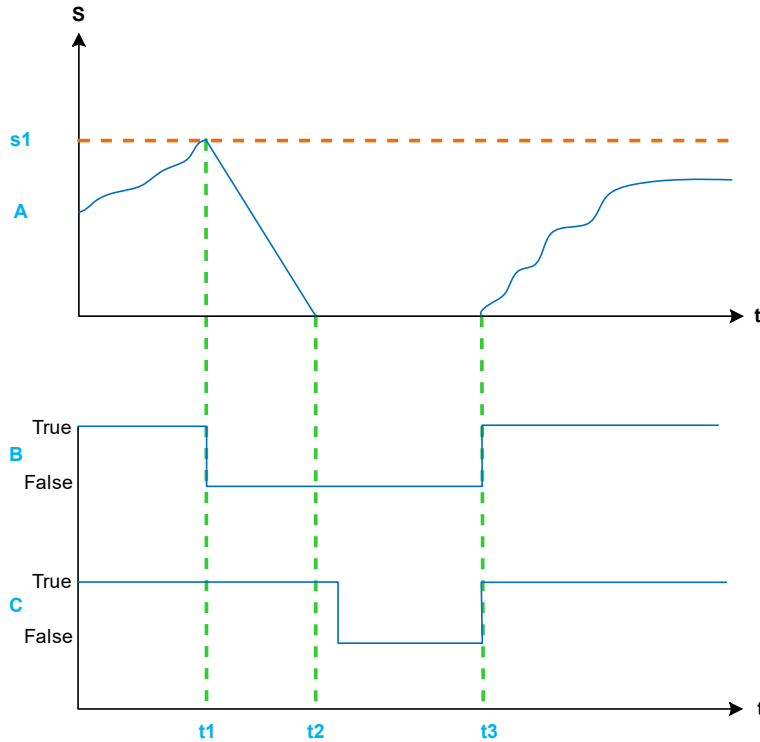


Figure 4-5: Freedrive mode TCP speed limit sequence

S: Motion speed

t: Time

A: Robot motion speed

B: Activate (True) and deactivate (False) freedrive mode

C: FREE button (True = press, False = release)

s1: TCP speed limit of 250 mm/s

t1: TCP speed limit achieved

t2: Maximum stopping time (350 ms)

t3: FREE or Pause/Resume button is pressed again

4.2.4.2 3-Position Switch Enable

Configure the “3-position enable input” function on the dedicated safety I/O interface in the JAKA App. Refer to the software user manual’s “Safety Settings” chapter for operation steps.

When the safety digital input signal is HIGH (3-position switch is in position 1 or position 3), the 3-position enable function is activated. In this state, the following restrictions apply:

- Jogging robot is restricted, which means you cannot move robot in “Manual Operation” interface of the JAKA App. If the robot is in JOG mode, it will stop moving.
- Debugging of programs in the “Programming” interface of the JAKA App is restricted. If in the debugging mode, the debugging process terminates, and the program terminates.

NOTE


The 3-position enable function does not take effect in automatic mode. When a program is executed in the “Programming” interface, the robot automatically switches to automatic mode. In this mode, the 3-position enable function cannot be triggered.

When the safety digital input signal is LOW (3-position switch is in position 2), the 3-position enable function is deactivated. In this state, the restrictions above are lifted.

If the safety digital input signal is inconsistent, the 3-position enable function is activated, and the App will prompt that the safety digital input signal is in error. If the two safety digital input signals differ, and the inconsistency lasts more than 1 second, the Cat. 1 Stop will be triggered.

The function description of 3-position enable switch is as follows:

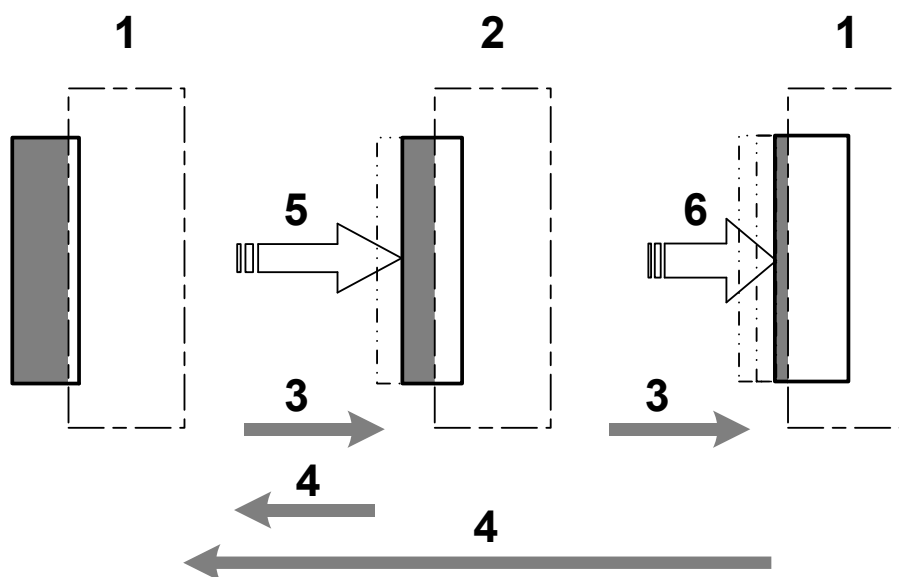


Figure 4-6: 3-position switch functions

- 1: Activate the 3-position enable function
- 2: Deactivate the 3-position enable function
- 3: Press
- 4: Release
- 5: Press lightly
- 6: Press tightly

The following graphic presents the 3-position enable sequence.

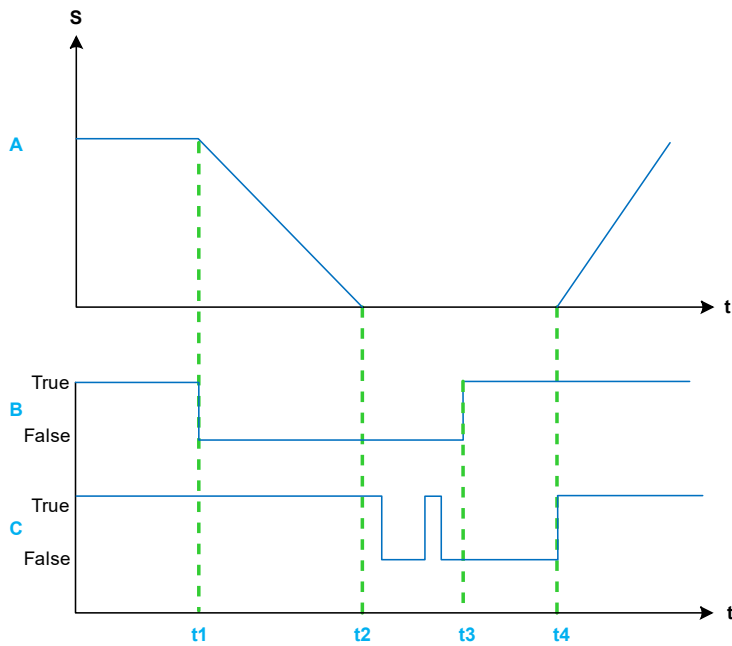


Figure 4-7: 3-position enable sequence

S: Motion speed

t: Time

A: Robot motion speed

B: 3-position enable limit (True = deactivate, False = activate)

C: Manual control robot (True = activate, False = deactivate)

t1: 3-position enable limit is triggered

t2: Maximum stopping time (350 ms)

t3: 3-position enable limit is lifted

t4: Activate manual control

4.2.5 Speed Monitoring Functions

4.2.5.1 Joint Speed Limit

The speeds of each robot joint and TCP are limited, thus the joint speed limit (SF5) needs to be configured in the “Joint Limit” interface, and TCP speed limit (SF9) needs to be configured in the “Motion Limit” interface of the JAKA App. For specific configuration steps, please refer to “Safety Settings” chapter in the software user manual. The robot system will monitor the speed of the robot, and when the motion speed exceeds the setup limit, the robot will decelerate until the speed is within the specified limit. If the robot speed does not decrease to the specified limit within 250 ms, according to IEC 60204-1:2009, the robot will execute a Cat. 1 Stop.

If the two safety digital input signals differ, the Cat. 1 Stop will be triggered.

The following graphic presents the speed monitoring sequence.

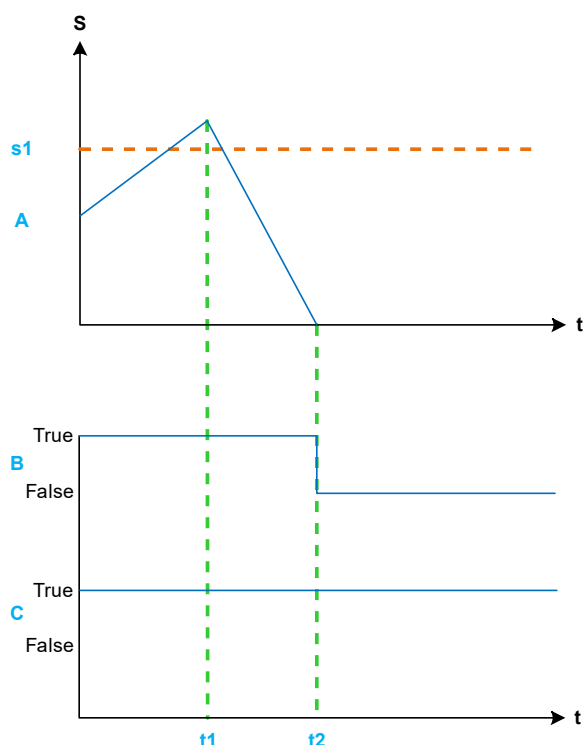


Figure 4-8: Speed monitoring sequence

S: Motion speed

t: Time

A: Robot joint speed/TCP speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or power off (False)

s1: Joint speed limit/TCP speed limit

t1: Joint speed limit or TCP speed limit is exceeded

t2: Maximum stopping time (250 ms)

4.2.5.2 Reduced Mode

Reduce mode can be entered through the following methods:

Configure the “Reduced mode input” in the “Dedicated Safety I/O” interface of the JAKA App. Connect external safety devices to the Pin7&9 or Pin11&13 of the I/O interfaces on the MiniCab to provide input signal (SF18).

Safety digital input signals are designed with dual redundancy. When the safety digital input signal is HIGH, it triggers the reduced mode. In this mode, the TCP speed, elbow speed, momentum and power, stopping time, and stopping distance of the robot are limited to the configured values. Refer to the software user manual’s “Safety Settings” chapter for instructions on configuring these limit values.

The limit data for each robot model are as follows:

Table 4-2: Limit data in reduced mode

Robot Model	Name	Value		Unit
		Min.	Max.	
MiniCobo	TCP speed limit	50 (1.97)	250 (9.84)	mm/s (in/s)

Robot Model	Name	Value		Unit
		Min.	Max.	
	Elbow speed limit	50 (1.97)	250 (9)	mm/s (in/s)
	Momentum limit	0.1	0.8	kg*m/s
	Power limit	0.1	35	W
	Stopping time	0.015	0.1	s
	Stopping distance	0.35 (0.01)	15 (0.59)	mm (in)
Mini 2	TCP speed limit	50 (1.97)	250 (9.84)	mm/s (in/s)
	Elbow speed limit	50 (1.97)	250 (9)	mm/s (in/s)
	Momentum limit	0.1	2.4	kg*m/s
	Power limit	0.1	50	W
	Stopping time	0.015	0.1	s
	Stopping distance	0.35 (0.01)	15 (0.59)	mm (in)

Configure the “Robot reduced mode output” (SF24) in the “Dedicated Safety I/O” interface of the JAKA App. After triggering SF11 or SF18, the robot will be in reduced mode, which the safety digital output signal is HIGH.

Similarly, activate the “Robot not in reduced mode output” (SF25) in the “Dedicated Safety I/O” interface of the JAKA App. When SF11 or SF18 is not triggered, the robot will not be in reduced mode, which the safety digital output signal is HIGH. Safety digital output signals are designed with dual redundancy.

The following graphic presents the reduced mode sequence.

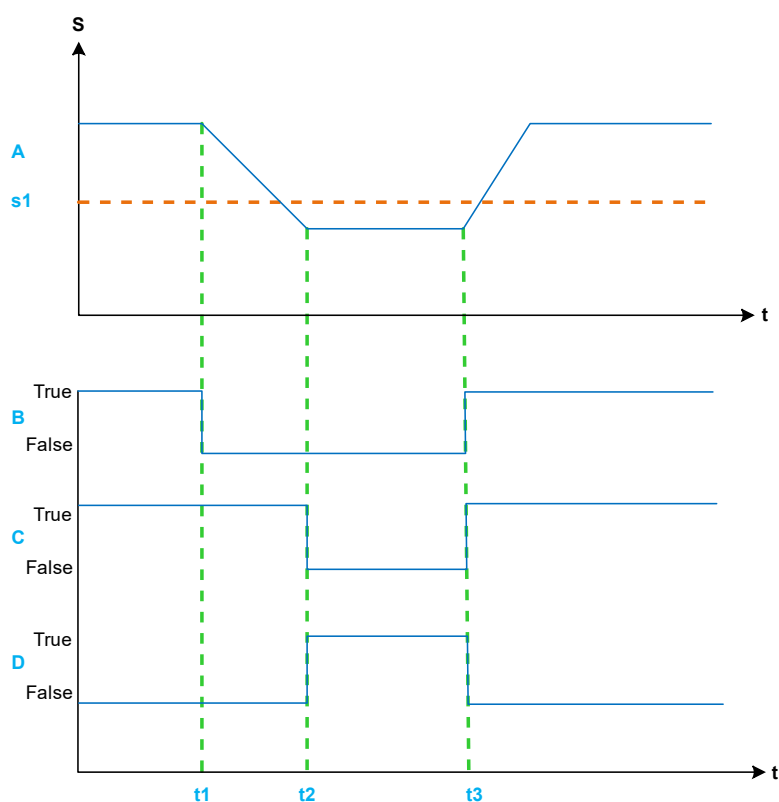


Figure 4-9: Reduced mode sequence

S: Motion speed

t: Time

A: Robot motion speed

B: Reduced mode signal input

C: Reduced mode signal output

D: Not in reduced mode output

s1: Joint speed limit/TCP speed limit

t1: Activate reduced mode

t2: Robot enters reduced mode

t3: Deactivate reduced mode

4.2.6 Torque and Power Limitations Functions

The robot system monitors and limits the robot power. The robot power is used to limit the total power of the robot (SF8), and its power limit value can be set in the “Motion Limit” interface of the JAKA App. Refer to the software user manual for specific instructions on how to set it up. Additionally, the robot system also monitors the torque (SF6) and power (SF7) of each joint. However, the torque and power limits are default values and cannot be configured or modified. If the actual values exceed the limit values, according to IEC 60204-1:2009, the robot will execute a Category 1 Stop.

The following graphic presents the torque and power limitations sequence.

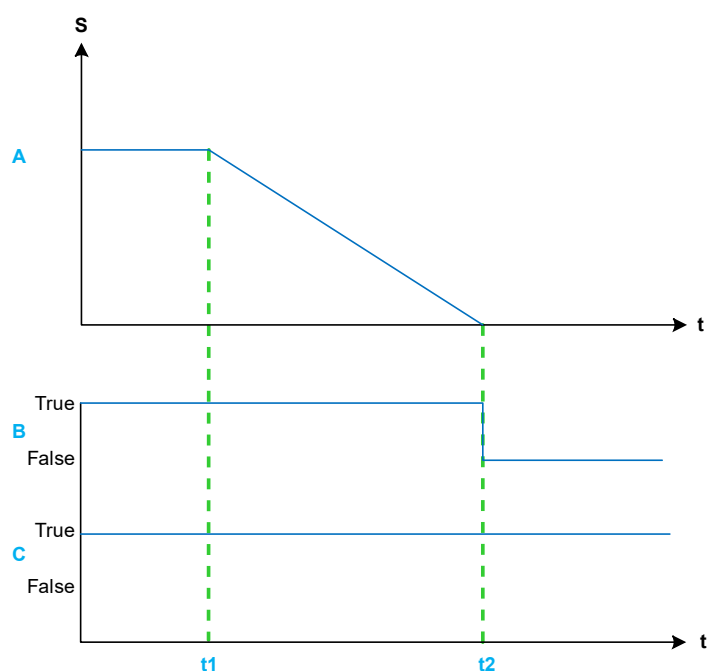


Figure 4-10: Torque and power limitations sequence

S: Motion speed

t: Time

A: Robot motion speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or power off (False)

t1: Reach or exceed the robot power, joint power, or joint torque

t2: Maximum stopping time (250 ms)

4.2.7 Collision Protection Safety Functions

The robot system provides the following methods to detect the collision:

- Monitor the actual deviation each joint position (SF12). When the limit is exceeded, the robot performs a Cat. 1 Stop.
- Monitor various parameters (force, momentum, TCP speed, power, stopping time and stopping distance) of the robot to detect collisions (SF14). When the limit is exceeded, the robot performs a Cat. 2 Stop.
- Estimate TCP force by monitoring robot joint angles and joint torques, compare the estimated force with TCP force limit values to detect collisions (SF26). When the torque limit is exceeded, the robot decelerates to stop following the programmed trajectory, then reverses along the original motion path, and finally remains stationary.

Based on the risk assessment results, configure the “Collision sensitivity” in the “Dedicated Safety I/O” interface of the JAKA App to activate the collision detection. A lower numerical value indicates higher sensitivity. You can also configure collision parameters in the “Motion Limit” interface of the JAKA App to activate the collision detection. Refer to the software user manual’s “Safety Settings” chapter for operational steps.

Table 4-3: Limit data in collision protection mode

Robot Model	Name	Value		Unit
		Min.	Max.	
MiniCobo	TCP speed limit	50 (1.97)	250 (9.84)	mm/s (in/s)
	Elbow speed limit	50 (1.97)	250 (9)	mm/s (in/s)
	Momentum limit	0.1	0.8	kg*m/s
	Power limit	0.1	35	W
	Stopping time	0.015	5	s
	Stopping distance	0.35 (0.01)	4000 (157.48)	mm (in)
Mini 2	TCP speed limit	50 (1.97)	250 (9.84)	mm/s (in/s)
	Elbow speed limit	50 (1.97)	250 (9)	mm/s (in/s)
	Momentum limit	0.1	0.8	kg*m/s
	Power limit	0.1	50	W
	Stopping time	0.015	5	s
	Stopping distance	0.35 (0.01)	4000 (157.48)	mm (in)

You can set the collision rebound angle in the “Collision Settings” interface of the JAKA App. If this angle is configured, when the robot system detects a collision, the robot will decelerate to stop and then rebound along the original path.

The following graphic presents the collision protection sequence (non-severe collision).

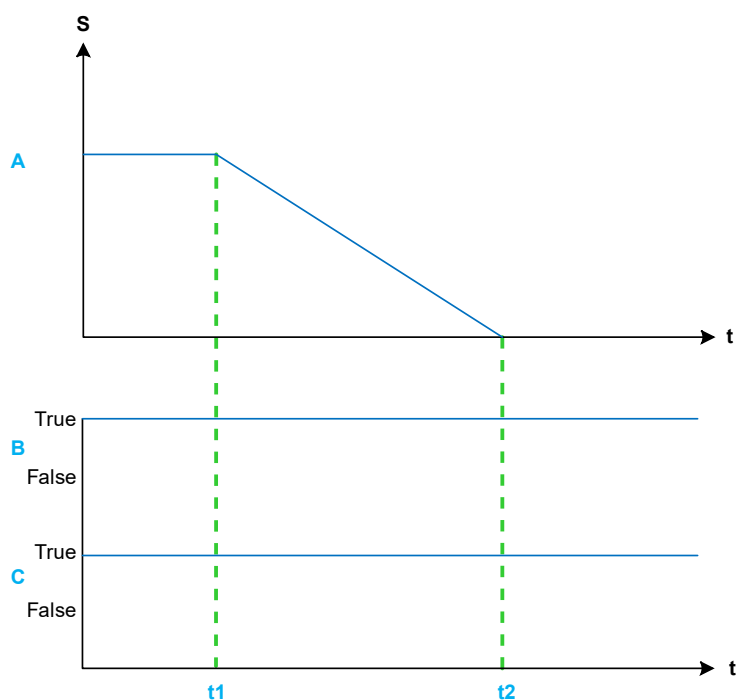


Figure 4-11: Collision protection sequence (non-severe collision)

S: Motion speed

t: Time

A: Robot motion speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or power off (False)

t1: Detect collision

t2: Maximum stopping time (350 ms)

If the robot system detects that the actual joint position deviates significantly from the target joint position, exceeding the deviation limit (SF12), it is considered a severe collision. According to IEC 60204-1:2009, the robot will execute a Cat. 1 Stop.



NOTE

The joint position deviation value is set to the factory default and cannot be configured or modified.

The following graphic presents the collision protection sequence (severe collision).

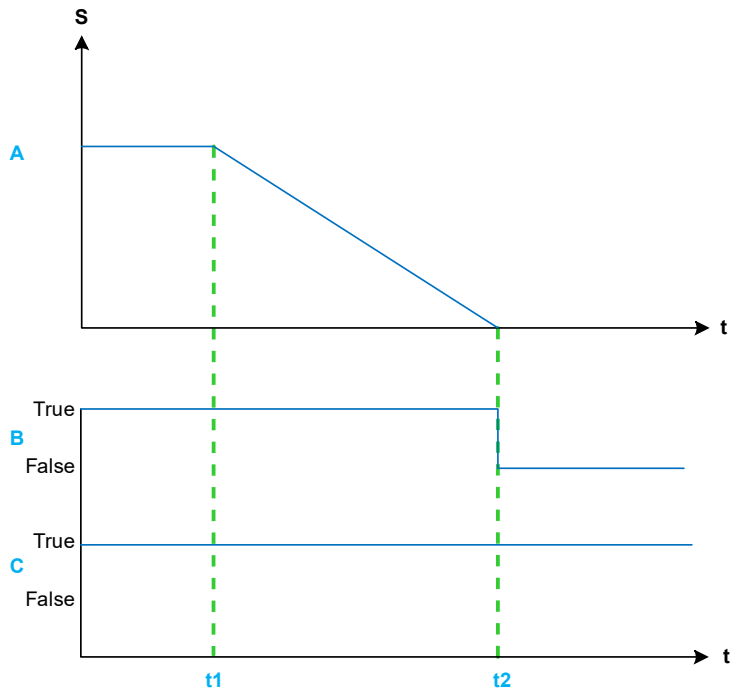


Figure 4-12: Collision protection sequence (severe collision)

S: Motion speed

t: Time

A: Robot motion speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or power off (False)

t1: Detect collision

t2: Maximum stopping time (250 ms)

4.2.8 Position Monitoring

The robot system provides the following methods to limit robot motion:

- Monitor if the robot each joint position exceed the position limit (SF4)
- Monitor if the robot tool orientation is within the set limit (SF10)
- Monitor if the robot position is within the set safety plane (SF11)

All limiting functions mentioned above can be activated simultaneously and can be configured in the JAKA App. Refer to the software user manual's "Safety Settings" chapter for operation steps.

If the robot joint positions reach or exceed the joint position limit (SF4) or the robot tool orientation reaches or exceeds the tool orientation limit (SF10), the App will display a pop-up window. According to IEC 60204-1:2009, the robot will execute a Cat. 1 Stop.

If the robot position reaches or exceeds the TCP position limit (SF11), the robot will execute the corresponding operation based on the robot state configured in the JAKA App.

- If configured to stop when the safety plane is triggered, the robot will stop, the program will be terminated and the robot will be disabled. The robot can be enabled again within the safety zone.
- If configured to safeguard stop when the safety plane is triggered, the robot will decelerate until it stops, and the "Safety plane safeguard stop" will pop up. After clicking "OK", the robot exits the safeguard stop and continues to move through the safety plane. When the robot triggers the safety plane again, it performs safeguard stop again. Apart from clicking "OK", the robot can also exit the safeguard stop and

continue moving through the safety plane by triggering the “Safeguard stop reset input” in the safety DI.

- If configured to reduce mode when the safety plane is triggered, robot enters the reduced mode (TCP speed ≤ 250 mm/s) and the “Robot enters the reduced mode” pops up. After clicking “OK”, the robot is still in the reduced mode, and only exits the reduced mode after the robot returns to the safety plane.

The following graphic presents the position monitoring sequence (the robot state of SF4, SF10, SF11 is configured to stop).

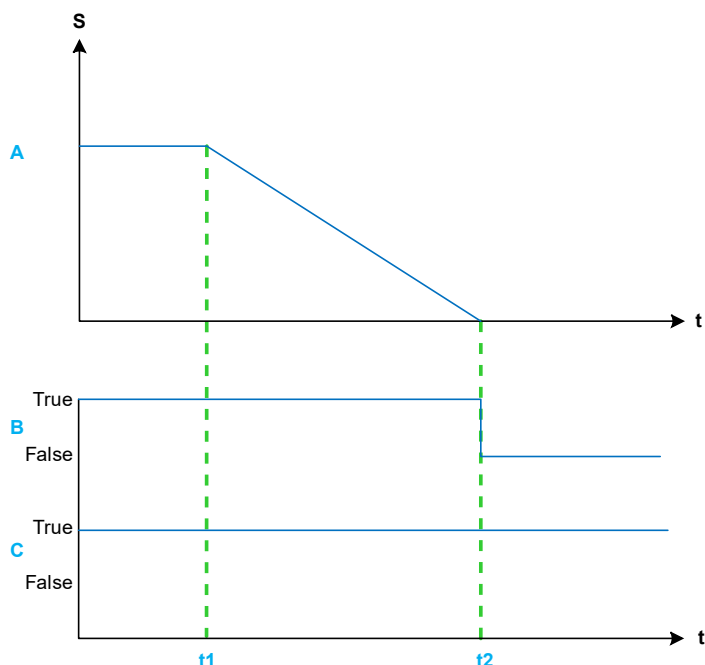


Figure 4-13: Position monitoring sequence (the robot state of SF4, SF10, SF11 is configured to stop)

S: Motion speed

t: Time

A: Robot motion speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or powered off (False)

t1: Reach or exceed the position limit

t2: Maximum stopping time (250 ms)

The following graphic presents the position monitoring sequence (the robot state of SF11 is configured to safeguard stop).

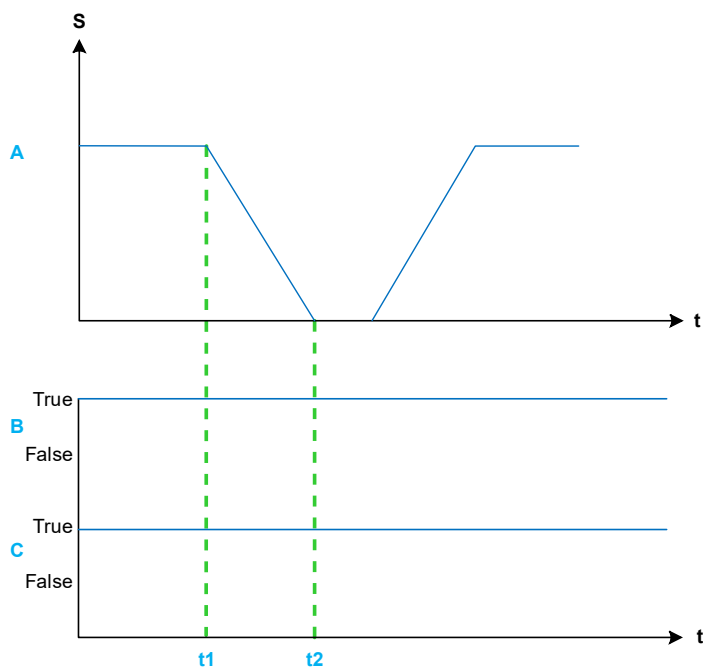


Figure 4-14: Position monitoring sequence (the robot state of SF11 is configured to safeguard stop)

S: Motion speed

t: Time

A: Robot motion speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or power off (False)

t1: Reach or exceed the position limit

t2: Maximum stopping time (250 ms)

The following graphic presents the position monitoring sequence (the robot state of SF11 is configured to reduced mode).

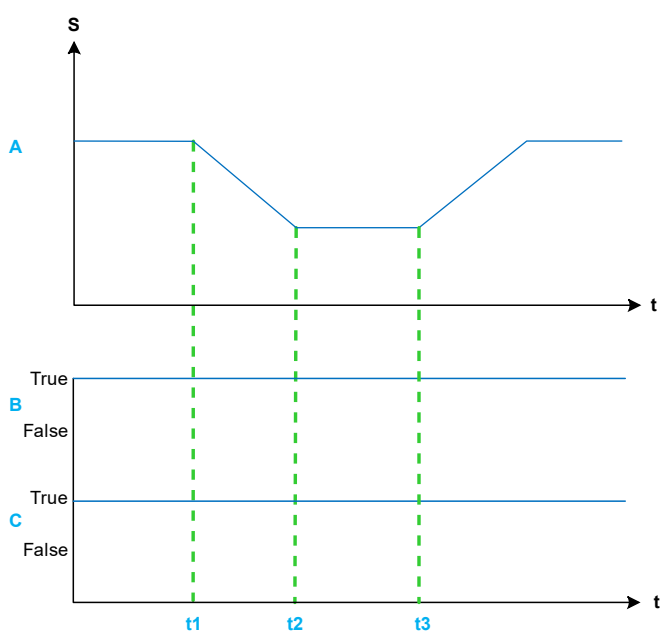


Figure 4-15: Position monitoring sequence (the robot state of SF11 is configured to reduced mode)

S: Motion speed

t: Time

A: Robot motion speed

B: Robot is disabled (True) or enabled (False)

C: Robot is powered on (True) or power off (False)

t1: Reach or exceed the position limit

t2: Robot is in the reduced mode

t3: Robot exceeds the reduced mode

4.3 Input Signal Redundancy

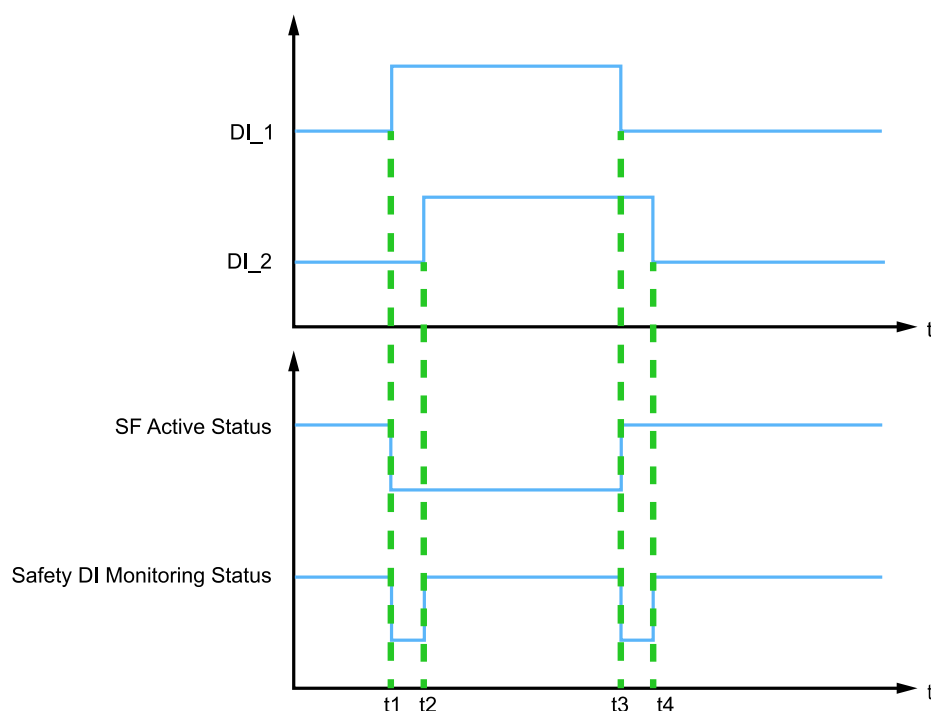


Figure 4-16: Input signal redundancy sequence

The channels of the safety digital input signal use redundancy with equivalent input signals. If one of the safety signals (taking DI_1 and DI_2 of MiniCab as examples) is HIGH, the safety function is activated (t1).

When the safety function is off and exits the safety state, DI_1 and DI_2 need to change from HIGH to LOW (falling edge) within the signal tolerance time. If the DI_1 and DI_2 signals differ, and the time exceeds the tolerance time for 1 second. For example, t2-t1 or t4-t3 time as shown in the figure above exceeds 1 second, the SCB determines that the safety DI is faulty and performs the safety DI fallback function. The safety DI error can only be eliminated by powering on the robot again after recovery. If the error is not cleared, it is forbidden to power on the robot. The signal used to activate the safety function must be stable, otherwise the robot will stop and need to be restarted.

Safety DI fallback function:

- The robot decelerates and stops. After stopping, the robot is powered off and the program terminates.
- Troubleshooting the two DIs with errors, and replace the actual input signal of the port with the low signal. The error will exist until the robot is powered on again and the error has been checked and cleared.

The App pops up that the safety DI has an error. After the error is troubleshooted, the robot needs to be restarted to exit the error status.

5 Technical Data

5.1 Technical Specifications

5.1.1 Robot Technical Specification

Table 5-1: Robot technical specifications

Model	MiniCobo	Mini 2
Payload	1 kg (2.2 lb)	2 kg (4.4 lb)
Weight (including cables)	9.4 kg (20.7 lb)	9.9 kg (21.8 lb)
Reach	580 mm (22.8 in)	
Repeatability	±0.1 mm (±0.004 in)	
Degree of freedom	6	
Programming	Graphical programming, and freedrive programming	
Demonstrator type	Mobile terminal (computer/pad/mobile phone)	
Collaborative operation	Collaborative operation as ISO 10218-1: 2011	
Range of action		
Joint 1	±360°	
Joint 2	±125°	
Joint 3	±130°	
Joint 4	±360°	
Joint 5	±120°	
Joint 6	±360°	
Tool typical speed	1 m/s (3.3 ft/s)	
Average power consumption	150 W	180 W
Temperature	0~50°C (32~122°F) ⁱ	
Humidity	0~90% RH, non-condensation	
IP	IP40	
Robot mounting	Mounting at any angle	
Tool I/O port	2 digital inputs	
	2 digital outputs	
	2 analog inputs	
Tool I/O power supply	12/24 V	
Tool I/O size	M8	
Base diameter	124 mm (4.9 in)	
Material	Aluminum alloy, PC	
Length of robot connection cable	6 m (236 in)	
Length of control stick cable	6 m (236 in)	

**NOTE**

At low environmental temperature $<10^{\circ}\text{C}$, a warm-up phase is recommended to be run with the robot. Otherwise, there is a risk that the robot stops or run with lower performance.

5.1.2 Control Cabinet Technical Specification

Table 5-2: Control cabinet technical specifications

Model	MiniCab
Weight	1.1 kg (2.4 lb)
Dimension (W×H×D)	180×46.6×128 mm (7.09×1.84×5.04 in)
Temperature	0-50°C (32-122°F)
IP classification	IP20
Material	Steel, aluminum alloy
Power supply	24 V DC (MiniCobo); 48 V DC (Mini 2)
Communication mode	TCP/IP, Modbus TCP, Modbus RTU, PROFINET, Ethernet/IP
I/O port	7 channels I/O multiplexing
I/O power supply	24 V DC

5.1.3 Power Adapter Technical Specification

The technical specification of 220V AC-DC power adapter is as follows:

Table 5-3: Power adapter technical specifications

	MiniCobo	Mini 2
Model	GST280A24-C6P, MW	HRP-300N3-48, MW
Illustration		
Rated voltage	24 V DC	48 V DC
Rated current	11.67 A	7 A
Current range	0~11.67 A	0~7 A
Rated power (maximum)	280.08 W	336 W
Ripple and noise (maximum)	200mVp-p	250mVp-p
Voltage accuracy	±3%	±3%
Linear regulation rate	±1%	±1%
Load regulation rate	±3%	±3%

5.2 Electrical Specifications

5.2.1 Electrical Warnings and Precautions

When designing and installing robot applications, you must obey the following warnings and precautions. The implementation of maintenance must also comply with these warnings and precautions.



WARNING

- Do not connect safety signals to non-safe PLCs with inappropriate safety levels. If you do not obey the warning, severe injuries or even death may be caused by the dysfunction of a safety stop function. Separate the safety I/O from the general I/O.
- All safety-related signals are constructed redundancy (two independent channels). Keeping two channels separate can ensure that the safety function will not be lost when a single failure occurs.



WARNING: ELECTRICITY

- Please ensure that all water-sensitive devices remain dry. If water gets into the product, cut off the power in time and contact your supplier.
- Use original cables to connect the robot. Do not use robots in applications that need to bend cables. Contact your supplier if longer or flexible cables are needed.
- For protective earth (PE), use the screw connectors in the control cabinet marking the earth symbol. The grounding conductor shall at least have the current rating of the highest current in the robot system.
- When connecting cables on the I/O of the control cabinet, ensure that the I/O cable avoids rubbing with the edge of the cable hole.



WARNING

- The robot has been tested according to international IEC standards for Electromagnetic Compatibility (EMC). Interference signals beyond the standard will cause abnormal behaviors of the robot. Very high signal levels or excessive exposure can damage the robot permanently. JAKA will not be held responsible for any damages caused by EMC problems.
- The length of the I/O cable to connect the control cabinet and other machinery and factory equipment must not be longer than 30 meters unless the result of the extension test shows the feasibility. Use shielding cables if necessary.

5.2.2 Introduction

This chapter mainly introduces the absolute maximum ratings and recommended usage conditions of MiniCab.



WARNING

It is imperative for users to adhere to the recommended electrical parameters when using the robot and MiniCab. Exceeding the maximum ratings may result in hardware damage.

5.2.3 Absolute Maximum Ratings

Table 5-4: Absolute maximum ratings

		Min.	Max. ¹
VL+	Control cabinet power voltage (V)	-0.3	40

		Min.	Max. ⁱ
VP+	Robot power voltage (V)	-0.3	40
V _{UDIO_COM+}	Integration interface common terminal voltage (V)	-0.3	30
I _{UDIO_24V}	Integration interface output current (A)	0	2.7
I _{UDIOx}	Integration interface single channel output current (A)	0	2
V _{busRS485}	RS485 bus withstand voltage (V) ⁱⁱ	-70	70

i: When the actual value exceeds the maximum value, it may cause permanent damage to the equipment. The maximum value represents the limit, and it is not recommended to use the equipment under this condition or any other condition exceeds the recommended conditions.

ii: Except the bus withstand voltage, all voltage values are related to grounding.

5.2.4 Recommended Conditions

Table 5-5: Recommended conditions of MiniCobo

MiniCobo		Min.	Typ.	Max.
VL+	Control cabinet power voltage (V DC)	20 ⁱ	24	30
VP+	Robot power voltage (V DC)	20 ⁱⁱ	24	30
I _{RMS}	Average operating current (A)			8
I _{peak}	Peak output current (A)			12 ⁱⁱⁱ
I _{UDIOx}	I/O interface single channel output current (A)		1	
	Ambient temperature (°C/°F)	0 / 32		50 / 122 ^{iv}
	Atmospheric pressure (Bar)		1	
	Altitude (m/in)		1000 / 39370	
	Humidity (%RH)	10		90

Table 5-6: Recommended conditions of Mini 2

Mini 2		Min.	Typ.	Max.
VL+	Control cabinet power voltage (V DC)	40 ⁱ	48	56
VP+	Robot power voltage (V DC)	40 ⁱⁱ	48	56
I _{RMS}	Average operating current (A)			3.75
I _{peak}	Peak output current (A)			12.5 ⁱⁱⁱ
I _{UDIOx}	I/O interface single channel output current (A)		1	

Mini 2		Min.	Typ.	Max.
	Ambient temperature (°C/°F)	0 / 32		50 / 122 ^{iv}
	Atmospheric pressure (Bar)		1	
	Altitude (m/in)		1000 / 39370	
	Humidity (%RH)	10		90

i: The minimum control cabinet voltage is the voltage that can activate functions of the control cabinet, but the output value of the UDIO_24V depends on the control cabinet supply voltage.

ii: The robot power supply provides power to the robot. Typically, the MiniCobo robot will trigger under voltage protection when the voltage is $\leq 20V$, the Mini 2 robot will trigger under voltage protection when the voltage is $\leq 30V$. When selecting a power supply, it is necessary to consider voltage loss and abnormalities caused by voltage drop.

iii: The peak output current value is related to the robot model and operating status.

iv: The surface of the control cabinet is hot during operation, which is required to be used in well-ventilated and heat-dissipating conditions.

5.2.5 Typical Power Consumption

Test condition: 25°C, 48V control cabinet power supply, 48V robot power supply, robot and integrated interface not connected, control stick connected.

Table 5-7: Typical power consumption

Parameters	Typ.	Max.
Power down the MiniCab (W)	1	5
Power up the MiniCab (W)	12	30
Power on the robot (W)		30

5.2.6 IPC Characteristics

Table 5-8: IPC characteristics

CPU	Intel(R) Celeron(R) CPU J1900 1.99GHz
RAM	DDR3L 2G
Hard drive	64G
Interfaces	HDMI*1、USB3.0*1、USB2.0*1、Ethernet*2

5.3 Robot Buttons and Interfaces

5.3.1 Ring-Shaped Light

A ring-shaped light is positioned at the robot base, its color and corresponding functions are shown in table below.

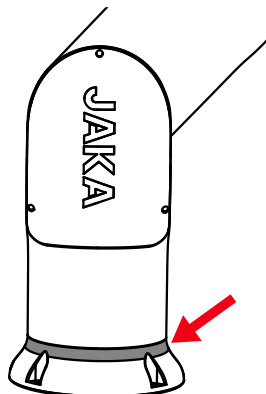


Figure 5-1: Ring-shaped light

Table 5-9: Meaning of ring-shaped light color

Color	Robot State
Blue	Powered on but disabled
Green	Enabled
Red	Failure/Protective stop
Yellow	Freedrive mode
Yellow flash	Pause
Yellow flash	Single-step program debugging

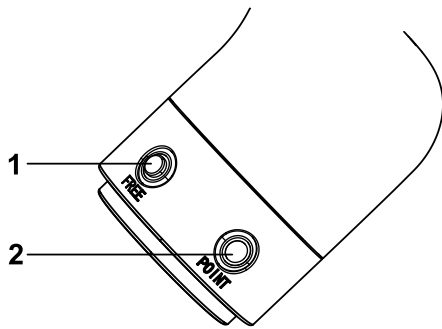


WARNING

User must first assess the risks of a sudden stop or start before using the Pause/Resume button. Sudden stops or starts may cause injury to personnel or damage to the equipment.

5.3.2 Buttons

The robot is equipped with a tool I/O interface (TIO) and two buttons on the side of the flange. The two buttons are a freedrive button (Free) and a point recording button (Point), as shown in figure below.



- 1 Free button
- 2 Point button

Figure 5-2: Free and point button

When the Free button is pressed, the robot enters the freedrive mode. The robot can be dragged to the desired position manually.

The Point button is used with the JAKA App. When this button is pressed, the corresponding position will show up in the programming interface of the App as a command (see JAKA App Software User Manual for details).



WARNING

User must first assess the risks before using the free drive and point buttons. Ensure that the robot's mounting orientation, end payload, TCP, and other parameters are correctly set, otherwise it may cause injury to personnel or damage to the equipment.

5.4 Robot Rotation Direction

Refer to [5.1.1 Robot Technical Specification](#) for robot rotation range. The rotation direction is as follows:

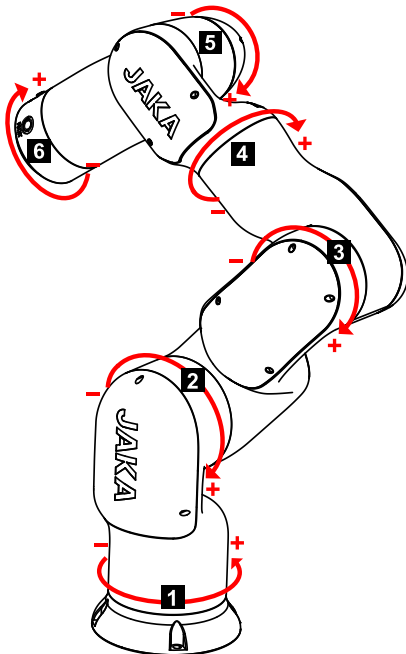


Figure 5-3: Robot rotation direction

5.5 Robot Singularity

Singularities are positions described in Cartesian space that have no impact on MoveJ. When the robot is at a singularity, replace the Cartesian movement by MoveJ if the motion can be achieved by MoveJ. If there is

singularity in the middle of the robot's motion path, the robot's trajectory can be changed to bypass the singularity. You can also change the robot's mounting position or the size of the end effector to change the value of Cartesian points in the robot's joint space. A singularity is a robot arm configuration where one or more degrees of freedom are lost causing the robot to be stuck in a position and unable to move. This usually occurs when joint axes align in one way or another.

When a robot is at a singularity, the following effects may occur:

- Robots cannot move in one or more directions.
- The robot cannot find a suitable set of joint angles for the TCP to move to the desired position.
- The robot joint should move at high speed to achieve the desired TCP speed.
- Intense movements near singularities will result in accidents and shorten the lifespan of robot joints.

The JAKA Mini series robots have three types of singularities: shoulder singularities, elbow singularities and wrist singularities. Descriptions and avoidance methods of the singularity are as follows.

Shoulder singularity: When the axes intersection of the joint 5 and joint 6 lies on the plane of axes of joint 1 and joint 2, the robot is at shoulder singularity.

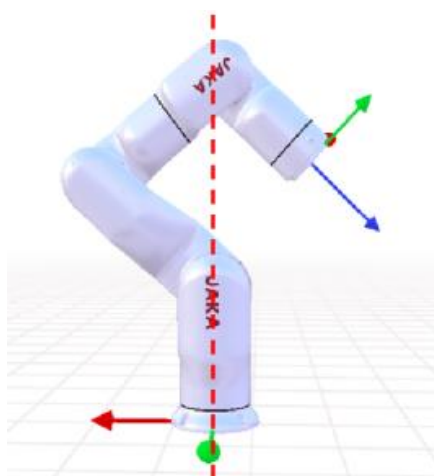


Figure 5-4: Shoulder singularity

Ways to avoid shoulder singularity:

- Ensure that the robot TCP is not directly above the robot base. TCP cannot reach the positions directly above the robot. A shoulder singularity will occur if the TCP approaches a position directly in line with the base Z axis.
- Do not use MoveL when the angle of joint 1 is 180° between two points in space. When the difference is about 180°, the robot reaches shoulder singularity. You can use MoveJ to avoid singularity or set an intermediate transition point outside the shoulder singularity range.

Elbow singularity: When joints 2, 3 and 4 are coplanar or when joint 3 is 0°, the robot is at elbow singularity.

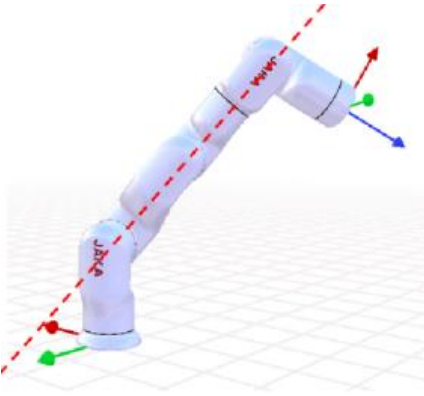


Figure 5-5: Elbow singularity

Ways to avoid elbow singularity:

- When the robot is at an elbow singularity, it indicates that the target position is close to the limits of the working range of the robot. At this point, it is necessary to adjust the robot mounting position or extend the length of the end effector.

Wrist singularity: When joint 4 and joint 6 are collinear or when joint 5 is 0° , the robot is in wrist singularity.

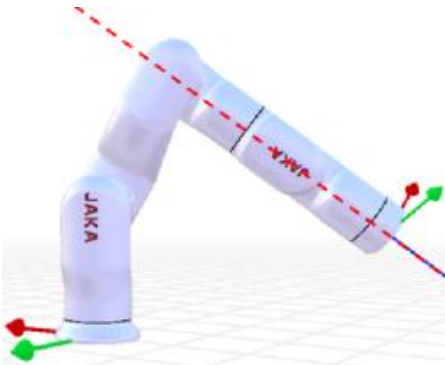


Figure 5-6: Wrist singularity

Ways to avoid wrist singularity:

- The wrist singularity typically occurs in movements with orientation changes. Therefore, in cases of considerable orientation change, it is advisable to prioritize MoveJ and not use Cartesian motions such as MoveL or MoveC.

5.6 Control Stick Buttons

The JAKA robot comes with a control stick, which can be used to control both the robot and the control cabinet. The functions of the control stick buttons are described as follows:

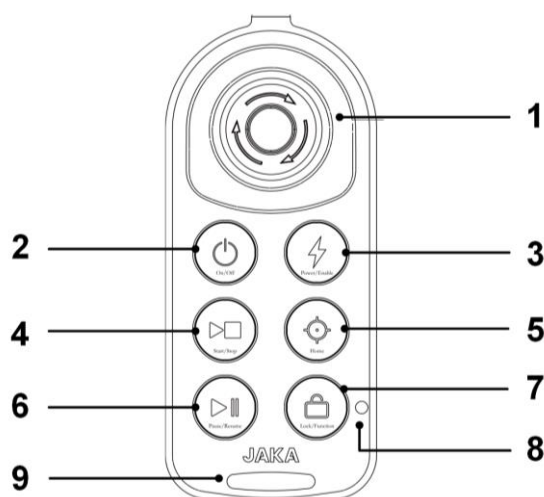


Figure 5-7: Control stick

Table 5-10: Control stick functions

	Name	Description
1	Emergency stop button	For emergency stop. The emergency stop button is limited to emergencies and should not be used as regular power off equipment.
2	Power up/down button	Power up: Press the power button for 1 second and release. The buzzer beeps once, and the control cabinet is powered up. Power down: Press and hold the power button for 3 seconds or more. The control stick beeps 6-7 times, and the control cabinet is powered down.
3	Enable button	Robot power on: Press the enable button once and wait for the ring-shaped light to turn blue, indicating that the robot is powered on. Robot power off: Press the enable button once and wait for the ring-shaped light to go off, indicating that the robot is powered off. Enable robot: When the robot is powered on, press and hold the lock button, and then press the enable button at the same time until the ring-shaped light turns green and a click is heard, which means the robot is enabled. Disable robot: When the robot is enabled, press and hold the lock button, and then press the enable button at the same time until the ring-shaped light turns blue and a click is heard, which means the robot is disabled.
4	Start/stop button	Start the program: Press the button to start the default program, execute the default program after the robot reaches the initial position of the program. Stop the program: Press to stop the program when the robot runs the program.
5	Reset button	Reset: After the robot is enabled and the robot is not executing the program. Press and hold the reset button to control the robot to return to initial orientation set in the JAKA App. Continue holding the button when the robot has moved to initial orientation, the lock indicator will turn blue.
6	Pause/resume button	Pause: During the automatic operation of the robot, press the button to pause the program. Resume: Press the button to resume the program when the program is paused.

	Name	Description
7	Lock button	Lock the control stick: Press and hold the lock button for until the indicator turns orange. Unlock the control stick: Press and hold the lock button until the indicator turns off. Combination function: Other buttons can be used with the lock button.
8	Lock indicator	Lock state: In the lock state, the indicator is orange, and all buttons except the lock button and power button are disabled. The App can now control the robot. Unlock state: In the unlock state, all lights turn off, and the control stick can be used. The App cannot be used to control the robot in this state.
9	Control stick indicator	When powering on the control cabinet, the control stick indicator first shows red, blue and green alternately with three beeps, and then it turns orange, waiting for the control cabinet to power up. After activating, the control stick indicator will flash blue. After powering on the robot, the control stick indicator will flash blue. After enabling the robot, the control stick indicator will flash green. When the robot is in the emergency stop status and protective stop status, the control stick indicator flashes red.



NOTE

- Once the control cabinet is powered up, the control stick will beep twice per second if any button is pressed.
- When using the control stick to operate the robot, ensure that the robot you are operating is within your sight and follow the relevant safety guidelines to avoid any injury to personnel or equipment near the robot.

5.7 Dimensional Drawings

5.7.1 Dimensional Drawings of Robot

5.7.1.1 Robot Outline Dimensions

The sizes of the Mini series robots are shown below. During the installation, the motion range of the robot must be considered to avoid injury to or collision with surrounding personnel and equipment.

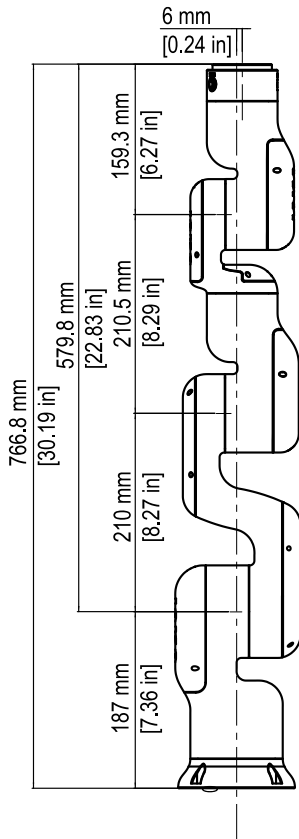


Figure 5-8: Outline dimensions

5.7.1.2 Robot Working Space

The workspaces of the Mini series robots are shown below. When you mount the robot, the reachable space of the robot must be considered.

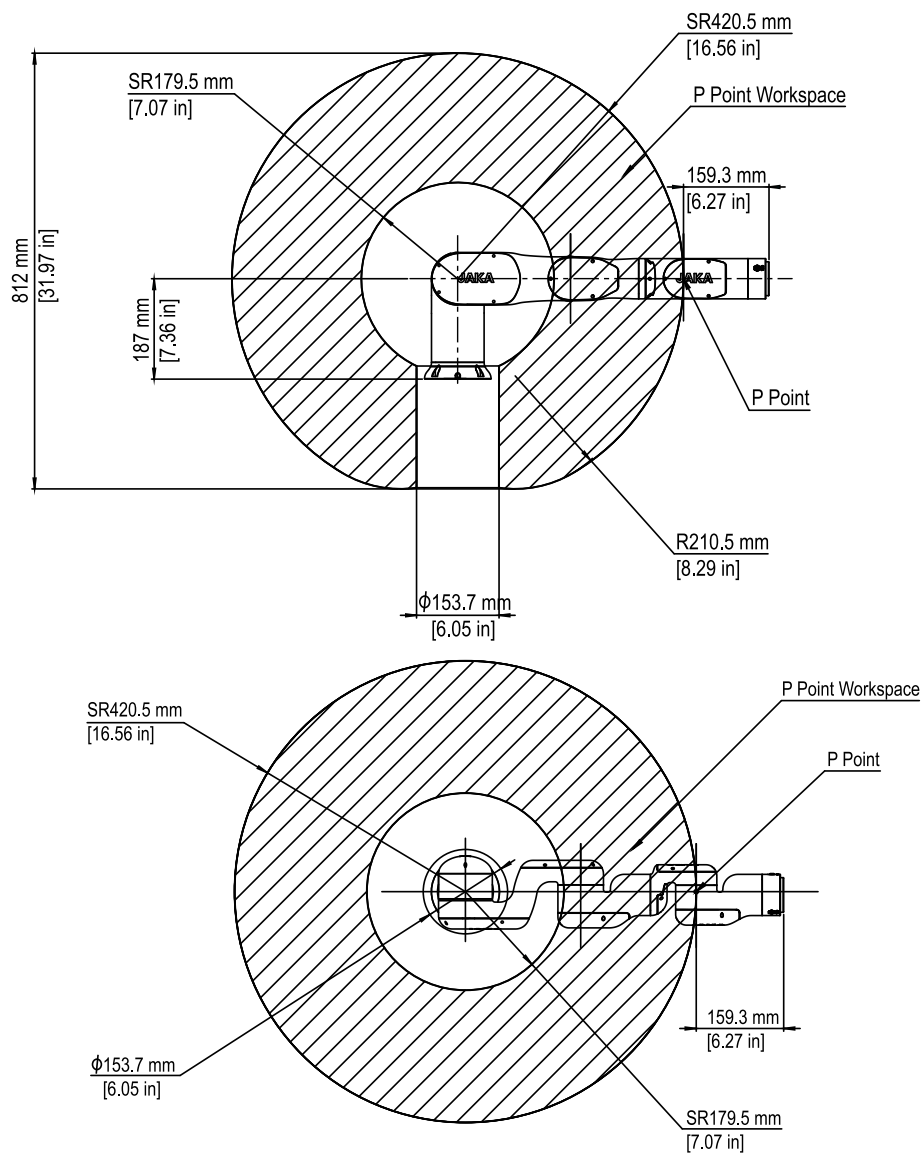


Figure 5-9: Working space

5.7.2 Dimensional Drawings of Control Cabinet



Figure 5-10: Width, height and depth of control cabinet

Table 5-11: Width, height and depth of control cabinet

Dimension	Description	Data	Unit
W	Width	180 (7.09)	mm (in)
H	Height	46.6 (1.84)	mm (in)
D	Depth	128 (5.04)	mm (in)

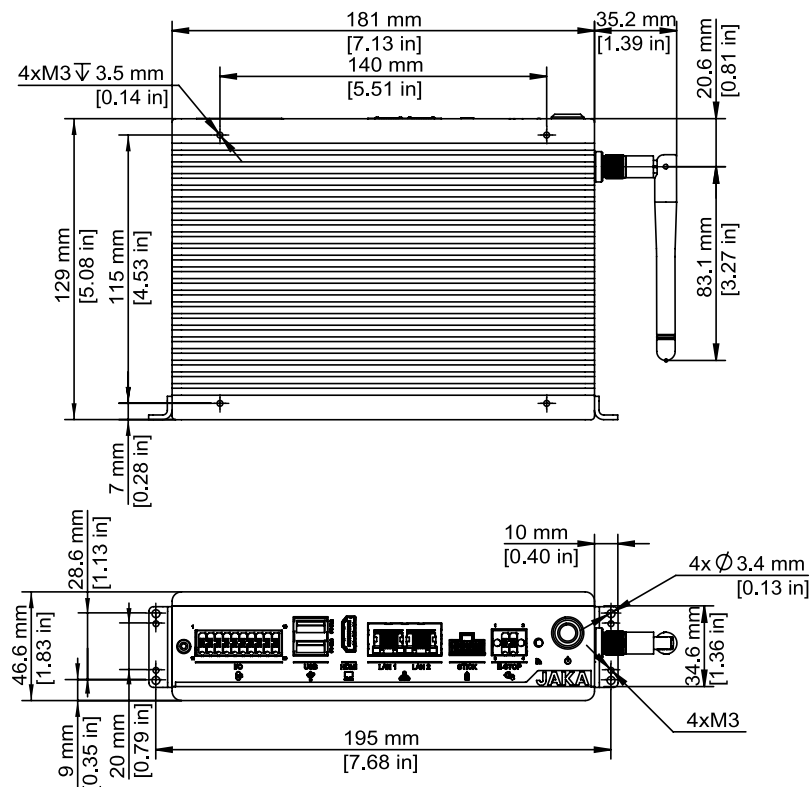


Figure 5-11: Control cabinet outline dimension

5.8 Performance Data

5.8.1 Load Capacity of the Robot

The maximum payload of the robot is related to the offset of the gravity center, and degree of the offset is related to the distance between the center of robot end flange and the payload centroid. Figures below show the relation between the payload and the offset of the gravity center:

X: Center of gravity offset

Y: Payload

1. Load capacity of the MiniCobo

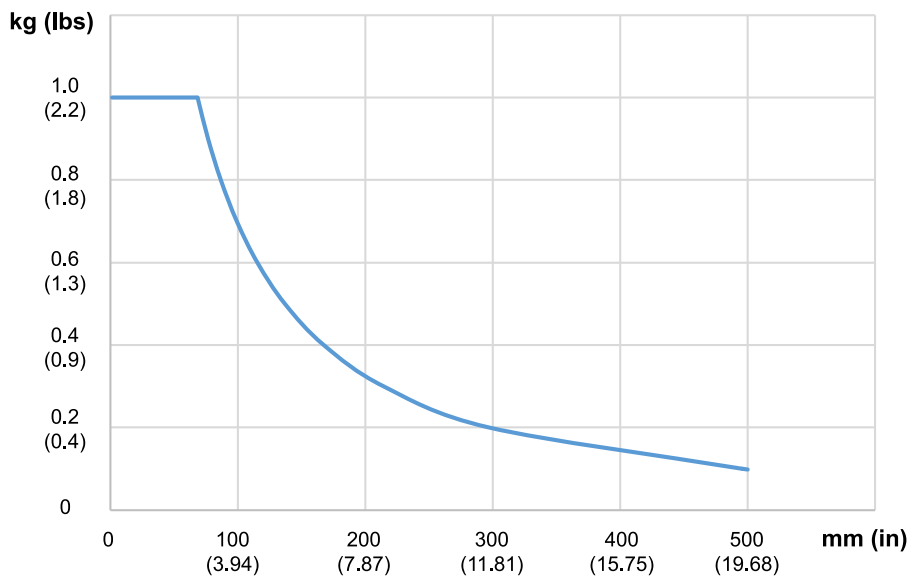


Figure 5-12: Load capacity of the MiniCobo

2. Load capacity of the Mini 2

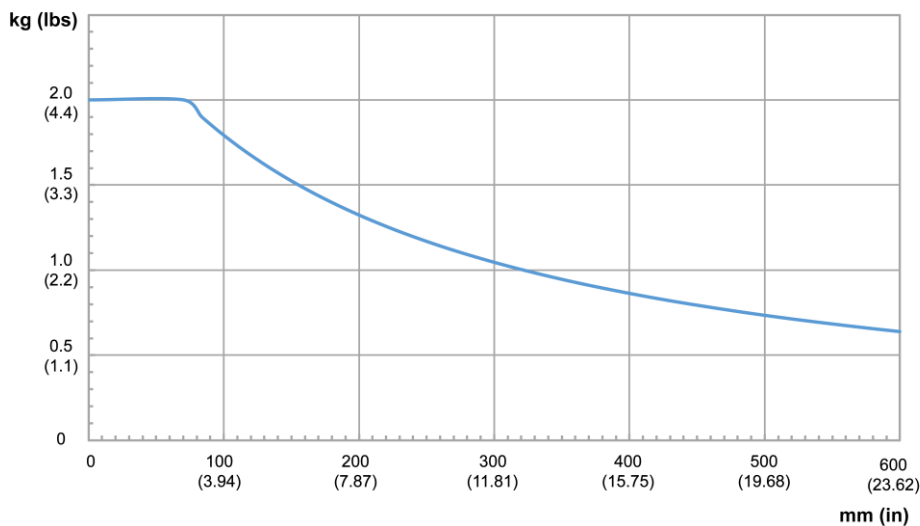


Figure 5-13: Load capacity of the Mini 2



NOTE

Offset distance is the center of gravity distance from the center of the flange.

5.8.2 Stopping Time and Distance

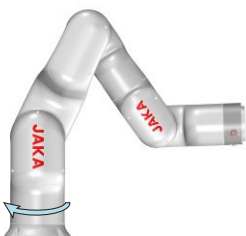
The time from the application of a stop signal to the standstill of the robot is measured. This measurement is carried out for various reaches and velocities (measurement according to ISO 10218-1).

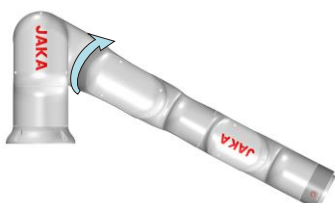
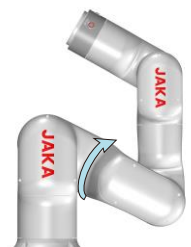
Safety stopping time is the time it takes to stop the robot from the moment when the emergency stop button is pressed or a safety protection function is triggered. The stopping distance is the distance the end of the robot moves during the safety stopping time. Among them, pressing the emergency stop button falls into Cat.1, while triggering the safety protection function falls into Cat.2. During this period, the robot is still moving and may harm the personnel or other equipment. Therefore, users and integrators should consider this time and distance in

risk assessment.

The test conditions are as follows:

- Speed: 100%, 66%, 33%
- Reach: 100%, 66%, 33%

	100% Reach	66% Reach	33% Reach
Joint 1			

	100% Reach	66% Reach	33% Reach
Joint 2			

	100% Reach	66% Reach
Joint 3		

- Load: Refer to the [Table 5-12](#)

Table 5-12: Load of stopping time and distance test

Model	Payload
MiniCobo	1 kg (2.2 lb)
Mini 2	2 kg (4.4 lb)

See [Appendix 1: Stopping Time and Distance](#) for the respective stopping distance and time of Cat.1 and Cat.2.



WARNING

Breakdown of the internal joint holding brake:

- Take into account a possible breakdown of the internal joint holding brake during your risk assessment.
- Take into account that the internal joint holding brake of the JAKA robot only withstands a limited number of brake operations.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

If there is a power interruption of the control system, the brakes are applied and the robot mechanics may leave the planned trajectory.



WARNING

Leaving the planned trajectory of the robot mechanics:

- Use the buffering of the 24 V supply (UPS) in order to enable a controlled stop of the mechanics, in accordance with stop category 1, by making use of the stored residual mechanical and electrical energy.
- Use a synchronous stop on the path to avoid collisions with obstacles.
- Observe the extension of the run-on path while performing your risk assessment.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

5.8.3 Forces and Torques on Robot Base

The illustration shows the directions of the robot stress forces.

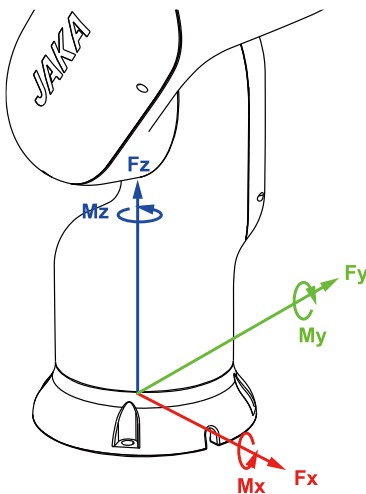


Figure 5-14: Directions of forces and torques on robot base

Fx: Force in X direction

Mx: Bending torque in X direction

Fy: Force in Y direction

My: Bending torque in Y direction

Fz: Force in Z direction

Mz: Bending torque in Z direction

The table below shows the maximum forces and torques in various directions working on the robot base.

Table 5-13: Forces and torques on robot base

	F_x	F_y	F_z	M_x	M_y	M_z
MiniCobo, Mini 2	120 N (27 lbf)		140 N (31 lbf)	100 Nm (885 lbf·in)		90 Nm (797 lbf·in)



NOTICE

These forces and torques are maximum values that are rarely encountered during operation. The values also never reach their maximum at the same time!

5.8.4 Torques on Robot Flange

The table below shows the maximum torques working on the robot flange.

Table 5-14: Torques on robot flange

Robot Model	MiniCobo	Mini 2
Torque	14 Nm (124 lbf·in)	14 Nm (124 lbf·in)

6 Transport and Commissioning

6.1 Quick Start Guide

The following table provides a brief overview for robot usage:

Table 6-1: Quick start guide

No.	Steps	Operations	References
1	Robot mounting	Unpack the robot.  NOTE If you want to move the robot later, please keep the original packaging.	
2		Remove robot from packaging.	Refer to 6.3 Unpacking
3		Transport the robot to the intended mounting location.	Refer to 6.4 Handling
4		Lift and secure the robot to the platform or foundation.	Refer to 6.5.2 Mounting
5		Secure the tools on the end of the robot, if any.	Refer to 6.9 Mounting the End-Effector
6	Control cabinet installation	Unpack the control cabinet.	Refer to 6.3 Unpacking
7		Take out the control cabinet from the packaging materials and carry it to the vicinity of the robot.  NOTE The length of robot connection cable is 6 m (36.220 in), so the distance between control cabinet and robot should be less than it.	Refer to 6.5.3 Mounting the Control Cabinet
8	Connect the cables	Connect the robot connection cable, control cabinet power cord, control stick cable, Ethernet cable (if any), and TIO cable (if any) in sequence.	Refer to 6.6.1 Robot Connection Interface 6.7.5.1 Power Interface 6.7.4.6 Control Stick Interface (STICK) 6.6.2 Tool I/O Port
9		The emergency stop button on control stick is released by default, if it is pressed, please release it first.	Refer to 5.6 Control Stick Buttons
10	Connect the robot	Power up the control cabinet, power on and enable the robot with the control stick.	Refer to 6.8.1 Power On and Shut Down the Robot System
11		Connect the robot using a computer or pad or mobile phone.	Refer to JAKA App Software User Manual
12	Configure the robot	Set the mounting orientation, payloads and collision sensitivity of the robot in the JAKA App.	Refer to JAKA App Software User Manual
13	Control the robot	Enable robot to move by pressing the FREE button on the end of the robot or entering programming or	Refer to 5.3.2 Buttons

No.	Steps	Operations	References
		manual operation interface of the JAKA App.	JAKA App Software User Manual

6.2 Transport and Storage

6.2.1 Transport Requirements

Please use the original packaging to transport the robot. If you want to move the robot later, please keep the original packaging.

When the robot is lifted, corresponding measures should be taken for positioning to avoid damage caused by accidental movement.

When moving the robot from its packaging to the installation position, it should be supported by at least 2 persons until all screws at the robot base are securely fastened.



WARNING

- Ensure that the back or other body parts of the operators are not overloaded when the equipment is lifted. Use appropriate lifting equipment. JAKA is not responsible for damage incurred during the transport of the equipment. The robot must be supported by at least two persons depending on the model.
- Please comply with the relevant lifting regulations in each region and country.
- Ensure that installation instructions are strictly followed when the robot is installed.



NOTICE

If the robot is transported without using its original packaging, all warranties will be voided.

6.2.2 Storage Conditions

1. Storage temperature: -10~50°C (14~122°F)

For long-term storage, to maintain the robot system reliability, it is recommended to keep the temperature within 25±10°C (59~95°F). Avoid sudden temperature changes if possible. (10°C/h (50°F/h) and above).

2. Storage humidity: 20% RH~85% RH

For long-term storage, to maintain robot system reliability, it is recommended to keep the humidity within 45%~65%. Keep away from dew condensation or mildew.

3. Anti-static

It is easy to generate static when kept in extremely dry conditions. The shock of electrostatic discharge may damage the semiconductor. Please store the robot system in an anti-static bag.

4. Other environmental conditions

Please keep robot system in an environment that does not produce poisonous gas, dirt, and dust. Do not place heavy objects on it during storage.

6.3 Unpacking

The following figure presents the packaging of the JAKA robot and the control cabinet.



Figure 6-1: Robot or control cabinet box

1. Open the box top side.
2. Lift up and remove the upper inner packaging part.
3. Inspect the robot and the control cabinet for transport damage.

6.4 Handling

The table below shows the joint angle of packaging orientation. The robot can automatically change to this orientation by configuring it in the JAKA App. It is recommended to transport the robot in this orientation.

Table 6-2: Handling joint angle

Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6
0°	120°	-120°	0°	-90°	0°

At least two persons are required for lifting and securing the robot during transportation. One person should hold the connection places between joint1 and joint 2 with both hands, while the other holds the joint 2 with both hands, as shown in the figure below.

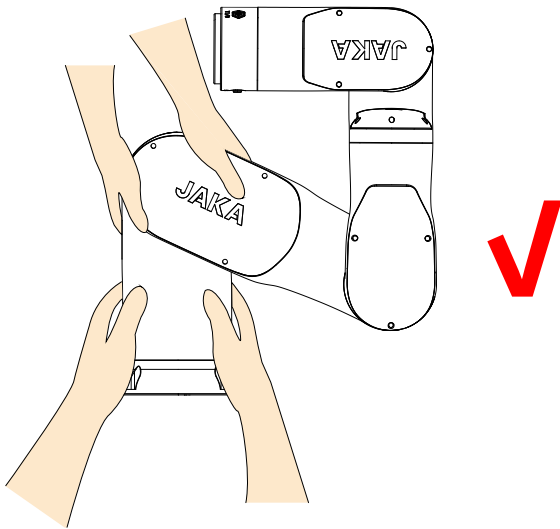


Figure 6-2: Correct handling method

Lifting the robot by grabbing the robot's upper arm or joint 4/5/6 is strictly prohibited, as shown in the figure below. This action may cause damage to the internal structure of the robot.

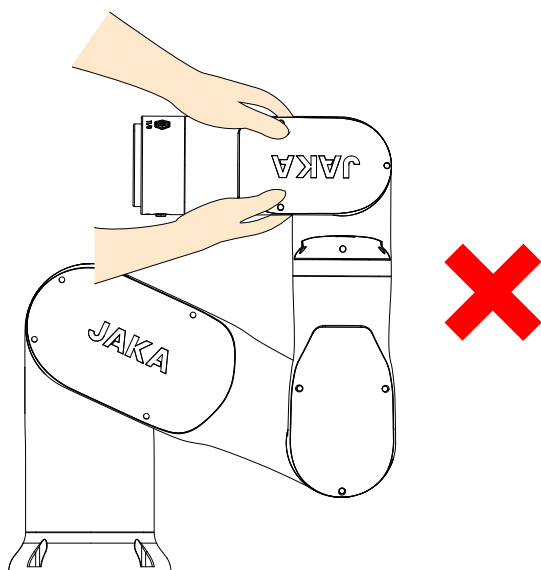


Figure 6-3: Incorrect handling method

6.5 Mechanical Installation

6.5.1 Safety Instructions



WARNING

- Ensure that the robot is installed correctly and safely.
- The installation surface must be vibrant proof with enough loading capacity.



WARNING

- Ensure that the end effector is installed correctly and safely.
- Ensure the tool's safety to prevent any accidental falling of parts that could pose a danger.



WARNING

- Ensure that the control cabinet and cables are not exposed to liquids. A damp control cabinet can pose a risk of electric shock or fatal injury.
- The control cabinet must not be exposed in dusty or humid environments exceeding its IP level. Pay close attention to environments with conductive dust.



WARNING

If the robot is submerged in water for an extended period, it may be damaged. Robots should not be installed in water or a humid environment.

6.5.2 Mounting the Robot

The robot can be mounted in various different positions. Several typical mounting methods are shown in figure below:

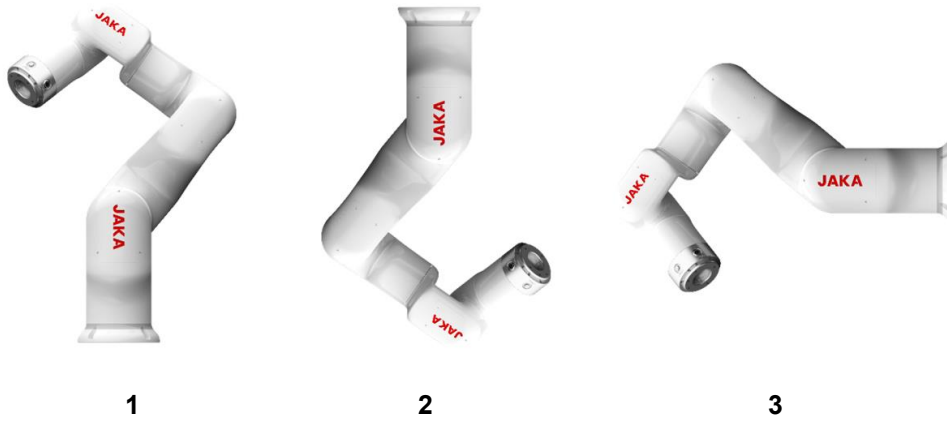


Figure 6-4: Robot mounting positions

- 1 Floor mounting
- 2 Inverted mounting
- 3-4 Wall mounting

6.5.2.1 Mounting Surface Requirements

The robot is intended to be mounted on a sturdy surface. The surface must not only withstand the torque stated below, but also have sufficient stiffness so that the deformations and vibrations which occur do not lead to any major deviations on the TCP. It is advisable to avoid directly mounting the robot on an empty box, as this can easily induce resonance and generate abnormal noise. When the robot is mounted on a linear actuator or mobile platform, the acceleration of the mounting base should be minimized. High acceleration may trigger a false collision alarm, causing the robot to stop.

Note the torque to be taken up by the mounting surface during normal operation:

Table 6-3: Torque of robot mounting surface

Robot Model	Mounting Surface Torque	Mounting Surface Force
MiniCobo, Mini 2	612 Nm (5417 lbf·in)	500N (112 lbf)



WARNING

Do not connect the power during the robot mounting process.



NOTICE

- When mounting the robot at different angles, the mounting app in the App must be modified. See JAKA App Software User Manual for details.
- It is normal that the mounting holes of the robot are rusted or have traces of use. The JAKA robot needs to accept a series of performance tests before leaving the factory. Therefore, there will be traces in the mounting hole, which is not a quality problem and does not affect its use.
- If the robot is not fixed on the foundation firmly, the mechanical structure of the robot may be unstable, and it may tip over.
- When inverted mounting or wall mounting is adopted, the rigidity and stability of the mounting plane should be ensured, it is very dangerous to mount the robot on the wall or ceiling with insufficient intensity and rigidity, which may cause the robot to fall or vibrate, leading to severe injuries or serious damage to the robot.

- The robot needs to be used in an environment that matches its IP level, which helps reduce the failure and extend the service life of the robot.

6.5.2.2 Dimensions of Robot Base

The following figure presents the installation dimensions of robot base.

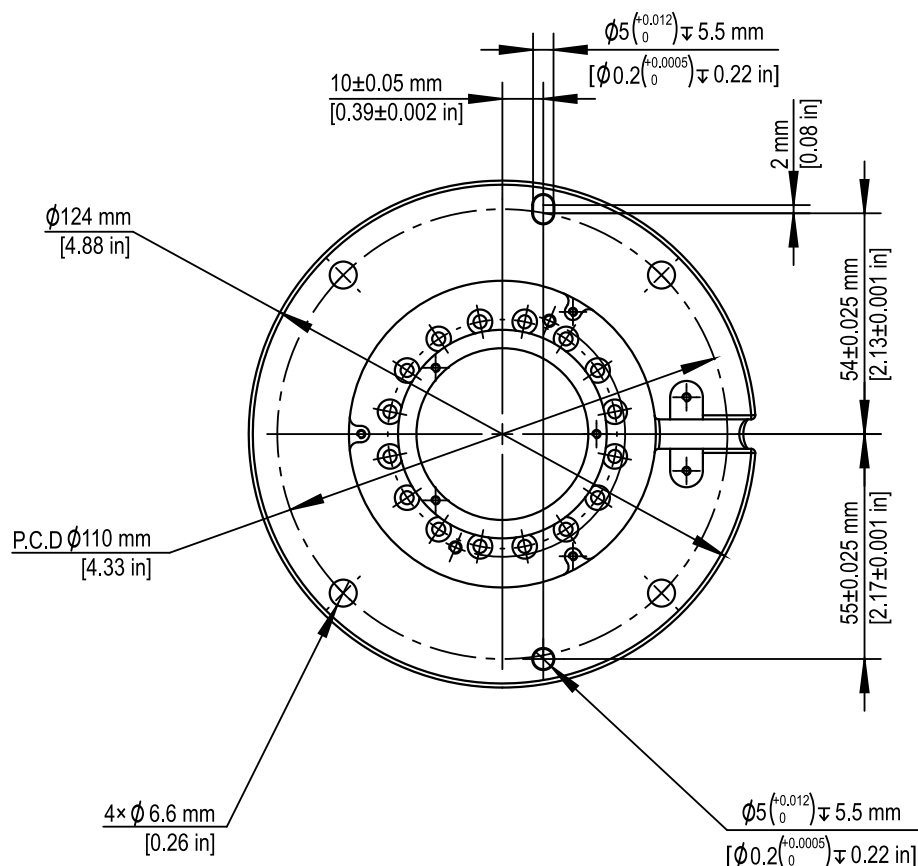


Figure 6-5: Base dimensions

6.5.2.3 Mounting the Robot

- Use a lifting device when necessary for lifting up the robot out of the transport packaging.
- Place the robot on the mounting surface and hold it in place while completing the next step.
- Fasten the robot base to the mounting surface by using the four through holes. For additional fixing, centering pins can be added. The following table presents the tightening torque and size.

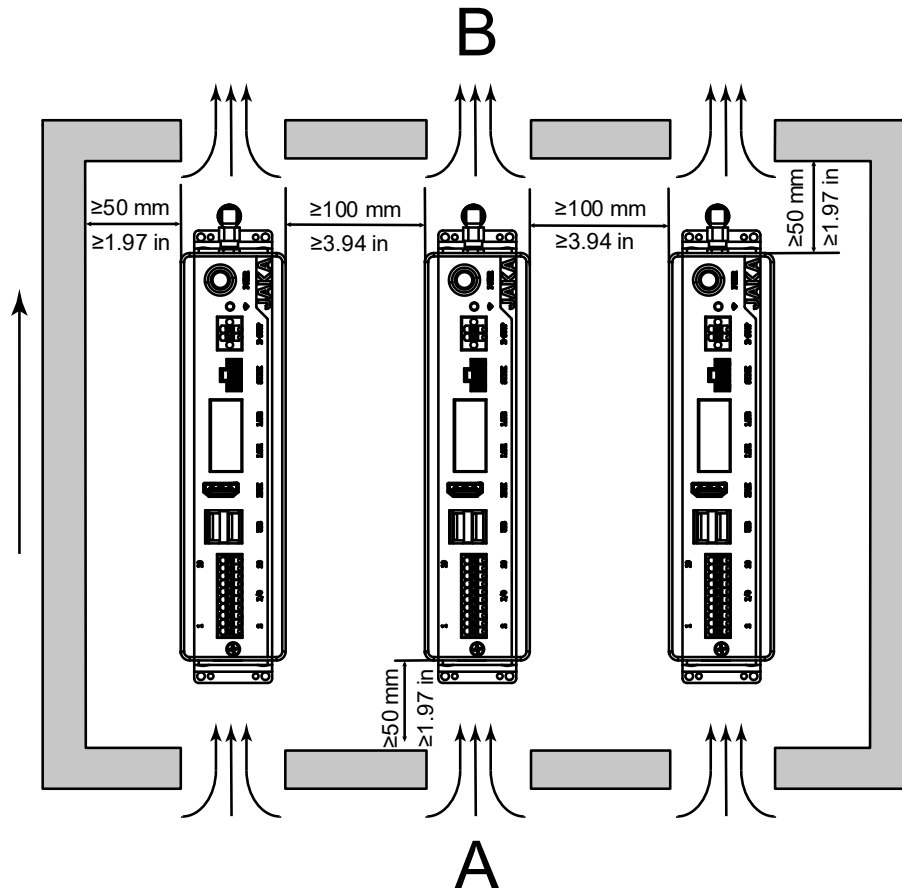
Table 6-4: Robot mounting data

Model	Screw Size ⁱ	Screw Quantity	Recommended Tightening Torque	Through Hole	Pin Hole
MiniCobo Mini 2	M6	4 pcs	15 Nm (133 lbf-in)	Φ6.6 mm (0.3 in)	Φ5 mm (0.2 in)

i: Class 12.9 hex socket head cap screws are recommended for the base mounting. The recommended screw engagement depth is 1.5 to 2 times the nominal thread diameter.

6.5.3 Mounting the Control Cabinet

MiniCab needs to be installed in a dry place with good ventilation and cooled by natural convection. In situations where significant heat is generated, such as when the robot operates at high speeds, carries heavy loads, or brakes frequently, an external fan should be used to cool MiniCab. In order to ensure the MiniCab can be cooled by a fan or natural convection, please refer to the distance shown in the diagram for mounting the control cabinet. To meet the cooling requirements of the control cabinet, ensure a gap of at least 100 mm (3.94 in) between two MiniCabs, and a gap of at least 50 mm (1.97 in) on each side of the control cabinet.



A: Air inlet

B: Air outlet

Figure 6-6: Control cabinet mounting space



NOTICE

Inoperable equipment/reduction of lifetime:

Ensure sufficient airflow for the control cabinet, especially if the device is installed in an enclosure.

Failure to follow these instructions can result in equipment damage.

6.5.3.1 Bracket Mounting

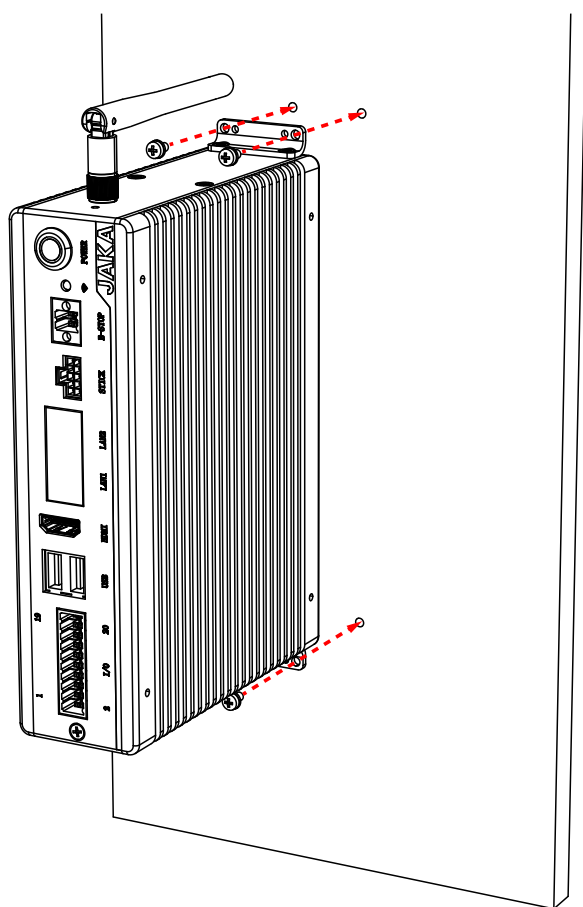


Figure 6-7: Bracket mounting diagram

The mounting steps are as follows:

	Steps	Tools and Specifications
1	According to the control cabinet's outline dimensions, drill four holes on the mounting plane.	
2	Align the holes of the brackets and mounting plane, and secure two brackets by screws.	Screw type: M3 Screw quantity: 4 pcs Tightening torque: 0.6-0.8 Nm (5.31-7.08 lbf in)
3	Ensure that the mounting surface is smooth and free of vibration and confirm that the control cabinet is securely mounted.	When the mounting plane is mobile, to reduce the impact of vibration on MiniCab during movement, it is recommended to add a rubber gasket between MiniCab and the mounting plane.

6.5.3.2 Bottom Mounting

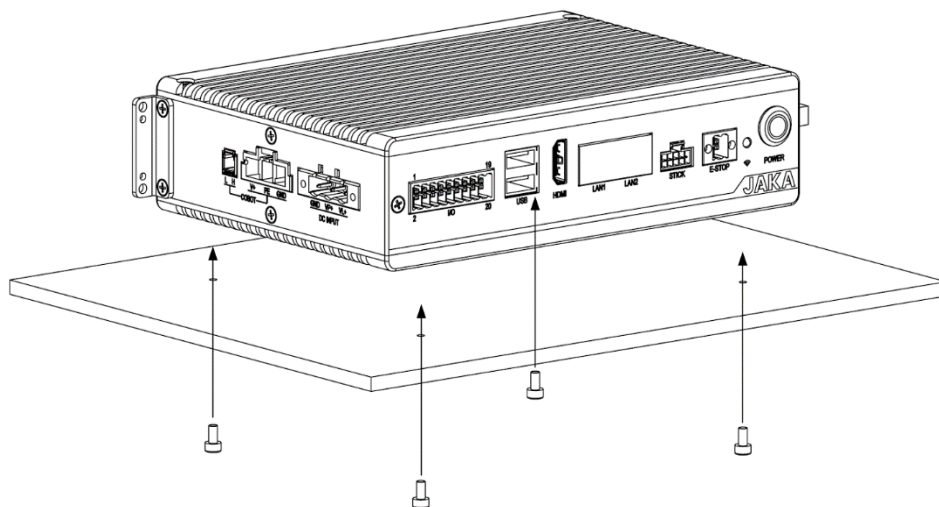


Figure 6-8: Bottom mounting diagram

The mounting steps are as follows:

	Steps	Tools and Specifications
1	According to the control cabinet's outline dimensions, drill four holes on the mounting plane.	Aperture: 3.4 mm (0.13 in)
2	Align the holes of the control cabinet bottom and mounting plane, and secure the control cabinet by screws.	Screw type: M3 Screw quantity: 4 pcs Tightening torque: 0.6-0.8 Nm (5.31-7.08 lbf in)
3	Ensure that the mounting surface is smooth and free of vibration and confirm that the control cabinet is securely mounted.	When the mounting plane is mobile, to reduce the impact of vibration on MiniCab during movement, it is recommended to add a rubber gasket between MiniCab and the mounting plane.

6.5.3.3 Guide Rail Mounting

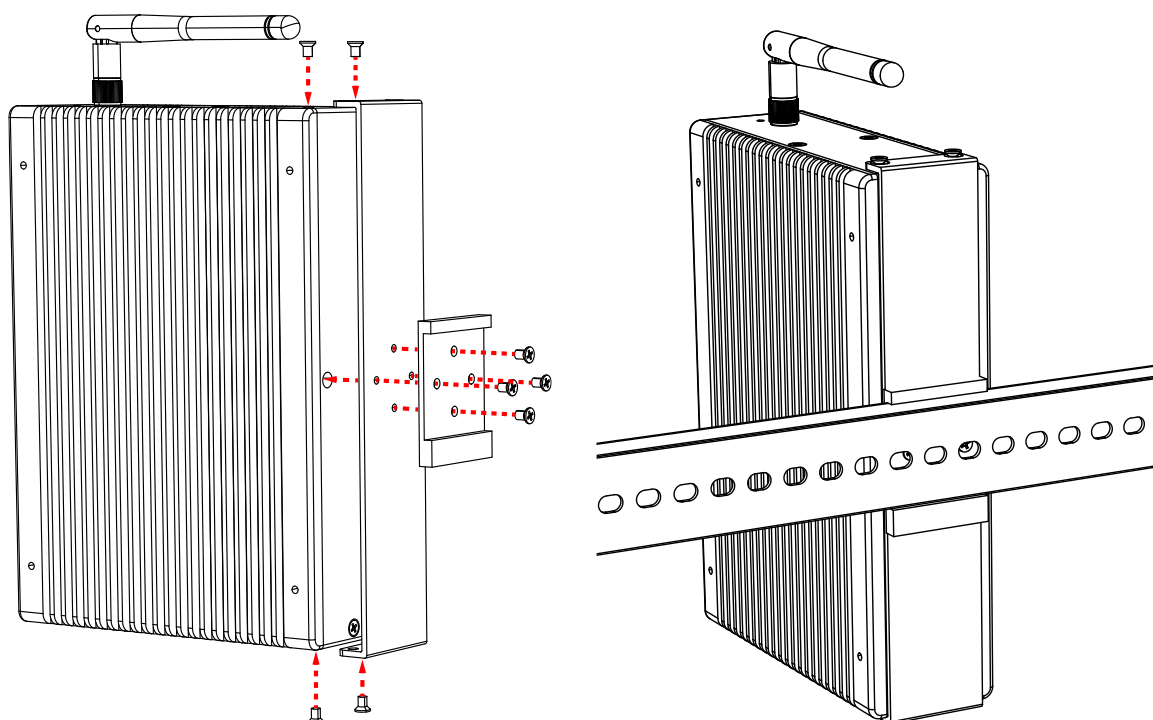


Figure 6-9: Guide rail mounting diagram

The rail mounting accessories and rail clamp are optional accessories. If you need to mount MiniCab on the guide rail, please contact the JAKA technicians to acquire these accessories. The mounting steps are as follows:

	Steps	Tools and Specifications
1	Align the mounting holes on the rail mounting accessories with the four mounting holes on the side panels of the control cabinet, then use screws to secure them together.	Screw type: M3 Screw quantity: 4 pcs Tightening torque: 0.6-0.8 Nm (5.31-7.08 lbf in)
2	Align the mounting holes on the rail clamp with the four mounting holes on the rail mounting accessories, then use screws to secure them together.	Screw type: M3 Screw quantity: 4 pcs Tightening torque: 0.6-0.8 Nm (5.31-7.08 lbf in)
3	Mount the MiniCab with the rail mounting accessories onto the guide rail.	

6.6 Electrical Connections of the Robot

6.6.1 Robot Connection Cable Interface

Use the robot connection cable provided by JAKA to connect the robot and the control cabinet. Before powering on the robot, ensure that the connector is locked firmly. Before disconnecting the robot connection cable, the robot must be powered off.



WARNING

- Do not disconnect the robot connection cable while the robot is not powered off.
- Do not extend or modify the original cable.

6.6.1.1 Bending Radius of Robot Connection Cable

Robot connection cables cannot be used with cable carriers and are designed for fixed installation only. The following table shows the diameter (OD) of the robot connection cables and its minimum bending radius.

Table 6-5: Specifications of robot connection cable

Model	Cable Diameter (OD)	Min Bending Radius
MiniCobo	7.1 mm (0.28 in)	42.6 mm (1.68 in)
Mini 2		

6.6.2 Tool I/O Port

The tool input and output interface are referred to as TIO, installed on the side of the end flange of the robot, which contains two digital inputs, two digital outputs and two analog inputs, and can multiplex the two-channel RS485 signal at the same time. For the definition of the interface, see [6.6.2.1 Definition of End Effector Side](#).

The TIO cable connector features a foolproof design to ensure correct installation. Align the raised part of the TIO connector with the concave groove on the TIO connector of the flange and insert the cable. The TIO position is as shown in the figure below.

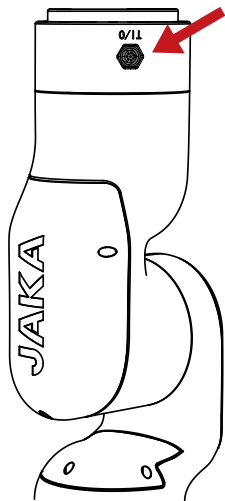


Figure 6-10: Robot tool I/O

The definition and specifications of the TIO cable are as follows.

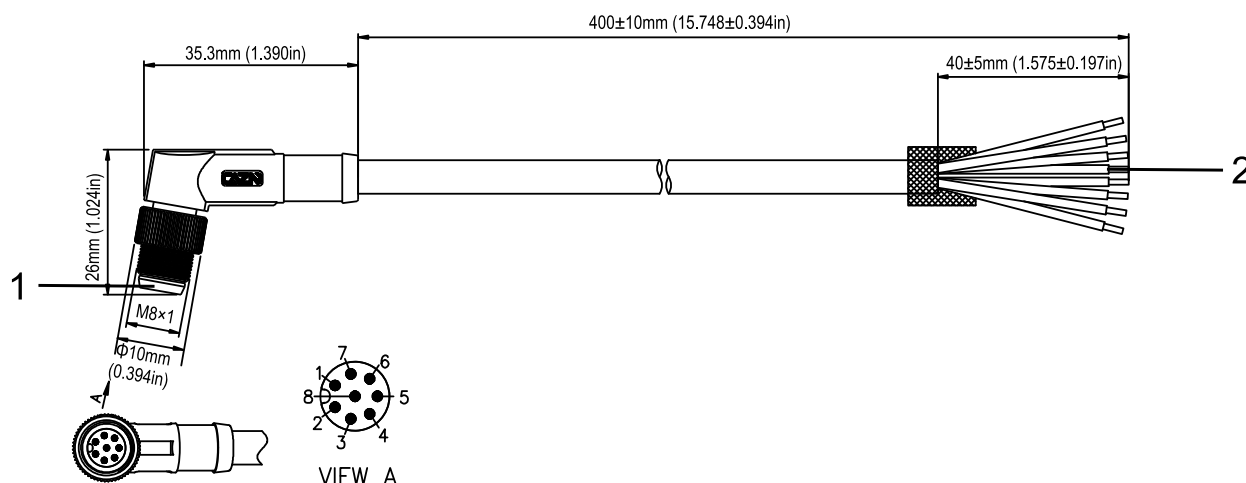


Figure 6-11: TIO cable dimensions

1 Connect the robot flange

2 Connect the end effector

6.6.2.1 Definition of End Effector Side

The definition table of the TIO V3.0 interfaces is as follows:

Table 6-6: Specifications of TIO pins

Pin	Definition	I/O	Color	Description
1	+24V	-	Red	Positive electrode, 24V/12V (switchable); configurable for enabling or disabling; continuous current capacity 1A; peak output current up to 2A.
2	DI1	I	Blue	Digital input 1: sink or source configurable by application
3	DI2	I	Green	Digital input 2: sink or source configurable by application
4	DO1/RS485A_1	O	Yellow	Digital output 1: sink, source, or push-pull configurable by application; current output capability $\leq 1A$ Multiplexed as RS481 communication A+
5	DO2/RS485B_1	O	Pink	Digital output 2: sink, source, or push-pull configurable by application; current output capacity $\leq 1A$ Multiplexed as RS481 communication B-
6	AIN1/RS485A_2	I	Brown	Analog input 1: detection range 0-10V Multiplexed as RS482 communication A+
7	AIN2/RS485B_2	I	White	Analog input 2: detection range 0-10V Multiplexed as RS482 communication B-
8	GND	-	Gray	24V negative electrode

6.6.2.2 Wiring of End effector Side

6.6.2.2.1 Digital Input

TIO supports 2 DI digital input interfaces, compatible with sink and source. It can be configured in the App. For the details of operation, see JAKA App Software User Manual.

1. Dry contact input

When the DI input is configured as sink:

The dry contact input (switch-type input) is connected to GND of TIO (gray wire) at one end, and to the digital input (blue or green wire) at the other end.

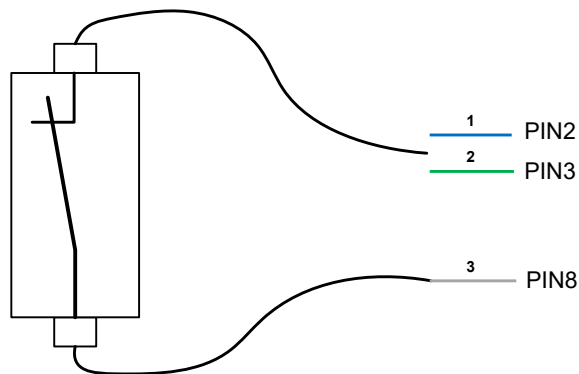


Figure 6-12: Sink wiring of digital input

- 1 Blue
- 2 Green
- 3 Gray

When the DI input is configured as source:

The dry contact input (switch-type input) is connected to GND of TIO (red wire) at one end, and to the digital input (blue or green wire) at the other end.

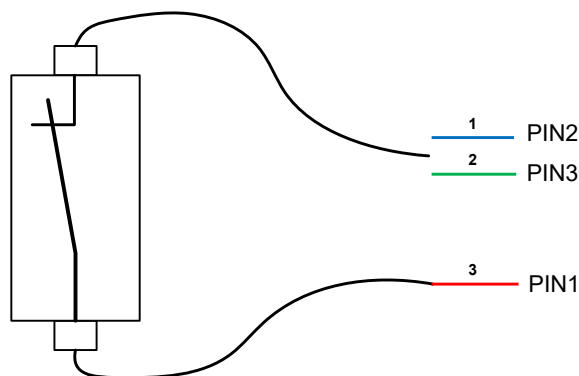


Figure 6-13: Source wiring of digital input

- 1 Blue
- 2 Green
- 3 Red

2. Connect sink or source device

The connection method of sink and source type digital input devices: V+ is connected to 24V of TIO (red wire), 0V is connected to GND of TIO (gray wire), and the signal wire is connected to the digital input of TIO (blue or green wire).

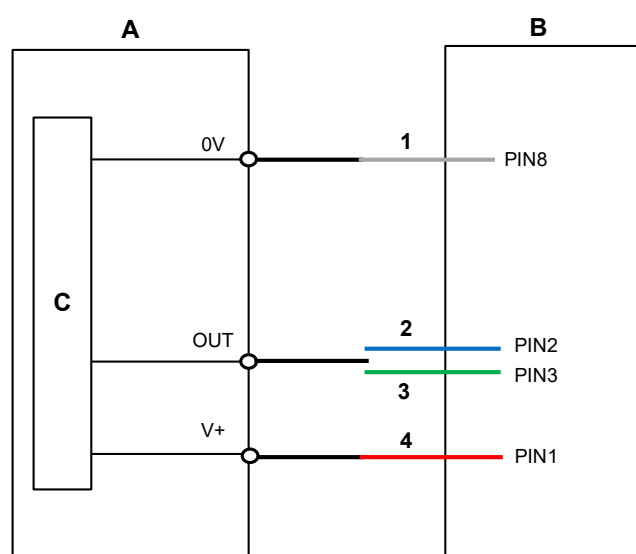


Figure 6-14: Wiring of sink device for digital input

A: External sink device

B: TIO V3

C: Main circuit

1 Gray

2 Blue

3 Green

4 Red

6.6.2.2.2. Digital Output

When the digital output interface is sink or source output, it adopts open drain output and the maximum continuous current output is 1A. Connection method: the external input interface is connected to the digital output of TIO (yellow or pink wire), V+ side of external device is connected to 24V of TIO (red wire), and 0V side of external device is connected to GND of TIO (gray wire).

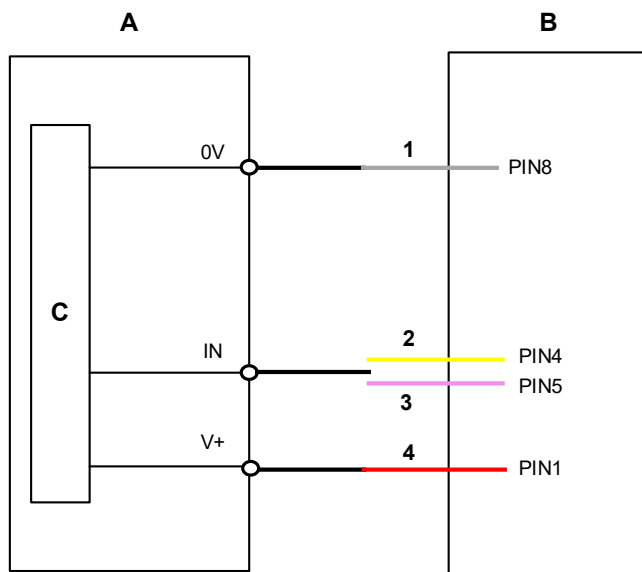


Figure 6-15: Wiring of sink device for digital output

A External sink device

B TIO V3

C Main circuit

1 Gray

2 Yellow

3 Pink

4 Red

6.6.2.2.3. RS485

Using the RS485 function, the wiring method: external RS485+ is connected to RS485+ of TIO (yellow or brown wire), external RS485- is connected to RS485- of TIO (pink or white wire), the V+ of external device is connected to the 24V of TIO (red wire), and 0V of external device is connected to the GND of TIO (gray wire).

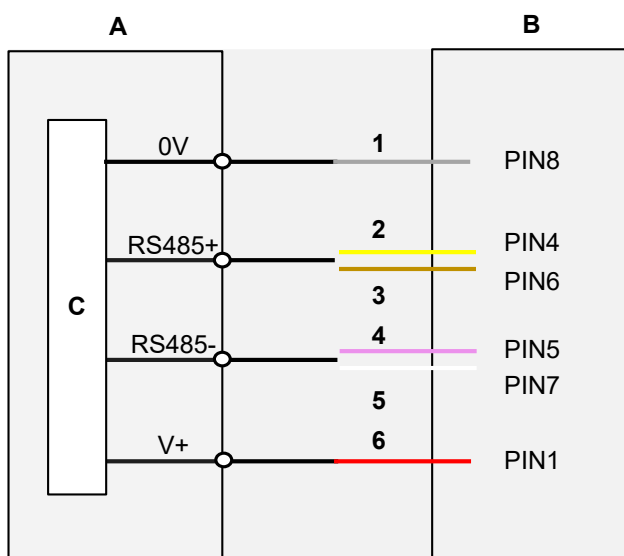


Figure 6-16: Wiring of RS485

A External device

B TIO V3

C Main circuit

1 Gray

2 Yellow

3 Brown

4 Pink

5 White

6 Red

6.6.2.2.4. Analog Input

TIO supports 2 analog voltage input interfaces, and the voltage input range is 0-10V. The wiring method: the external analog voltage positive electrode is connected to AIN1/AIN2 of TIO (white and brown wire), and the internal circuit of the negative electrode on the TIO board is grounded. The V+ of external device is connected to the 24V of TIO (red wire), and 0V of external device is connected to the GND of TIO (gray wire).

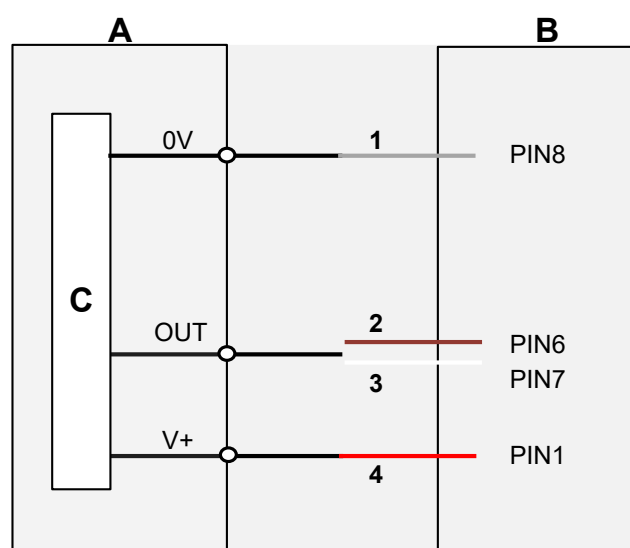


Figure 6-17: Wiring of analog input

A External device

B TIO V3

C Main circuit

1 Gray

2 Brown

3 White

4 Red

6.7 Electrical Connections of the Control Cabinet

6.7.1 Electrical Warnings and Precautions

When designing and installing robot applications, you must obey the following warnings and precautions. The implementation of maintenance must also comply with these warnings and precautions.



WARNING

- Do not connect safety signals to non-safe PLCs with inappropriate safety levels. If this warning is not obeyed it may lead to severe or fatal injury. Separate the safety I/O from the general I/O.
- All safety-related signals are constructed redundancy (two independent channels). Keeping two channels separate can ensure that the safety function will not be lost when a single failure occurs.
- For the introduction of the I/O function in the control cabinet, refer to [6.7.4.1 I/O Interfaces](#).



WARNING: ELECTRICITY

- Please ensure that all water-sensitive devices remain dry. Cut off power immediately if there is a leakage.
- Use original cables to connect the robot. Do not use robots in applications that need to bend cables. Contact your supplier if longer or flexible cables are needed.
- For protective earth (PE), use the screw connectors in the control cabinet marking the earth symbol. The grounding conductor shall at least have the current rating of the highest current in the robot system.
- When connecting cables on the I/O of the control cabinet, open the cabinet door, remove the metal board of the cable hole, and ensure that the I/O cable avoids rubbing with the edge of the cable hole.



WARNING

- The robot has been tested according to international IEC standards for Electromagnetic Compatibility (EMC). Interference signals beyond the standard will cause abnormalities. Very high signal levels or excessive exposure can damage the robot permanently. JAKA will not be held responsible for any damages caused by EMC problems.
- The length of the I/O cable to connect the control cabinet, machinery and factory equipment must not be longer than 30 meters unless the result of the extension test shows the feasibility. Use shielding cables if necessary.

6.7.2 Grounding

6.7.2.1 Grounding Requirements

The controller cabinet ground must come from the mains power supply PE.

- The grounding cable color shall be green-yellow.
- The ground for the controller cabinet, robot manipulator and peripheral devices must be the same, preferably an equipotential ground grid (mesh).
- Ground connection points must have stable inter-metallic bonding, like screw fixation. Paint, dirt, rust, and other insulating material must be removed from the contacting surfaces.

For requirements on the marking of the supply ground connection inside the control cabinet refer to UL 508C. For further details on how grounding systems should be designed refer to IEC 61000-5-2. For details of cross-sectional area of PE refer to IEC 60204-1.

6.7.2.2 Grounding Positions

Grounding positions and methods are as follows.

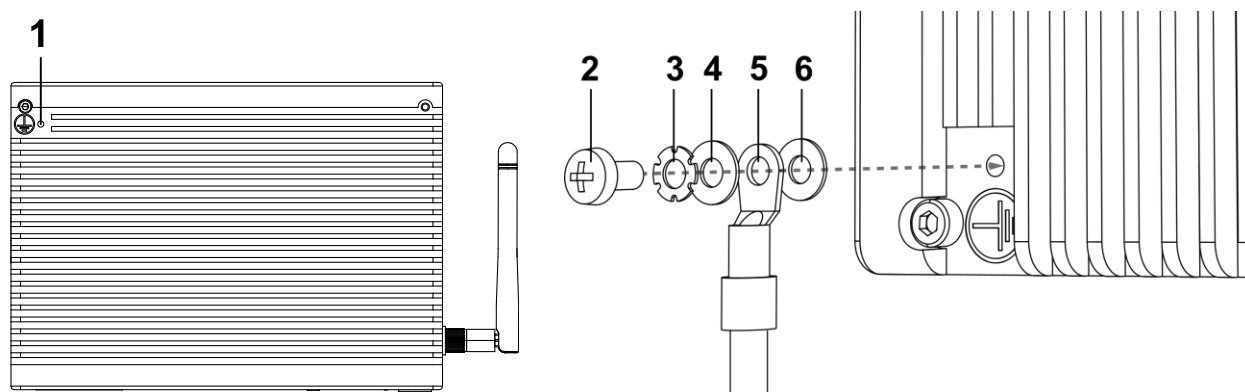


Figure 6-18: MiniCab grounding

1. Feed and fasten the ground cable (protective earth ground) to the ground connection (1) of the robot housing. The ground connection is marked with the $\oplus$ symbol.

2. Mount the ground cable (protective earth ground)

in the following order:

Screw M3 (2)

Lock washer M3 (3)

Washer M3 (4)

Ring cable lug M3 (5)

Washer M3 (6)

Tightening torque: 1.4 Nm (12.4 lbf-in), property class of the screw: 8.8 or greater.

Ground cable cross-section 2.5 mm² (AWG 13.2) or greater.

3. Verify the correct routing and fastening of the cables to help prevent any collision of cables and moving parts.



WARNING

ELECTRIC SHOCK DUE TO IMPROPER GROUNDING

Ground robot components in accordance with local, regional and/or national standards and regulations at a single, central point.

Failure to follow these instructions will result in death or serious injury.

6.7.3 Electrical Interfaces

MiniCab has user interfaces on the front panel and side panel. The front panel includes I/O, two USB, a HDMI, two LAN, a control stick and an E-STOP interface, a Wi-Fi status indicator and a POWER button. The side panel includes a power input interface, a robot connection cable interface, a 2.4G Wi-Fi antenna, and a network reset button.

6.7.4 Front Panel Interfaces

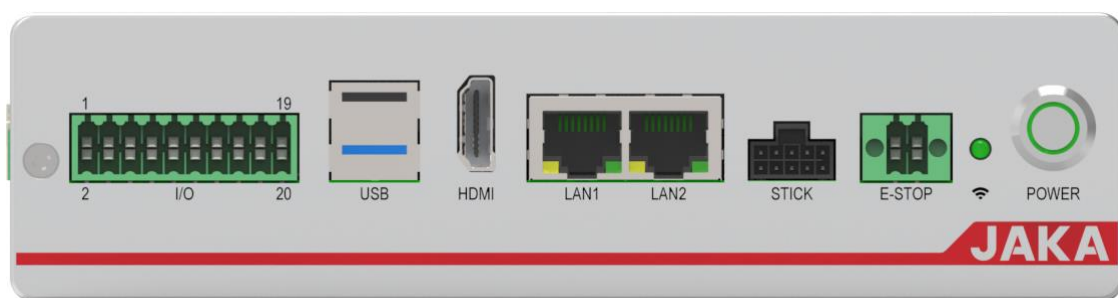


Figure 6-19: Front panel interfaces

6.7.4.1 I/O Interfaces

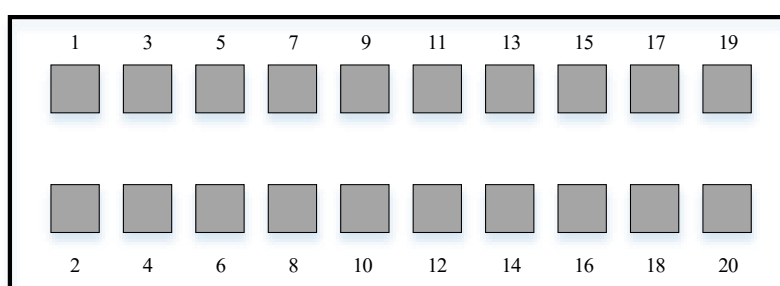


Figure 6-20: I/O interfaces

The front panel of the control cabinet has a 20 Pins I/O interface, the specifications of these Pins are as the table below.

Table 6-7: Specifications of I/O interfaces

PIN	Signal Name	Signal Type	Description
1	UDIO_24V	PO	Integrated interface 24V power output, internal integrated 2.7A over current protection function
2	VUDIO_COM+	PI	Common terminal of user interface power, default external short connect to PIN1
3	UDIO_24V	PO	Integrated interface 24V power output, same as PIN1
4	GND	PO	User interface power logic ground
5	Remote_OFF	I	Remote powering down input, connecting to 24V to power down the MiniCab
6	GND	PO	User interface power logic ground, same as PIN4
7	I/O_4	I/O	I/O multiplexing channel 4, or safety I/O channel 2 (Sink)

PIN	Signal Name	Signal Type	Description
8	Remote_ON	I/O	Remote powering up input, connecting to +24V to power up the MiniCab
9	I/O_3	I/O	I/O multiplexing channel 3, or safety I/O channel 2 (Sink)
10	I/O_7	I/O	I/O multiplexing channel 7, or safety input channel 3 (Sink)
11	I/O_2	I/O	I/O multiplexing channel 2, or safety I/O channel 1 (Sink)
12	I/O_6	I/O	I/O multiplexing channel 6, or safety input channel 3 (Sink)
13	I/O_1	I/O	I/O multiplexing channel 1, or safety I/O channel 1 (Sink)
14	I/O_5	I/O	I/O multiplexing channel 5, Sink
15	Reserved_CANH	I/O	For internal debugging only
16	Reserved_CANL	I/O	
17	MasterBus_RS485A	I/O	Master station interface of RS485 that usually used to extend the RS485 interface of I/O board
18	MasterBus_RS485B	I/O	
19	SlaveBus_RS485A	I/O	Slave station interface of RS485 that usually used to communicate with PLC and other external devices
20	SlaveBus_RS485B	I/O	

When wiring, select the wire that matches the terminal. The requirements for the wires are as follows:

Table 6-8: Wire specifications of I/O interfaces

Terminal Model	Recommended PIN Connector	Wires
DEGSON : 15EDGKNHB-3.5-20P-14-18A(H)	China: E0512 Overseas: PHOENIX AI1-12RD-Ferrule	0.75 mm ² & 17AWG Length < 30 m (1181 in)

6.7.4.1.1. Digital I/O Interfaces

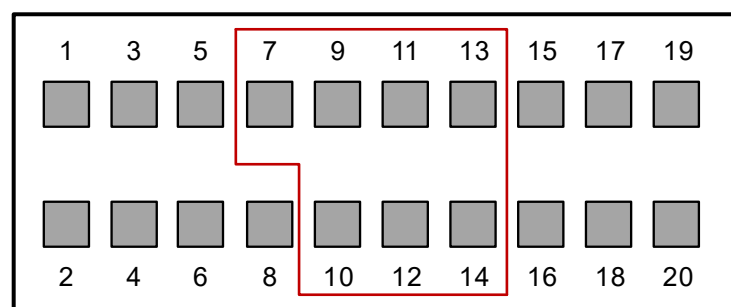


Figure 6-21: Digital I/O interfaces

Digital I/O has 7 digital signal interfaces, each I/O channel simultaneously has NPN input and output functions.

You can set the function of each interface in the App. Only one interface type can be selected for each channel at the same time. For example, if I/O_1 is set as DI, it cannot be set as DO at the same time.

Table 6-9: Level range

UD IO_COM	Low Level Range
24V	0~7V

1 Digital input:

When configured as DI, it is an Sink input, and IO_1 (PIN13) is valid when shorted to GND (PIN6). When using internal UDIO_24V, please short PIN1 and PIN2. Taking DI_1 as an example, the wiring diagram for other DI interfaces is the same. The wiring diagram is as follows:

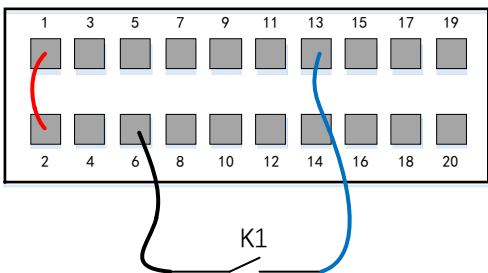


Figure 6-22: DI wiring

2 Digital output:

When configured as DO, it is an Sink output, which uses an open collector output internally and is connected with a freewheeling diode. It supports a maximum output current of 1A. When using internal UDIO_24V, please short PIN1 and PIN2. Taking DO_1 as an example, the wiring diagram for other DO interfaces is the same. The wiring diagram is as follows:

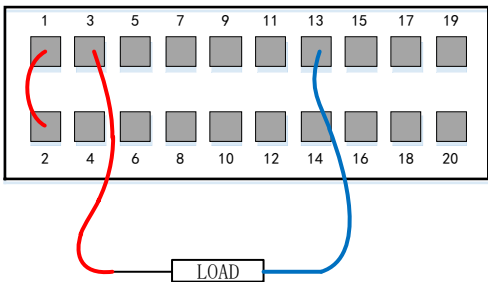


Figure 6-23: DO wiring

6.7.4.1.2. Safety I/O Interfaces

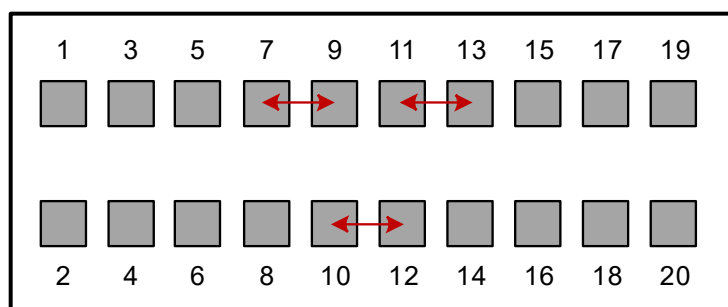


Figure 6-24: Safety I/O interfaces

The safety I/O function is only supported in software versions 1.7.1.37 and above. Therefore, before using the safety I/O function, an upgrade is required. The version requirements are as follows:

Table 6-10: Version limit of safety I/O

Type	Version
JAKA App	1.7.1.37 and above
Controller	1_7_1_36 and above
PSCB	03_02_PR and above
Servo	R3196 and above

For update steps, please refer to the software user manual. Contact JAKA technician to obtain the update installation package.

To ensure the safety function of the robot, the digital I/O interfaces I/O_1, I/O_2, I/O_3, I/O_4, I/O_6, and I/O_7 in the control cabinet can be configured as dedicated safety I/O. Among them, I/O_6 and I/O_7 can only be configured as safety DI, and can only be configured for protective stop function. The other four channels can be configured as safety DI or safety DO. The electrical specifications of the safety I/O interfaces are the same as the digital I/O interfaces. The safety I/O is designed with dual redundancy, so if one channel fails, the safety function still works. Therefore, when wiring, a pair of safety I/O should be connected simultaneously. For example, when connecting I/O_1, I/O_2 must be connected at the same time. The pairing relationship of the safety I/O is as follows:

Table 6-11: Pairing of safety I/O interfaces

DI	DO	Interface
DI1 & DI2	DO1 & DO2	PIN13 & PIN11
DI3 & DI4	DO3 & DO4	PIN9 & PIN7
DI6 & DI7		PIN12 & PIN10

Taking the DI1 & DI2 as an example, the wiring diagram is as follows (When using internal UDIO_24V, please short PIN1 and PIN2):

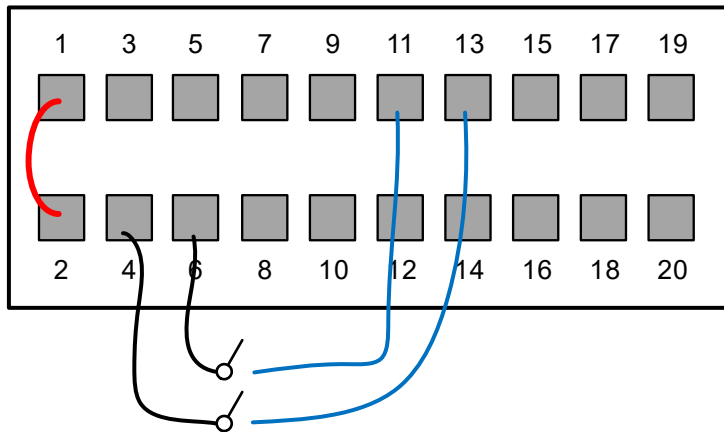


Figure 6-25: Safety I/O wiring

6.7.4.1.3. Remote On/Off

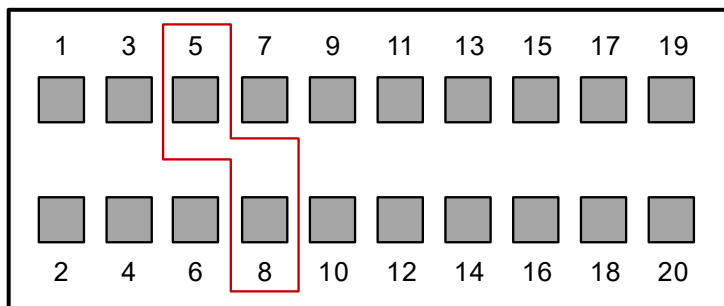


Figure 6-26: Remote on/off interfaces

MiniCab can be powered on after power is supplied for 4 seconds. Besides using the power on/off button on the control stick and the POWER button on the front panel of the control cabinet, MiniCab can also be remotely powered on and off by PIN5 and PIN8 of the I/O interface on the front panel of the control cabinet.

When using an external power on/off button, a self-resetting/momentary push button switch must be used, otherwise the remote power on/off operation may be ineffective.

1. Power Up

The external DC 24V power supply is required. Connect the positive of the power supply to the PIN8 (Remote_ON) interface and the negative to PIN6 (GND). The diagram below shows the wiring diagram for remote powering up the MiniCab. Press K1 for about 1 second and release it to power up the control cabinet.

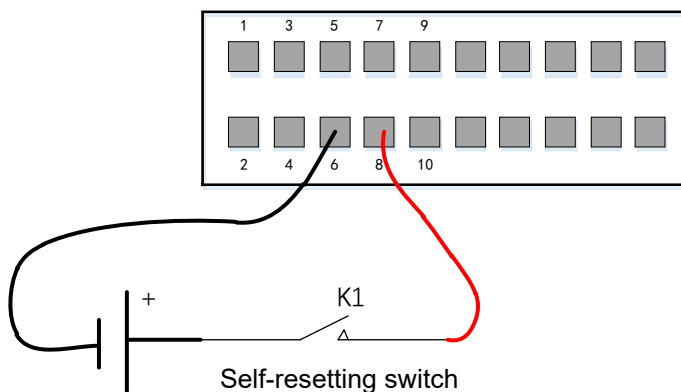


Figure 6-27: Remote on wiring

2. Power Down

If the external DC 24V power supply is used, connect the positive of the power supply to the PIN5 (Remote_OFF) interface and the negative to PIN6 (GND). It is also available when using internal UDIO_24V. Press K2 for more than 3 second to power down the control cabinet.

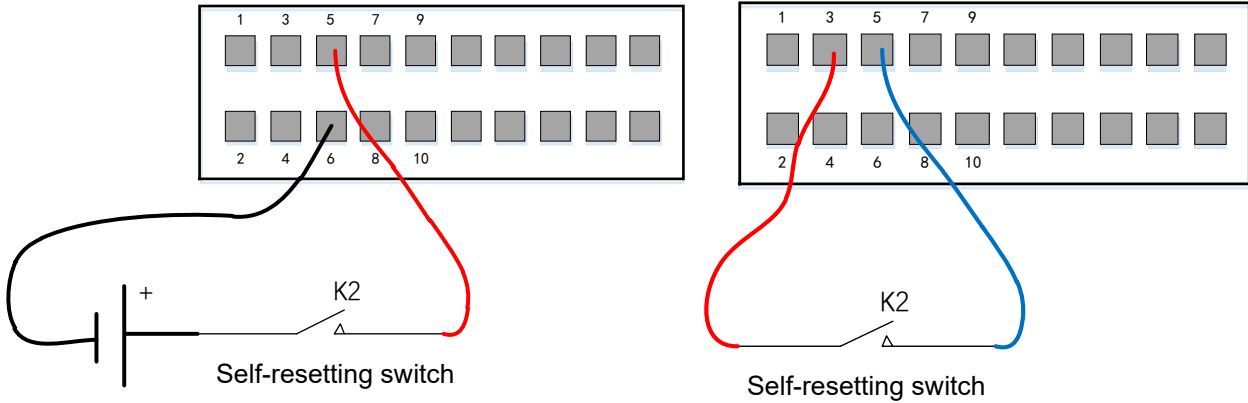


Figure 6-28: Remote off wiring

6.7.4.1.4. RS485 Interface

Pin 17 to Pin 20 on the MiniCab front panel I/O interface are RS485 communication interfaces, which can be connected to external devices for communication. Pin 17 and Pin 18 are master interfaces and Pin 19 and Pin 20 are slave interfaces. The diagram below shows the wiring diagram.



NOTE:

It is recommended to connect a 120Ω resistor at the terminal during wiring, and the YAGEO MF0207FTE52-120R resistor model is recommended.

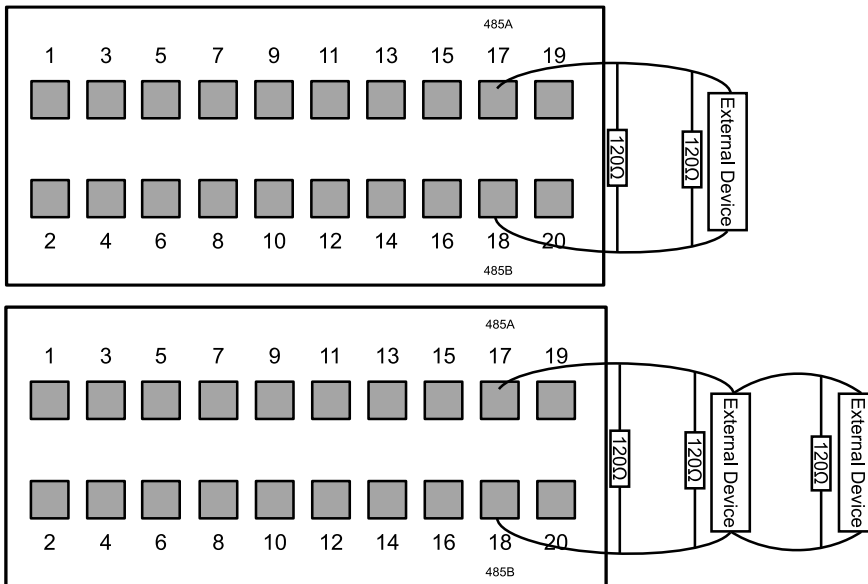


Figure 6-29: RS485 wiring (MiniCab acting as Client)

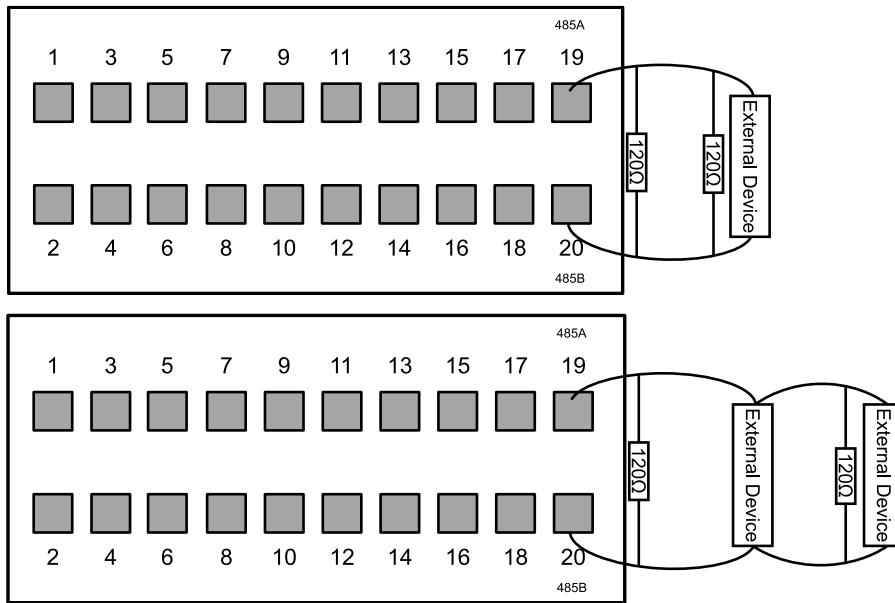


Figure 6-30: RS485 wiring (MiniCab acting as Server)

6.7.4.2 Protective Stop Interface

The MiniCab provides configurable safety function input interfaces in the I/O interfaces, which the protective stop function is valid by configuring the interface. This interface is designed with dual redundancy, and the function can be activated when either signal is valid. If external safety equipment is required, select devices that with dual redundancy.

Users can connect safety doors, safety light curtains, and sensors according to requirements.

6.7.4.2.1 Factory Settings

The Pin 4 & Pin 12 and Pin 6 & Pin 10 are shorted by default, which means the protective stop interfaces are shorted to the GND. The wiring diagram is as follows.

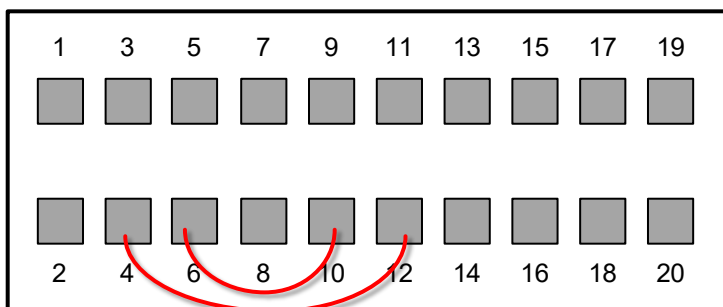


Figure 6-31: Protective stop wiring (factory settings)

6.7.4.2.2 Connect External Protective Stop Switch

For safe operation, users can remove the jumper and use one or more external protective stop switches. The wiring diagrams for single switch and multiple switch connections are as follows.

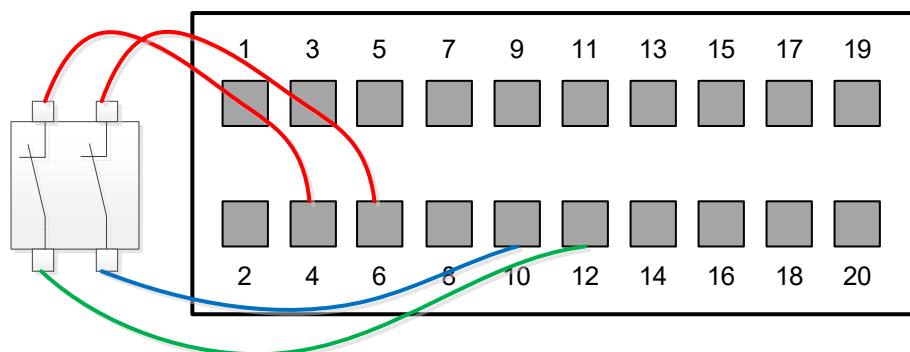


Figure 6-32: Protective stop wiring (single switch)

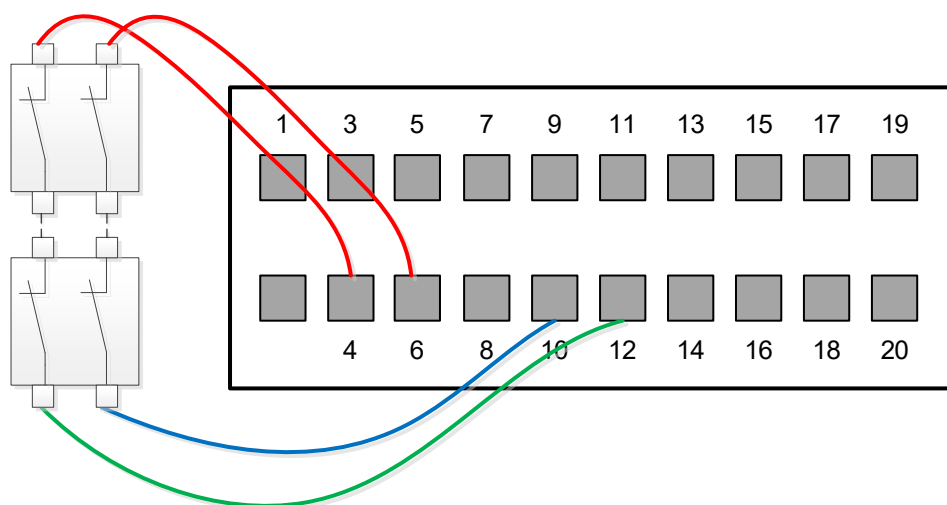


Figure 6-33: Protective stop wiring (multiple switch)



NOTE:

Before connecting the protective stop switches, you must configure the DI6 & DI7 as protective stop in the "I/O" page of the JAKA App.

6.7.4.3 USB Interface

The USB interface on the MiniCab front panel is for internal debugging use only. If you need to use it, please contact JAKA technicians.

6.7.4.4 HDMI Interface

The HDMI interface on the MiniCab front panel is for internal debugging use only. If you need to use it, please contact JAKA technicians.

6.7.4.5 Network Interface (LAN)

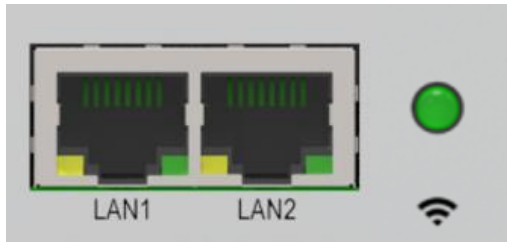


Figure 6-34: Network interfaces

6.7.4.5.1. Network Configuration

MiniCab has two network ports on its front panel, LAN1 and LAN2. Insert the network cable into the port to connect. In addition, there is a network status indicator on the front panel, which is used to display the current network status.

1. LAN1

The LAN1 port is a 10M/100M adaptive port, which is configured as 10.5.5.x by default. When connecting to the LAN1, the device's IP address should be configured within the range of 10.5.5.101 to 10.5.5.254, or set the device to obtain an IP address automatically.



NOTE:

If the device used does not support IP modification, it is recommended to connect the device to LAN2 and configure the IP address of LAN2 to be in the same network segment as the device. If the LAN2 is already occupied, please contact JAKA technician to modify the LAN1 network settings.

2. LAN2

The LAN2 port is a 10M/100M/1000M adaptive port, which is configured as obtaining IP address automatically by default. It can be configured as a static IP through the JAKA App. Refer to the software user manual for configuration steps.



NOTE:

The IP for the LAN2 cannot be the 10.5.5.x. If it is configured as this IP, the "The segment is disabled" will pop up.

6.7.4.5.2. Wi-Fi Connection

MiniCab has a Wi-Fi module, with a default setting of no password. Thus, the MiniCab can be connected via Wi-Fi. Once the MiniCab is powered up, the Wi-Fi indicator on the right side of the panel will illuminate. When connected to Wi-Fi, the indicator will flash. The Wi-Fi name is the MiniCab serial number, which is located on the label. If you need to change the Wi-Fi name, add a login password, etc., please contact JAKA technicians.

6.7.4.5.3. Network Reset

The MiniCab supports network reset. The reset button is located on the left side of the antenna (as shown in the figure below). In case you forget the Wi-Fi name and password, press and hold this button for more than 10 seconds to reset the Wi-Fi. At this point, the Wi-Fi gateway address will be restored to factory settings. For further instructions, please contact JAKA technicians.

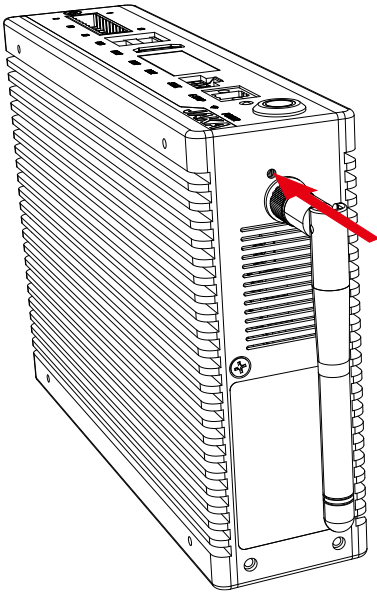


Figure 6-35: Network reset position

6.7.4.6 Control Stick Interface (STICK)

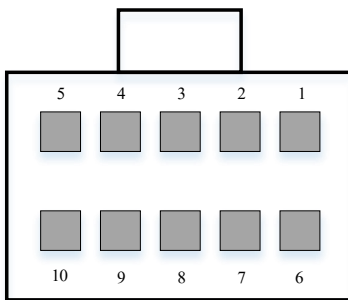


Figure 6-36: Control stick interfaces

The STICK interface is used to connect the control stick. Simply insert the control stick connection cable into this interface to connect. This interface is only for connecting the JAKA control stick and should not be modified.

6.7.4.7 Emergency Stop Interface (E-STOP)

The control cabinet provides safety function input interfaces, which the emergency stop function is valid by configuring the interface. This interface is designed with dual redundancy, and the function can be activated when either signal is valid. If external safety equipment is required, select devices that with dual redundancy.

Users can connect safety doors, safety light curtains, and sensors according to requirements.

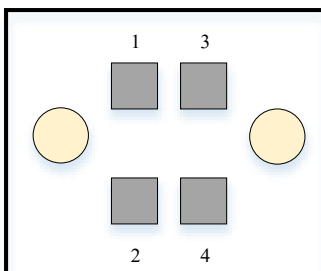


Figure 6-37: Emergency stop interfaces

The external emergency stop input interface is double row 3.5 mm (0.14 in) pitch pluggable terminal blocks, and the interface definition is as follows.

Table 6-12: Specifications of emergency stop interfaces

PIN	Signal	Signal Type	Description
1	VCC_24V	PO	Internal logic power 24V output
2	ESTOP1	I	E-STOP input 1, short connected to PIN1 by default
3	VCC_24V	PO	Internal logic power 24V output
4	ESTOP2	I	E-STOP input 2, short connected to PIN3 by default

When wiring, select the wire that matches the terminal. The requirements for the wires are as follows:

Table 6-13: Wire specifications of emergency stop interfaces

Terminal Model	Recommended PIN Connector	Wires
DEGSON : 15EDGKNHBM-3.5-04P-14-01A(H)	China: E0512 Overseas: PHOENIX AI1-12RD-Ferrule	0.75 mm ² & 17AWG Length < 30 m (1181 in)

6.7.4.7.1.Factory Settings

When not connected to an external emergency stop device, pins 1 & 2 and pins 3 & 4 should be shorted. By default, they are shorted to the internal 24V. The wiring diagram is as follows.

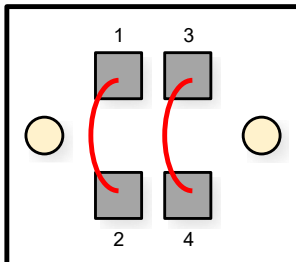


Figure 6-38: Emergency stop wiring (factory settings)

6.7.4.7.2.Connect External E-Stop Switch

For safe operation, users can remove the jumper and use one or more external emergency stop switches. The wiring diagrams for single switch and multiple switch connections are as follows.

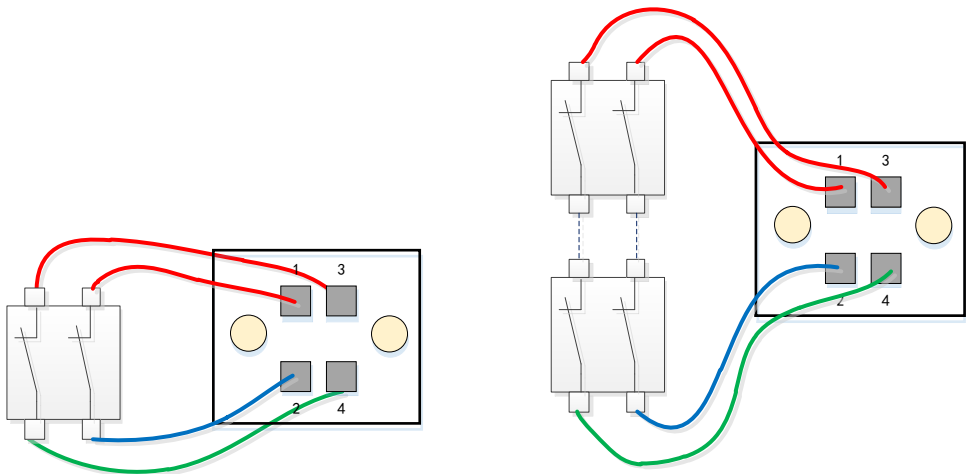


Figure 6-39: Emergency stop wiring

The robot can be used without the control stick, in which case additional emergency stop devices need to be connected. This interface can be used to connect an emergency stop switch to ensure safety.



WARNING

If the control stick is disconnected from the robot, the emergency stop button on the control stick will no longer be effective. The disconnected control stick must be moved away from the robot.

6.7.4.8 POWER Button



Figure 6-40: Power button

The MiniCab can be powered up and powered down by POWER button, the operation steps are as follows.

Power up: Press the POWER button for 1 seconds, and then the control cabinet will be powered up.

Power down: Press the POWER button for more than 3 seconds, and then the control cabinet will be powered down.

6.7.4.9 Status Indicator

There is a ring of status indicator around the POWER button on the front panel of MiniCab, which used to indicate the robot state.

Table 6-14: Color meaning of status indicator

Color	Robot Status
Blue	Robot is powered on and disabled
Green	Robot is enabled

Color	Robot Status
Red	Robot is in error
Yellow	Robot is in freedrive mode
Flash in yellow	Robot is paused

6.7.5 Side Panel Interfaces

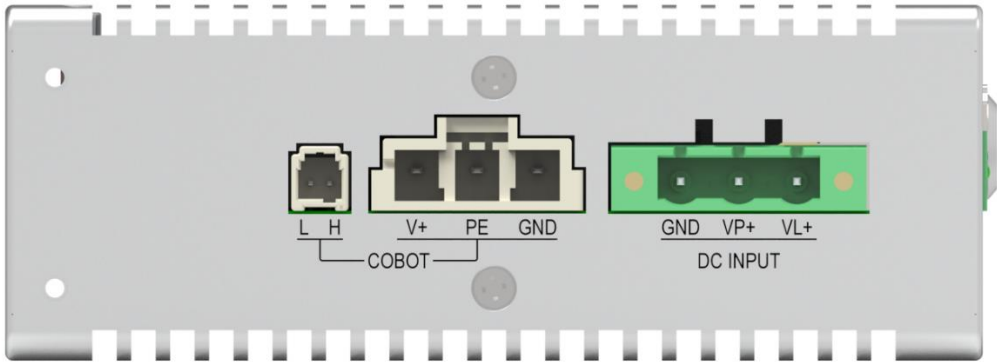


Figure 6-41: Side panel interfaces

6.7.5.1 Power Interface

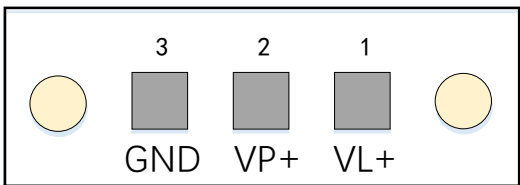


Figure 6-42: Power interfaces

MiniCab power interface contains three input interfaces: control cabinet power input VL+ (PIN1), robot power input VP+ (PIN2) and common negative GND (PIN3). Control cabinet power (VL+) supplies power to the control cabinet, while robot power (VP+) supplies power to the robot. Additionally, VP+ can also supply power to the control cabinet. Interface definitions are as follows.

Table 6-15: Specifications of power interfaces

PIN	Signal	Signal Type	Description
1	VL+	PI	Control cabinet power input
2	VP+	PI	Robot power input
3	GND	PI	0V input

When wiring, select the wire that matches the terminal. The requirements for the wires are as follows:

Table 6-16: Wire specifications of power interfaces

Terminal Model	Recommended Pin Connector	Wires
DEGSON: 2EDGKM-7.62-03P-14	China: E2508 Overseas: PHOENIX AI2,5-8GY-Ferrule	2 mm ² & 14AWG Length < 30 m (1181 in)

The robot power input (PIN2) supplies power to the robot and can also supply power to the control cabinet. When control cabinet power and robot power do not need to be separated, connect PIN2 and PIN3. If you need to disconnect VP+ in emergency situations while keeping the control cabinet powered, you can separately connect the power supply to VL+. The wiring diagram is as follows.

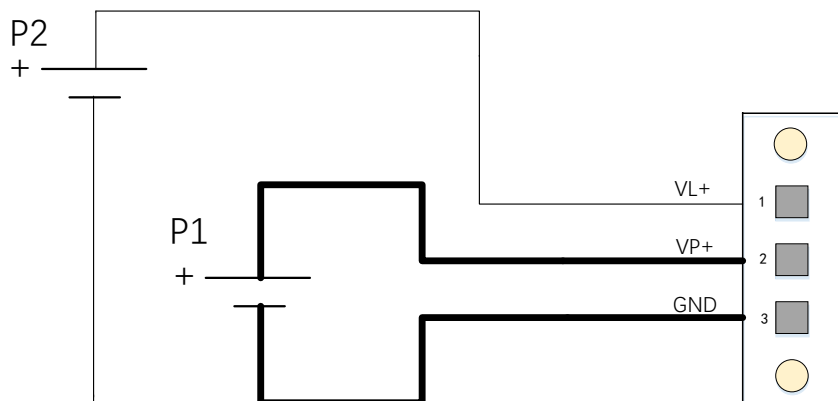


Figure 6-43: Power wiring

1. The robot power supply requirements are as follows.

Table 6-17: Robot power supply requirements

Robot Model	MiniCobo	Mini 2
P1	Rated voltage	24 V DC
	Voltage range ⁱ	20~30 V DC
	Current range	0~12 A
	Peak power consumption	280 W
	Recommended model ⁱⁱ	GST280A24, MW
		HRP-300N3-48, MW

2. The control cabinet power supply requirements are as follows.

Table 6-18: Control cabinet power supply requirements

Robot Model	MiniCobo	Mini 2
P2	Rated voltage	24 V DC
	Voltage range ⁱⁱⁱ	20~30 V DC
		40~56 V DC

Robot Model		MiniCobo	Mini 2
	Typical power consumption	12 W	12 W
	Max power consumption	≤30 W	≤ 30 W



NOTE:

i: 20V is not included; the MiniCobo robot under voltage threshold is 20V; 30V is not included; the Mini 2 robot under voltage threshold is 30V.

ii: The power supply models listed here are only for recommendation, you can purchase power supplies of equivalent specifications. Additionally, peak power consumption is dependent on the robot's load and usage scenario. The values listed here are the maximum value.

iii: If both VP+ and VL+ inputs are below 26V, the “UDIO_24V” output will be below 24V.

6.7.5.1.1. Brake Voltage Settings

MiniCab has a voltage braking circuit, which is used to bleed the electromotive force generated by the robot during deceleration and braking. When using an external power supply, it is necessary to set the braking voltage value in the JAKA App to avoid that the control cabinet is powered down by over voltage protection or damage to the control cabinet. For specific operations, please refer to the software user manual.



NOTE:

Before setting the braking voltage, the robot needs to be powered off and disabled.

6.7.5.2 Robot Connection Interface

The COBOT interface is used to connect the robot. This interface features a foolproof design with a locking mechanism. JAKA provides compatible cables for connection.

6.8 Initial Start-Up

6.8.1 Power On and Shut Down the Robot System

There are two options to power on and enable the robot:

- Powering on via control stick
- Powering on via JAKA App software

To power on and enable the robot via the control stick, proceed as follows:

1. Power up control cabinet: Press the **Power** button. The buzzer makes a noise, and the control cabinet is powered up.
2. Unlock the control stick: Press and hold the **Lock** button for 3 seconds, and the lock indicator goes out, which means the control stick is unlocked.
3. Power on robot: Press the **Enable** button, waiting for the ring-shaped light to turn blue, which means the robot is powered on.
4. Enable robot: When the robot is powered on, press and hold the **Lock** button, then press the **Enable** button at the same time until the ring-shaped light turns green, which means the robot is enabled.
5. Lock control stick: App cannot be operated when control stick is unlocked, you should lock control stick first. Press and hold the **Lock** button for 3 seconds, and the lock indicator turns orange, which means control stick is locked. You can use App now.

Steps of powering down control cabinet are as follows:

1. Unlock the control stick: Press and hold the **Lock** button for 3 seconds, and the lock indicator goes out, which means the control stick is unlocked.
2. Disable robot: When the robot is enabled, press and hold the **Lock** button, and then press the **Enable** button at the same time until the ring-shaped light turns blue, which means the robot is disabled.
3. Power off robot: Press the **Enable** button, waiting for the ring-shaped light to turn blue, indicating that the robot is powered off.
4. Power down control cabinet: Press and hold the **Power** button for 3 seconds or more. The control stick beeps 6-7 times, and the control cabinet is powered down. After powering off the control cabinet, please do not cut off the power immediately. Wait until the control stick light goes out, and after waiting for 5 to 10 seconds, you can disconnect the power.



NOTICE

Under non-emergency conditions, the robot system shall be operated strictly in accordance with the standard power-on/shutdown procedures. Forced power disruption is prohibited, as it will cause data loss and compromise system reliability.

6.8.2 Set the Limits

Before commissioning the robot, the following items should be verified:

- Verify the emergency stops and the operator protective devices.
- The robot complies with the relevant standards and protective devices are designed to stop the robot without leaving the path (category 1 stop).
- Robot functions successfully within preset limits Slowly move the robot beyond the limits of the preset working space in order to verify that this is prevented by the preset limits.
- Individually move the robot beyond the maximum/minimum angles in order to verify that this is prevented by the preset limits.

6.8.3 Start-Up

When the robot is operated for the first time, there is a risk of unintended equipment operation caused by possible wiring errors, improper mounting and fastening, or unsuitable parameters.



WARNING

Unintended equipment operation:

- Verify that the robot is properly and firmly fastened.
- Take all necessary measures to ensure that the moving parts of the robot cannot move in an unanticipated way.
- Verify that emergency stop equipment is operational and within reach of the zone of operation.
- Verify that the system is obstacle-free and ready for the movement before starting the system.
- Run initial tests at reduced velocity.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

If the robot power supply is disabled unintentionally, for example as a result of power outage or errors, the robot is no longer decelerated in a controlled way.



WARNING

Unintended equipment operation:

Verify that movements without braking effect cannot cause injuries or equipment damage.
Failure to follow these instructions can result in death, serious injury, or equipment damage.



WARNING

Hot surfaces:

- Do not allow flammable or heat-sensitive parts in the immediate vicinity of hot surfaces.
- Verify that the heat dissipation is sufficient by performing a test run under maximum load conditions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

6.8.3.1 Commissioning Procedure

To commission the robot, proceed as follows:

1. Comply with the instructions provided in this manual.
2. Verify that the load conforms to the specified payloads for the robot before operating.
3. Connect the robot in the JAKA App. For further Information, refer to the JAKA App Software User Manual.
4. Configure the installation setting of the robot. For further Information, refer to the JAKA App Software User Manual.
5. Verify the calibration and position and direction of the joints.
6. Verify the orientation of the world coordinate system in the manual operation interface. For further Information, refer to the JAKA App Software User Manual.



NOTE

The world coordinate system depends on the mounting position.

7. Configure the collision protection settings in accordance to the safety-related requirements of your application. For further information on the configuration, refer to the JAKA App Software User Manual.
8. Perform initial tests at reduced velocity and verify the functionality of the robot.
9. Verify that the operating condition of set cycles is lower than 80% of the total cycle time.



NOTE

For a total cycle time of 10 seconds take care of maximum 8 seconds cycle running time and 2 seconds waiting period. If you exceed the duty cycle proportion of 80%, you may reduce the lifetime of the gearboxes of the robot joints.

10. Optionally, configure the home position for the Home button operation of the control stick. For further information on the control stick operations, refer to [5.6 Control Stick Buttons](#), and for the configuration of the home position, refer to the JAKA App Software User Manual.
11. In case of using several robot systems, optionally assign unique names for the devices for easier identification. For further information, refer to the chapter Initial Settings in the JAKA App Software User Manual.

6.8.3.2 Starting the Default Program

There are two options for automatically starting the program set as default on the robot:

- Starting via control stick
- Starting via JAKA App software

To start the default program via the control stick, press the **Start/Stop** button on the control stick.

For further information on starting and setting the default program via JAKA App, refer to the JAKA App Software User Manual.

6.9 Mounting the End-Effector

6.9.1 Prerequisites

Observe the following prerequisites to help avoid incorrect operation of the robot:

- Verify that the load conforms to the specified payloads for the robot before operating.
- Limit the maximum payload of the robot in accordance with the maximum load capacity of the robot. For further information, refer to [5.8.1 Load Capacity of the Robot](#).

6.9.2 Dimensions of Robot Flange

The following figure presents the installation dimensions of robot flange.

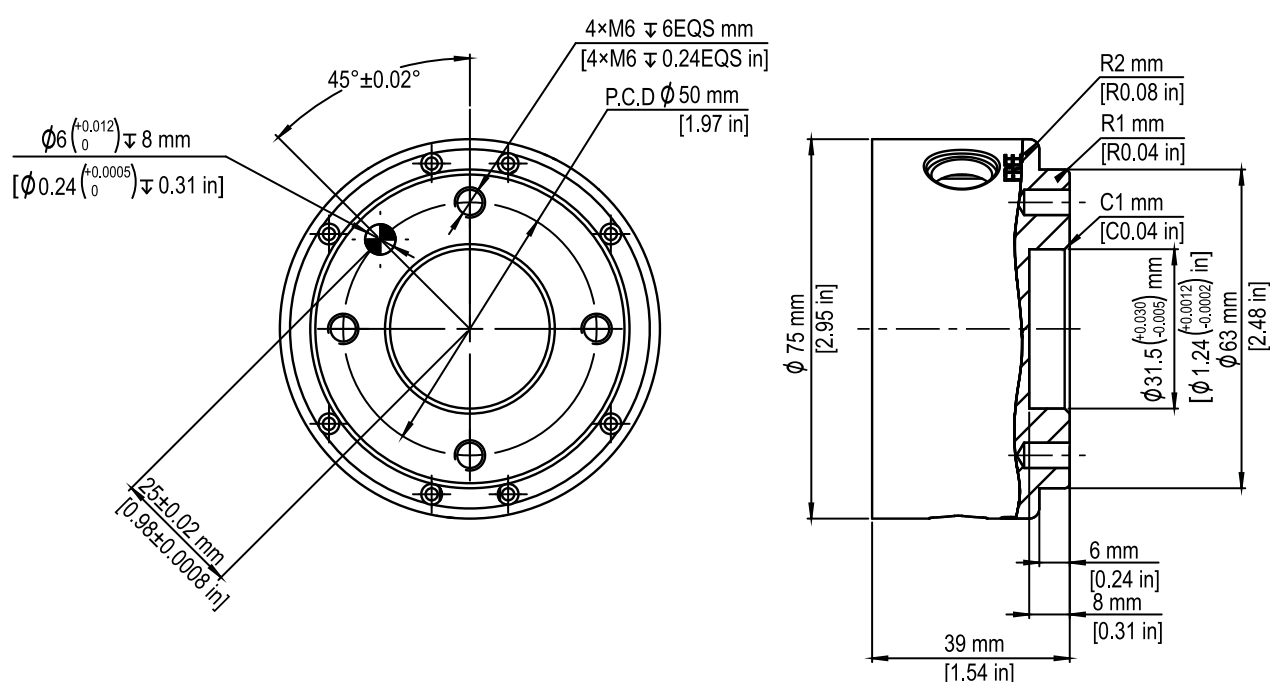
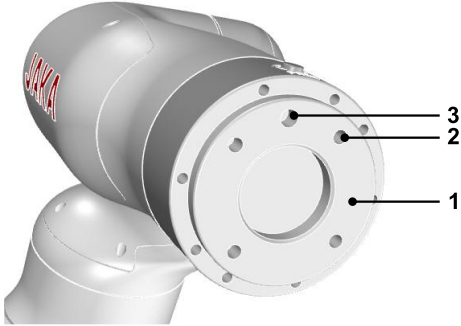


Figure 6-44: Flange dimensions

6.9.3 Mounting the End-Effector

Table 6-19: End-effector mounting steps

Step	Action
1	Fasten the end-effector to the mounting points provided for this purpose on the robot tool flange (1) • Pitch circle diameter DIN ISO 9409-1, 50 mm (1.97 in): 4x M6 (2), tightening torque: 15.3 Nm

Step	Action
	(135.42 lbf·in), property class of the screw: 12.9 or greater • Pitch circle diameter 50 mm (1.97 in): 1x fitting hole diameter 6 H7 (3)  Figure 6-45: Robot flange For further information on the flange dimensions, refer to 6.9.2 Dimensions of Robot Flange.
2	Set the payload in JAKA App. For further information, refer to the JAKA App Software User Manual.



NOTE

- Observe the permissible weights and distances that results in load capacity of the robot.
- According to ISO9409-1: 2004, the location pin hole center shall be aligned with the mechanical interface coordinate system (ISO 9787:2013)+Xm, and our product is offset 45° clockwise on this basis.

7 Maintenance

All safety instructions in this manual must be strictly followed for maintenance work. For more detailed maintenance instructions, please refer to the JAKA Service Manual.

The repair must be performed by an integrator authorized by JAKA or by JAKA's personnel.

After-sales service contact information: E-mail: support@jaka.com.

7.1 Safety Instructions

After maintenance, check to ensure the safety level required by the service. Valid national or local safety laws and regulations must be observed during the check. At the same time, check if all safety functions are functioning properly.

The purpose of maintenance is to ensure the normal operation of the system, or to help the system return to normal operation in the event of a failure. The maintenance includes fault diagnosis and actual maintenance.

The following safety procedures and warnings must be followed during the operation of the robot or the control cabinet:



WARNING

- It is forbidden to modify any information in the software safety configuration. If the safety parameters are changed, the entire robot system should be considered as a new system, which means that all safety examination processes, such as risk assessment, must be updated.
- Replace a failed component with the same component number or equivalent approved by JAKA.
- Reactivate all disabled safety measures immediately after maintenance is completed.
- Record all maintenance operations and save them in technical documents related to the entire robot system.



WARNING

- Remove the power cord from the bottom of the control cabinet to ensure that it is completely powered down. Disconnect other energy sources connected to the robot or the control cabinet. Take necessary precautions to prevent others from energizing the system during maintenance.
- Check the ground connection before re-powering the system.
- Observe the ESD regulations when disassembling the robot or the control cabinet.
- Avoid disassembling the power supplies inside the control cabinet. High voltage may remain in the power supply system for several hours after the control cabinet is powered down.
- Avoid water or dust entering the robot or the control cabinet.

For further information, refer to the JAKA Service Manual.

8 Troubleshooting

Table 8-1: Troubleshooting list

Issue	Probable Cause	Solution
Robot is not reachable via JAKA App.	Control cabinet is powered off.	Power on the control cabinet.
	Blocked by firewall.	Verify the firewall settings.
	Not connected to the Wi-Fi of the control cabinet.	Connect to the Wi-Fi of the control cabinet.
	Not connected to the control cabinet via Ethernet cable.	Verify the connection and functionality of the cable.
	Incorrect IP address.	Configure the IP settings accordingly.
Robot could not be enabled.	Joint x: self-learning flow or servo overcurrent.	Release the brake of joint x. Contact your local JAKA service representative
	Mechanical blocking of joint x.	
	Emergency stop is pressed.	Release the emergency stop button.
Robot is not powering on.	The simulation mode is active.	Switch to the real robot.
	Emergency stop is pressed.	Release the emergency stop button.
Emergency stop	Emergency stop is pressed.	Release the emergency stop button.
	No control stick connected.	Connect the control stick properly.
Default program is not executed.	Default program is not selected.	Select a default program in JAKA App.
Initial orientation is not as intended.	Initial orientation has not been adjusted on initial start-up.	Adjust the initial orientation in JAKA App.
Unexpected stop of robot during movement.	Overheating of the control cabinet.	Verify that there is sufficient airflow.

9 Disposal

This section contains information to handle potentially dangerous components and potentially hazardous materials.

JAKA products contain components in different materials. During decommissioning, all materials should be dismantled, recycled, or reused responsibly, according to the relevant laws and industrial standards.

EU RoHS

JAKA robots are produced with restrictions on the use of hazardous substances to protect the environment; they comply with the definition of the EU RoHS Directive 2011/65/EU. The substances restricted by RoHS include mercury, cadmium, lead, chromium VI, polybrominated biphenyls and polybrominated diphenyl ethers.

EU WEEE

Fee for disposal and handling of electronic waste of JAKA robots sold on the German market is prepaid to DPA-system by JAKA. Importers in countries covered by the European WEEE Directive 2012/19/EU must make their own registration to the national WEEE register of their country. A list of national registers can be found here: <https://www.ewrn.org/national-registers/national-registers>.

The following symbol indicates that the product must not be disposed of as common garbage. Handle each product according to local regulations for the respective content.



Figure 9-1: Disposal symbol

China RoHS

The following symbol shows the information to hazardous substances and the environmental protection use period of JAKA products according to Management Methods for the Restriction of the Use of Hazardous Substances in Electrical and Electronic Products (SJ/T 11364-2014).



Figure 9-2: Environmental protection use period symbol

The orange icon indicates that this product contains certain hazardous substances. The '20' in the icon represents the environmental protection period. It can be used during the environmental protection period and should be recycled after exceeding that time.

The table below shows the name and content of toxic and hazardous substances in the product.

Table 9-1: Toxic and hazardous substances list

Part Name	Pb	Hg	Cd	Cr (VI)	PBB	PBDE
Metal parts	x	o	o	o	o	o

Part Name	Pb	Hg	Cd	Cr (VI)	PBB	PBDE
Plastic parts	o	o	o	o	o	o
Electronic	x	o	o	o	o	o
Electrical contacts	o	o	o	o	o	o
Cables & cabling accessories	o	o	o	o	o	o



NOTE

- The table is made according to SJ/T 11364.
- o indicates that the concentration of hazardous substance in all the homogeneous materials for this part is below the limit as stipulated in GB/T 26572.
- x indicates that the concentration of hazardous substance in at least one of the homogeneous materials for this part is above the limit as stipulated in GB/T 26572.

10 Design Standards and Certification

10.1 Certification Description

10.1.1 Third-Party Certification

The robot is certified by the following inspection bodies.

Table 10-1: Third-party certification

	Body	Description
	SGS	The Mini series robots have passed the safety certification of the notified body, SGS, which complies with the EU machinery directive 2006/42/EC.
	KCs safety	The MiniCobo robot has been evaluated as compliant with the KCs safety standard.

10.1.2 Manufacturer Test Certification

Table 10-2: Manufacturer test certification

	Body	Description
	JAKA	The JAKA robot undergoes continuous internal factory tests and type testing procedures.

10.1.3 Declaration According to EU Directives

JAKA robot has been certified according to the following directives.

Table 10-3: Declaration according to EU Directives

Name	Description
2006/42/EC	Machinery Directive (MD)
2014/30/EU	Electromagnetic Compatibility (EMC)
2014/53/EU	Radio Equipment Directive (RED)



NOTE

- Robots are covered by both the MD and LVD directives. Based on the MD directive, only MD is required to be evaluated, but the harmonized standards of LVD are used simultaneously to ensure electrical safety when conducting MD assessments.
- EMC is mainly used to ensure compliance of robot without Wi-Fi module, EMC compliance of

robot with Wi-Fi module is included in RED.

- Certificates and declarations can be found on the official website www.jaka.com.

11 Warranties

11.1 Product Warranty

Without prejudice to any indemnity agreement that the user (customer) may reach with the distributor or retailer, the manufacturer shall give the user (customer) “product warranty” according to the following terms: If there is any defect due to manufacturing and/or material faults within the warranty period promised under the contract signed by JAKA, JAKA shall provide necessary spare components, the user (customer) shall send personnel to replace spare components, and replace or repair related components with new ones that reflect the latest technical level. If the equipment defect is caused by improper treatment and/or failure to follow the relevant information described in the user manual, this “Product Warranty” is invalid. This “Product Warranty” does not apply to or extend to maintenance carried out by authorized distributors or users (customers) themselves (e.g. installation, configuration, software download). The purchase receipt, together with the date of purchase, shall be required as evidence for invoking the Warranty. The warranty must be submitted within two months of the Warranty default becoming evident. JAKA has the ownership of the equipment or components that are replaced or returned to JAKA. Any other claim arising out of or relating to the equipment is excluded from the scope of the Warranty. Nothing in this Warranty shall attempt to limit or exclude a Customer’s Statutory Right, nor the manufacturer’s liability for death or personal injury resulting from its negligence. The duration of the Warranty shall not be extended by services rendered under the terms of the Warranty. Insofar as no Warranty default exists, JAKA reserves the right to charge the customer for replacement or repair. The above provisions do not imply a change in the burden of proof to the detriment of the customer.

11.2 Disclaimer

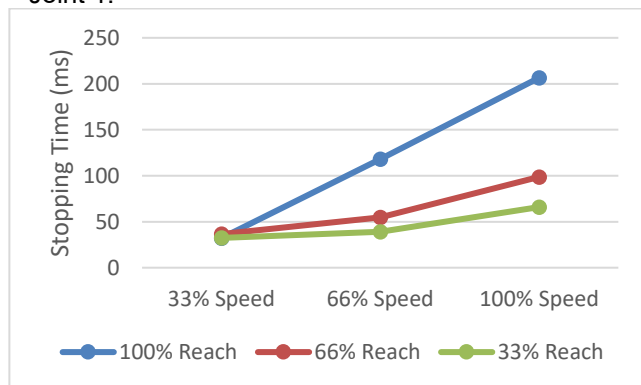
JAKA is committed to continuously improving the reliability and performance of the products, and therefore reserves the right to upgrade them. The products may be changed without notice. JAKA strives to ensure the accuracy and reliability of the contents in this manual but is not responsible for any error or omission herein.

Appendix

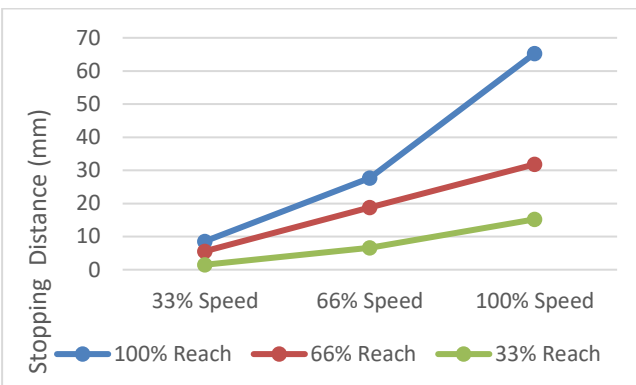
Appendix 1: Stopping Time and Distance

Table of type 1 stopping time and distance

Joint 1:

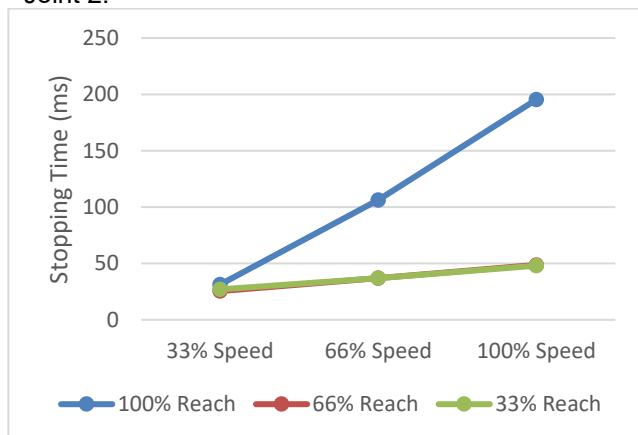


Joint 1 Type 1 Stopping time

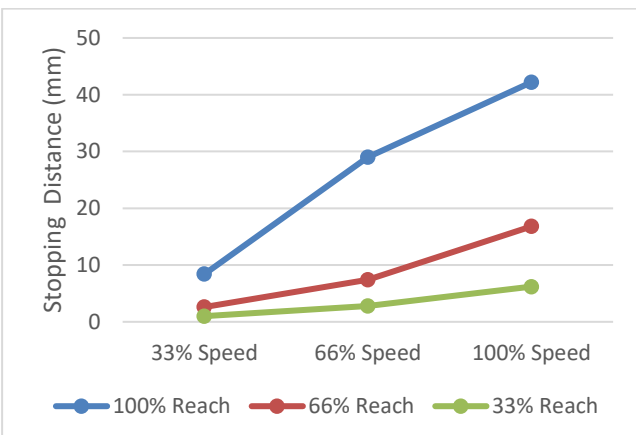


Joint 1 Type 1 Stopping distance

Joint 2:

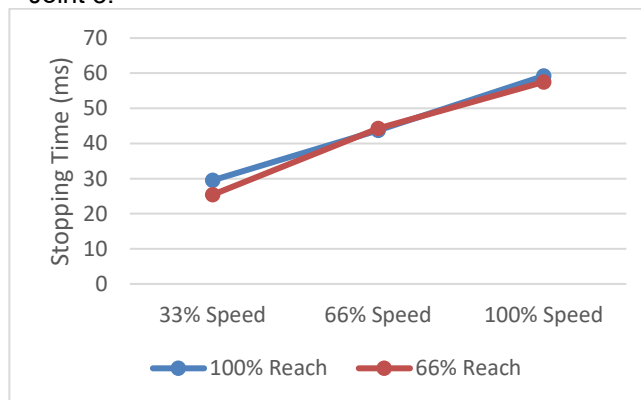


Joint 2 Type 1 Stopping time

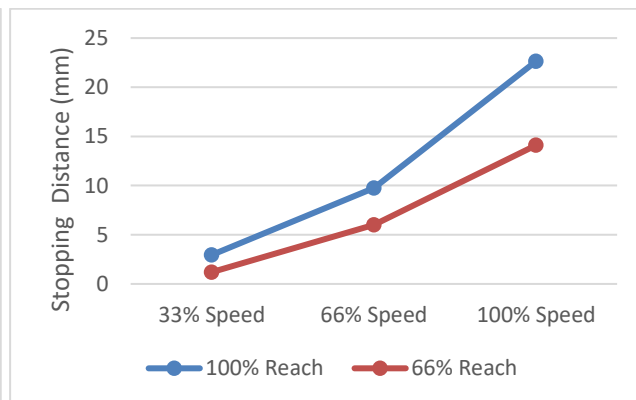


Joint 2 Type 1 Stopping distance

Joint 3:



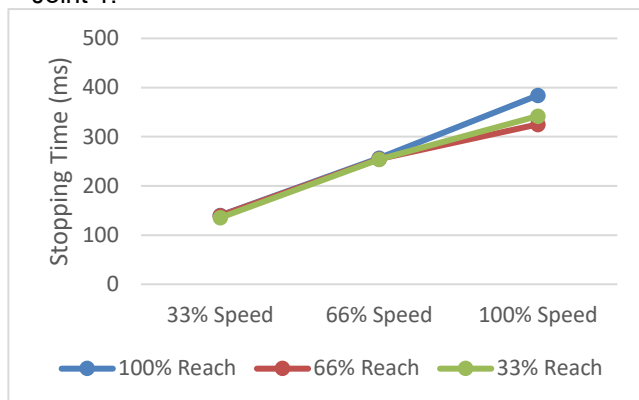
Joint 3 Type 1 Stopping time



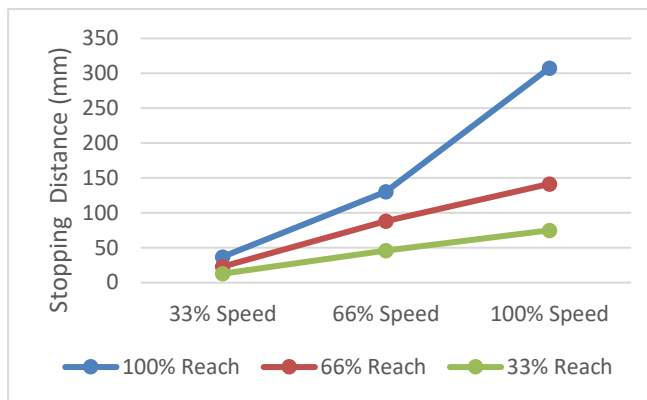
Joint 3 Type 1 Stopping distance

Table of type 2 stopping time and distance

Joint 1:

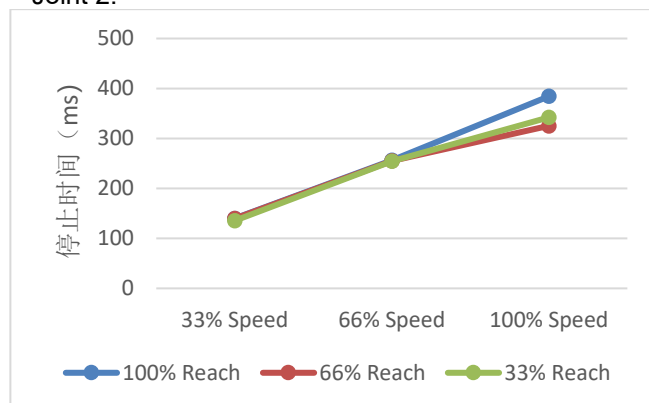


Joint 1 Type 2 Stopping time

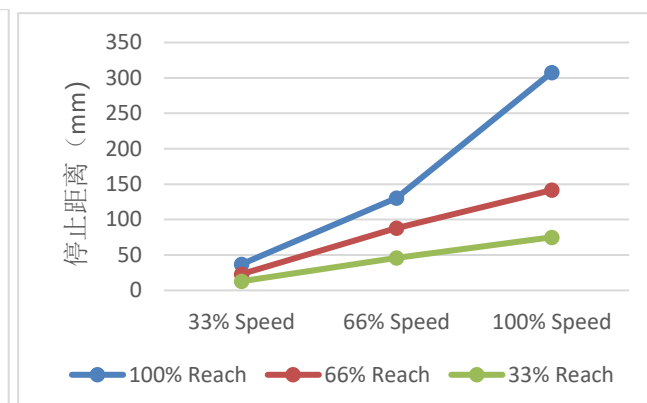


Joint 1 Type 2 Stopping distance

Joint 2:

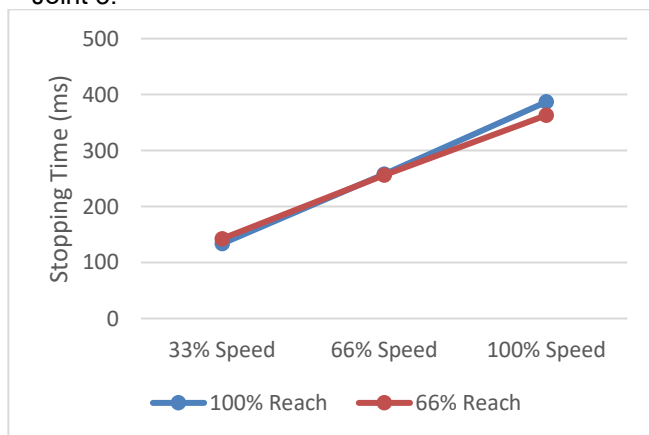


Joint 2 Type 2 Stopping time

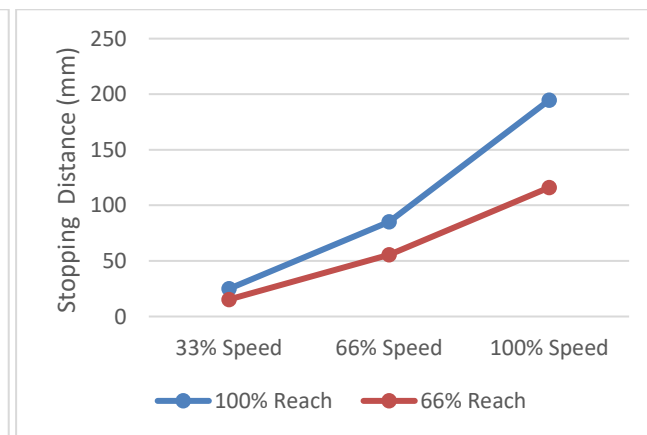


Joint 2 Type 2 Stopping distance

Joint 3:



Joint 3 Type 2 Stopping time



Joint 3 Type 2 Stopping distance



NOTE

- The unit of stopping distance is mm, 1 inch equals to 25.4 mm.
- This data is test data, the data results are affected by the version.

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